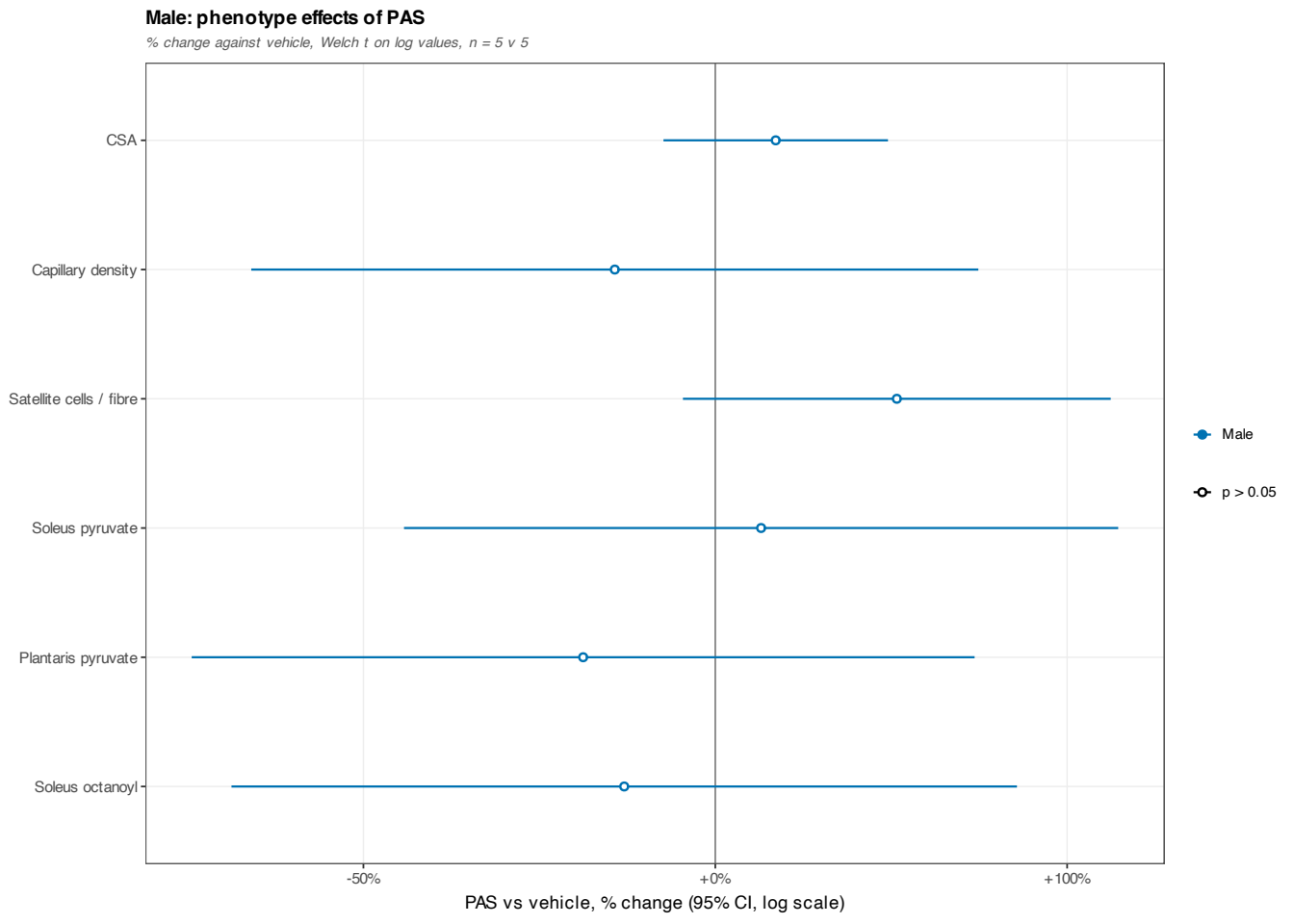
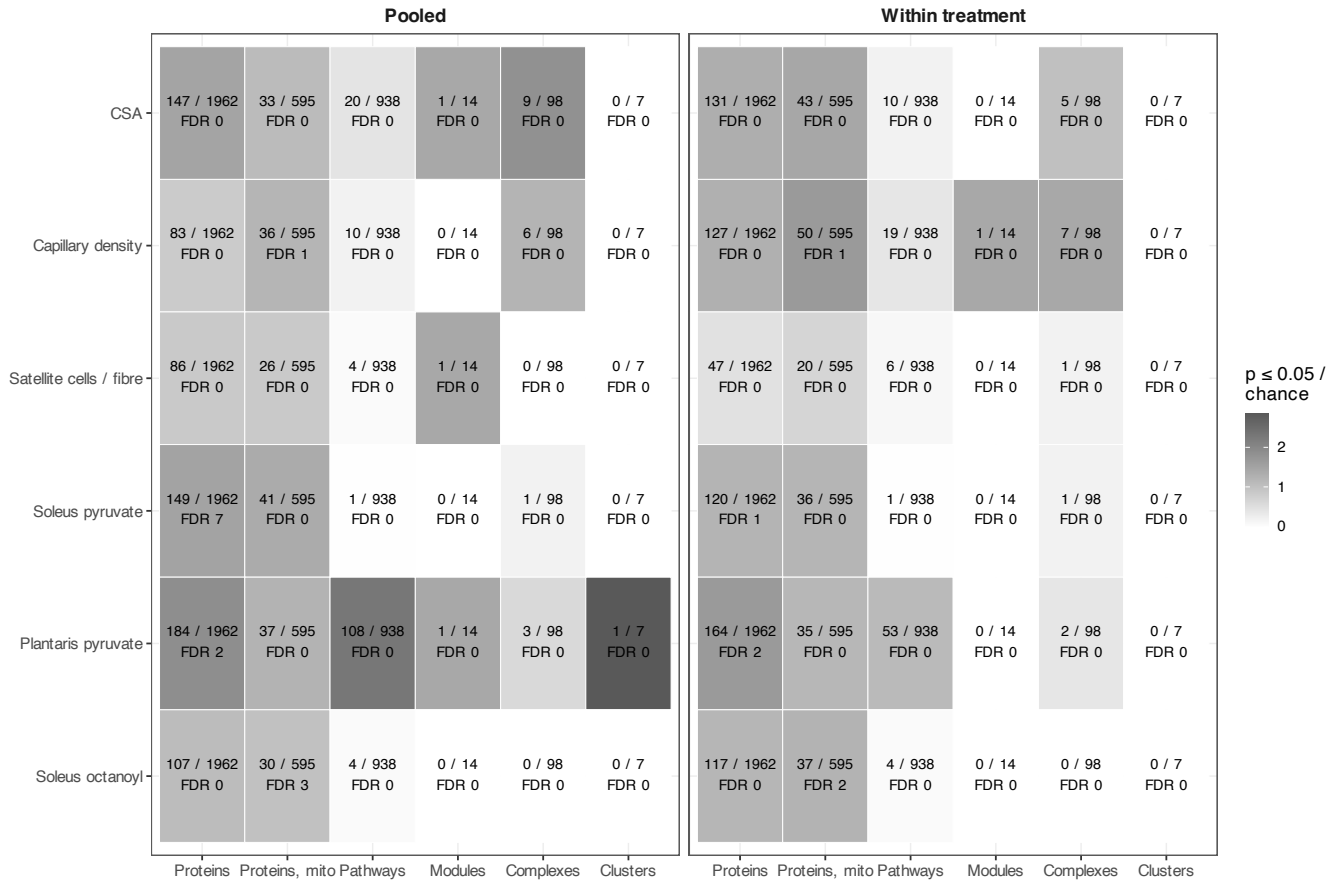

Male: phenotype effects of PAS
% change against vehicle, Welch t on log values, n = 5 v 5
CSA
Capillary density
Satellite cells / fibre
Soleus pyruvate
Plantaris pyruvate
Soleus octanoyl
-50%
+0%
+100%
PAS vs vehicle, % change (95% CI, log scale)
Male
p > 0.05

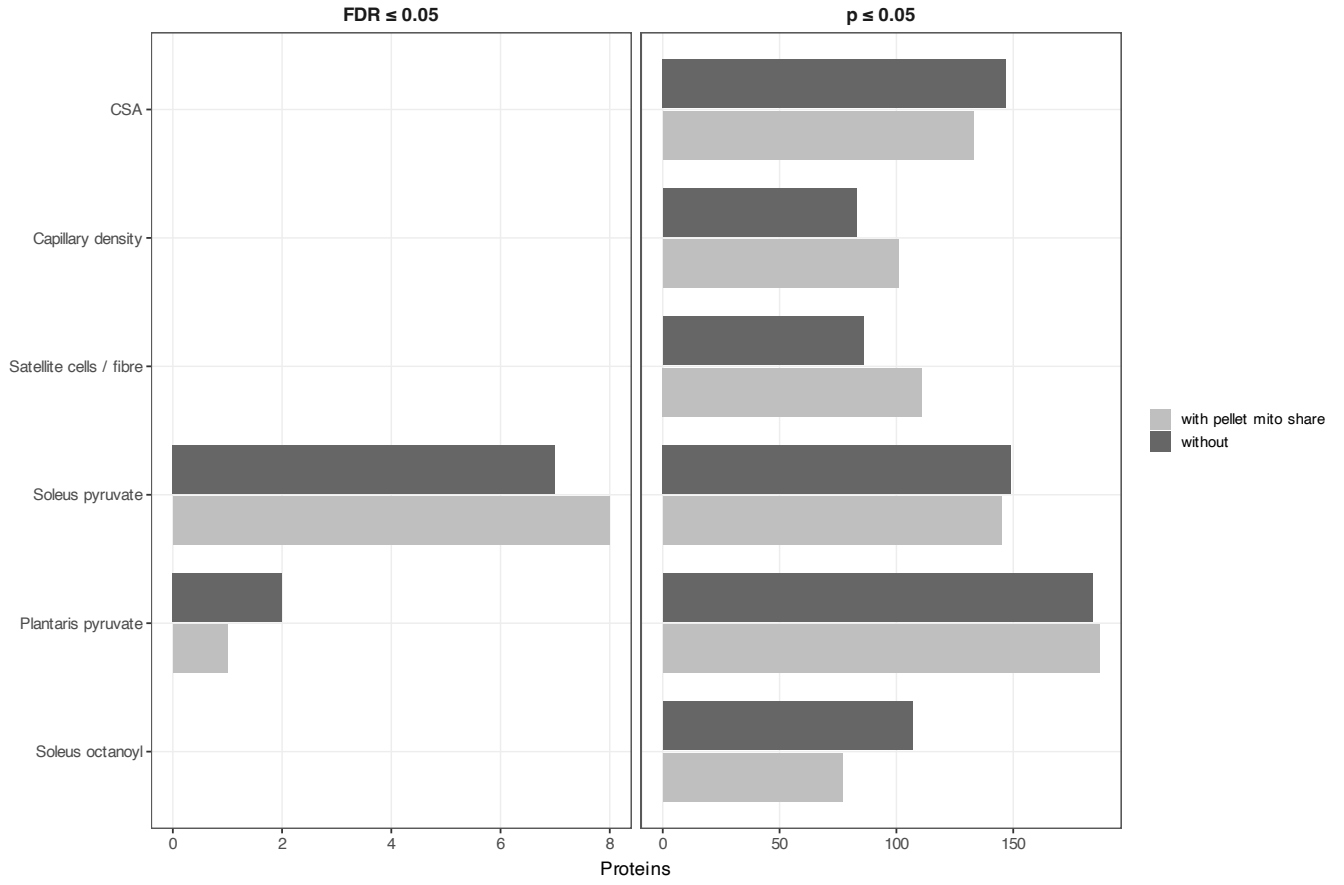
**Male: phenotype associations by level**

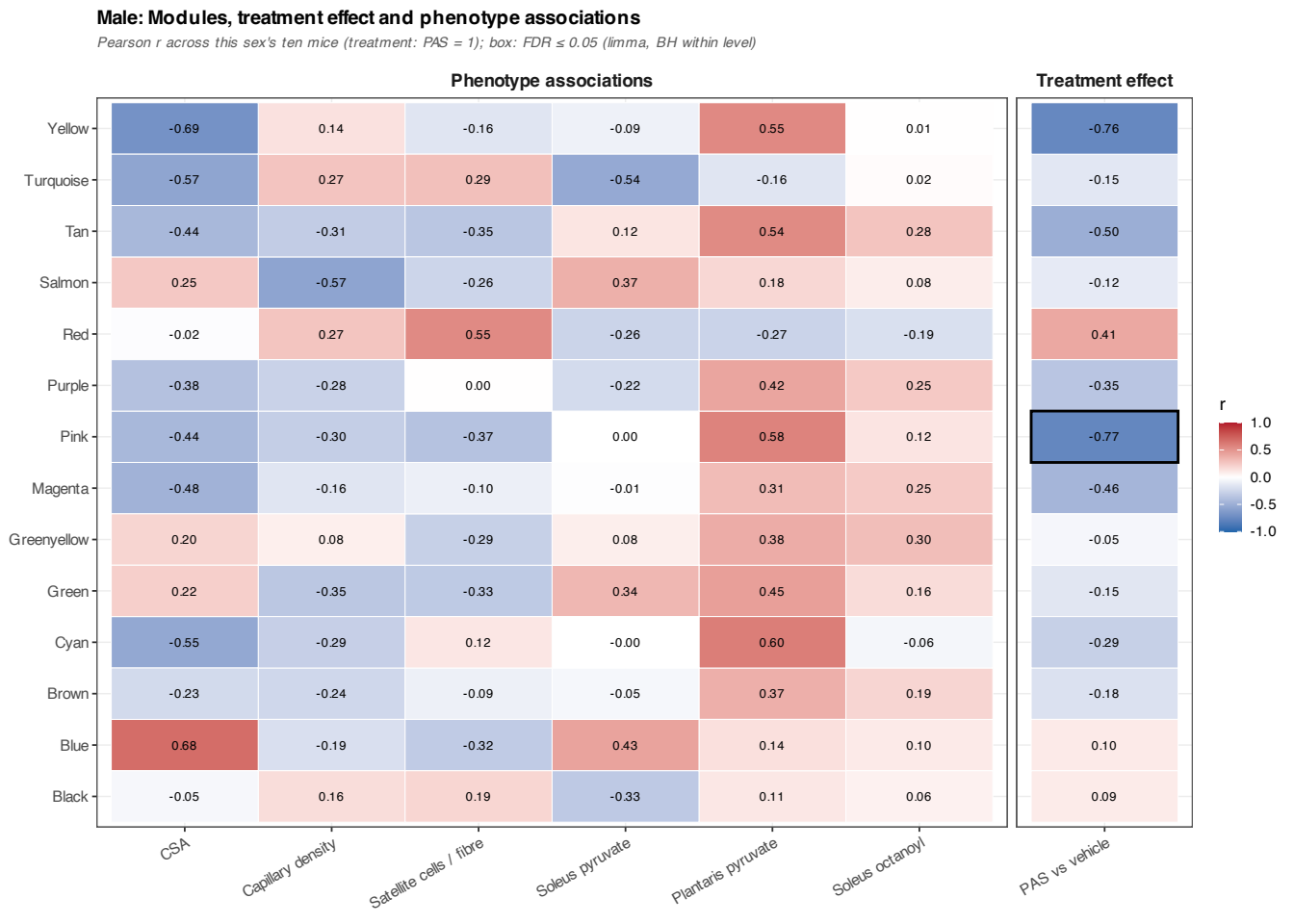
*This sex's ten mice, pooled across PAS and vehicle or within treatment; p ≤ 0.05 of tests; BH within level, phenotype and model*

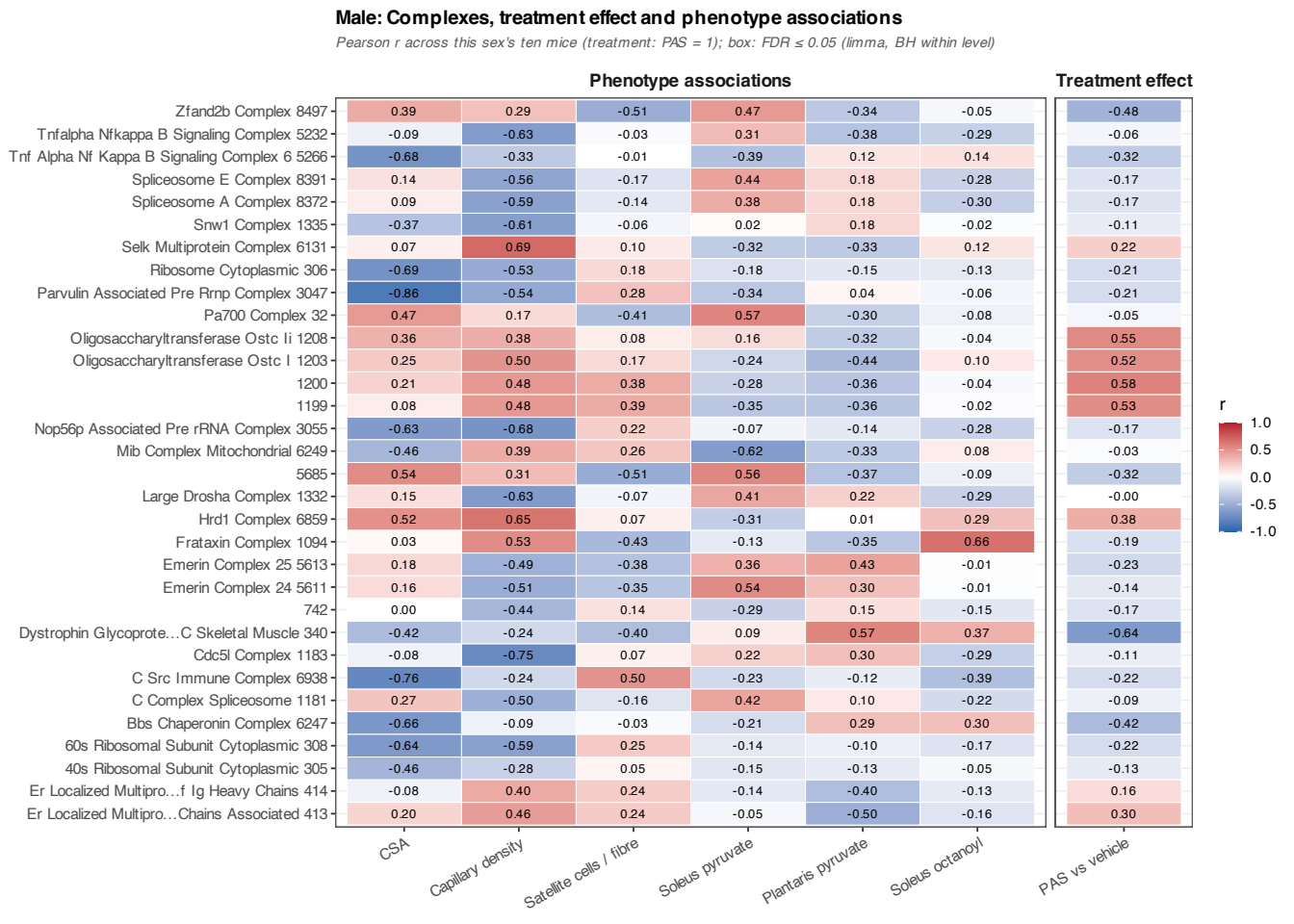


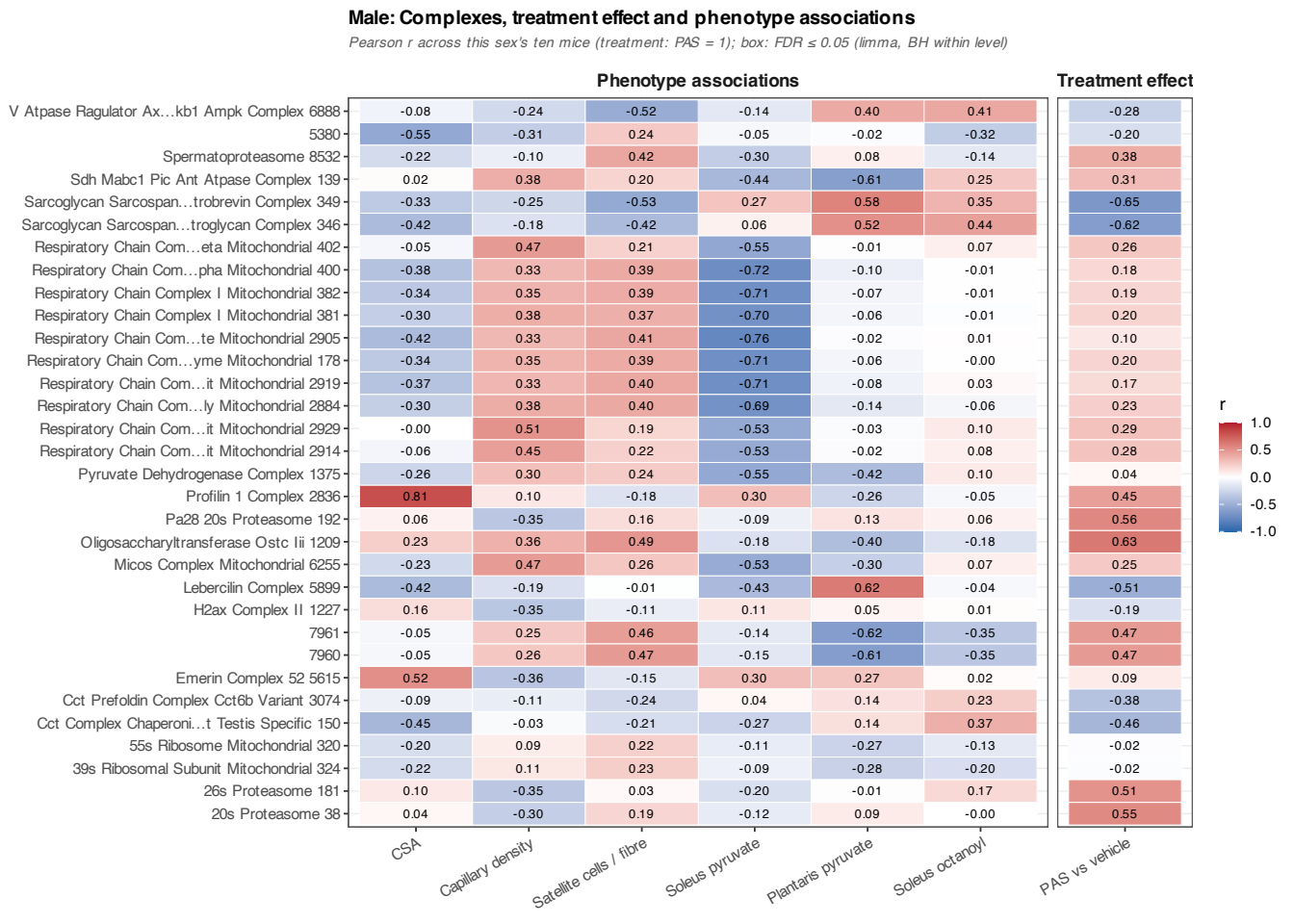
**Check: protein associations with and without pellet mitochondrial share**

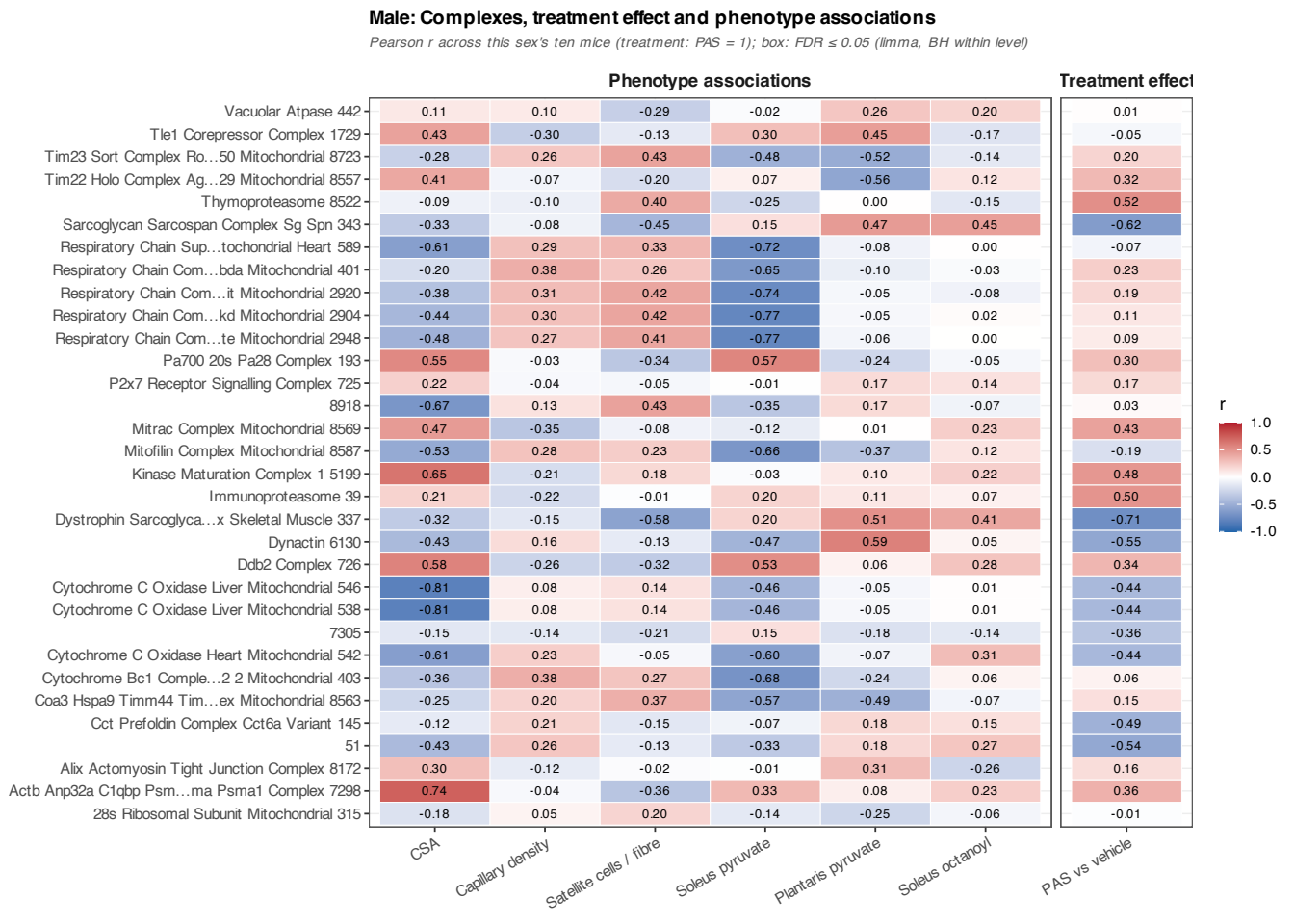
*Share centred within group, added to the association model*

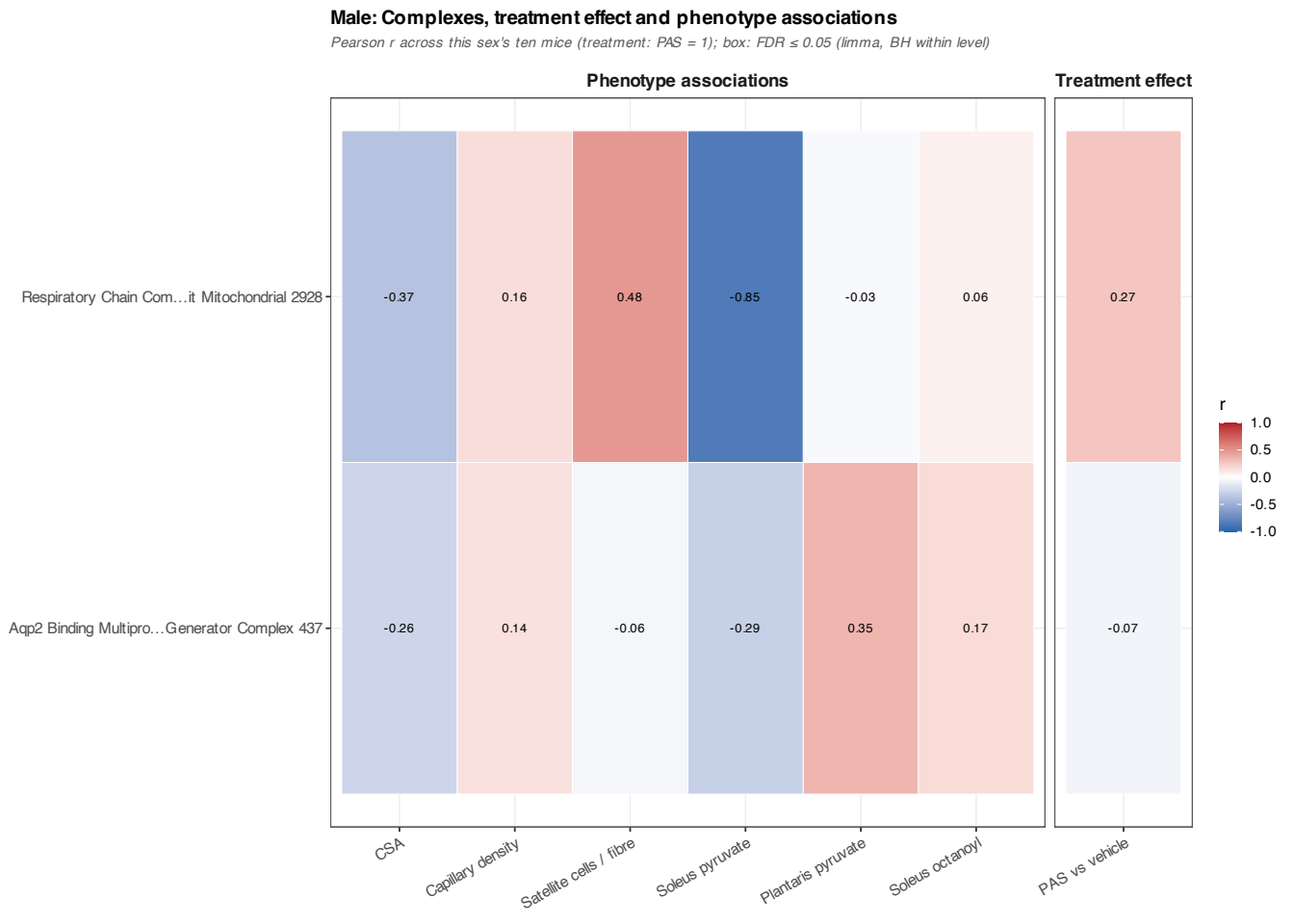


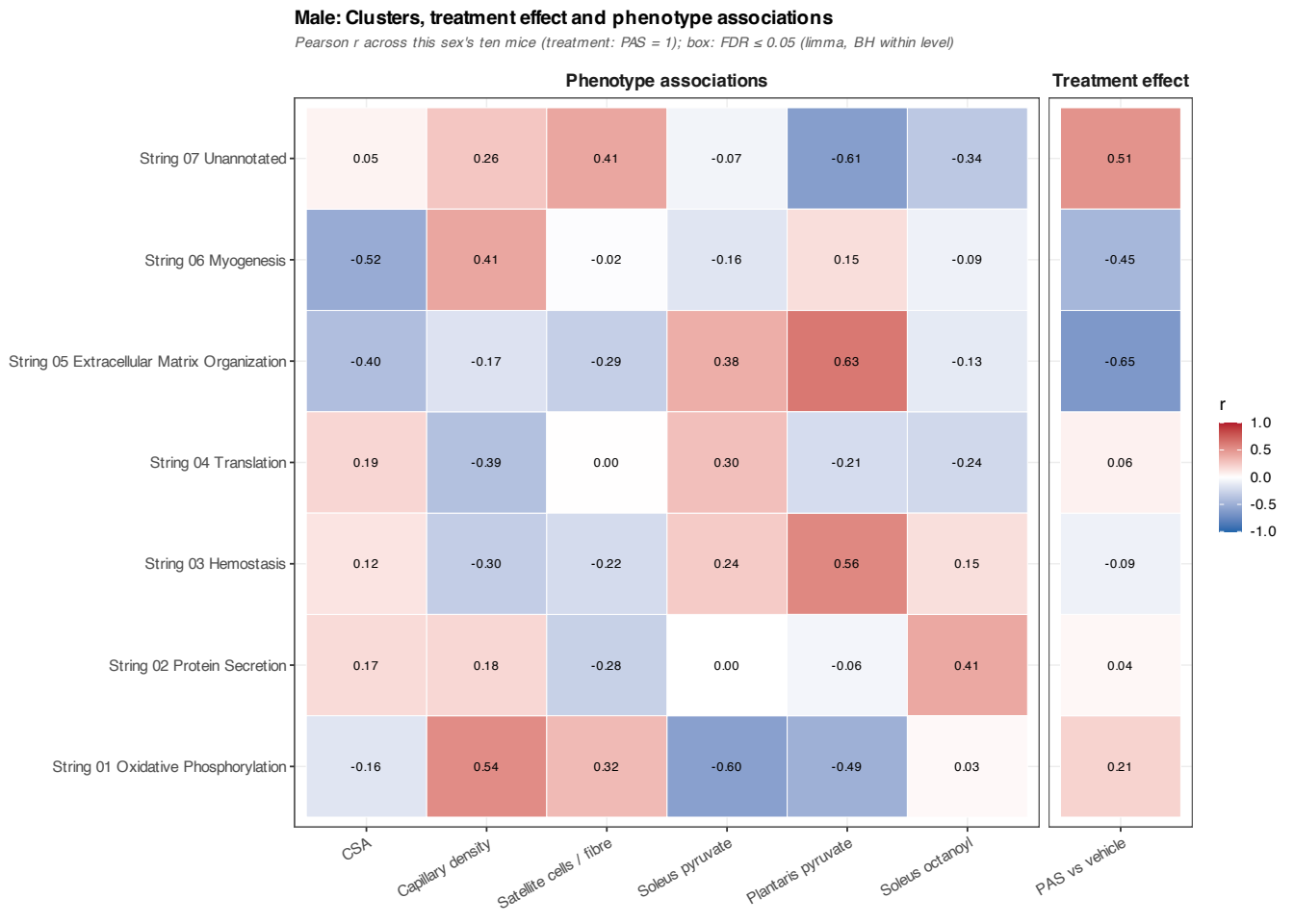

Male: Modules, treatment effect and phenotype associations
Pearson r across this sex's ten mice (treatment: PAS = 1); box: FDR ≤ 0.05 (limma, BH within level)
Phenotype associations
Treatment effect
Yellow -0.69 0.14 -0.16 -0.09 0.55 0.01 -0.76
Turquoise -0.57 0.27 0.29 -0.54 -0.16 0.02 -0.15
Tan -0.44 -0.31 -0.35 0.12 0.54 0.28 -0.50
Salmon 0.25 -0.57 -0.26 0.37 0.18 0.08 -0.12
Red -0.02 0.27 0.55 -0.26 -0.27 -0.19 0.41
Purple -0.38 -0.28 0.00 -0.22 0.42 0.25 -0.35
Pink -0.44 -0.30 -0.37 0.00 0.58 0.12 -0.77
Magenta -0.48 -0.16 -0.10 -0.01 0.31 0.25 -0.46
Greenyellow 0.20 0.08 -0.29 0.08 0.38 0.30 -0.05
Green 0.22 -0.35 -0.33 0.34 0.45 0.16 -0.15
Cyan -0.55 -0.29 0.12 -0.00 0.60 -0.06 -0.29
Brown -0.23 -0.24 -0.09 -0.05 0.37 0.19 -0.18
Blue 0.68 -0.19 -0.32 0.43 0.14 0.10 0.10
Black -0.05 0.16 0.19 -0.33 0.11 0.06 0.09
CSA
Capillary density
Satellite cells / fibre
Soleus pyruvate
Plantaris pyruvate
Soleus octanoyl
PAS vs vehicle
r
1.0
0.5
0.0
-0.5
-1.0


Male: Complexes, treatment effect and phenotype associations
Pearson r across this sex's ten mice (treatment: PAS = 1); box: FDR ≤ 0.05 (limma, BH within level)
Phenotype associations
Treatment effect
Zfand2b Complex 8497 0.39 0.29 -0.51 0.47 -0.34 -0.05 -0.48
Tnfalpha Nfkappa B Signaling Complex 5232 -0.09 -0.63 -0.03 0.31 -0.38 -0.29 -0.06
Tnf Alpha Nf Kappa B Signaling Complex 6 5266 -0.68 -0.33 -0.01 -0.39 0.12 0.14 -0.32
Spliceosome E Complex 8391 0.14 -0.56 -0.17 0.44 0.18 -0.28 -0.17
Spliceosome A Complex 8372 0.09 -0.59 -0.14 0.38 0.18 -0.30 -0.17
Snw1 Complex 1335 -0.37 -0.61 -0.06 0.02 0.18 -0.02 -0.11
Selk Multiprotein Complex 6131 0.07 0.69 0.10 -0.32 -0.33 0.12 0.22
Ribosome Cytoplasmic 306 -0.69 -0.53 0.18 -0.18 -0.15 -0.13 -0.21
Parvulin Associated Pre Rrnp Complex 3047 -0.86 -0.54 0.28 -0.34 0.04 -0.06 -0.21
Pa700 Complex 32 0.47 0.17 -0.41 0.57 -0.30 -0.08 -0.05
Oligosaccharyltransferase Ostc Ii 1208 0.36 0.38 0.08 0.16 -0.32 -0.04 0.55
Oligosaccharyltransferase Ostc I 1203 0.25 0.50 0.17 -0.24 -0.44 0.10 0.52
1200 0.21 0.48 0.38 -0.28 -0.36 -0.04 0.58
1199 0.08 0.48 0.39 -0.35 -0.36 -0.02 0.53
Nop56p Associated Pre rRNA Complex 3055 -0.63 -0.68 0.22 -0.07 -0.14 -0.28 -0.17
Mib Complex Mitochondrial 6249 -0.46 0.39 0.26 -0.62 -0.33 0.08 -0.03
5685 0.54 0.31 -0.51 0.56 -0.37 -0.09 -0.32
Large Drosha Complex 1332 0.15 -0.63 -0.07 0.41 0.22 -0.29 -0.00
Hrd1 Complex 6859 0.52 0.65 0.07 -0.31 0.01 0.29 0.38
Frataxin Complex 1094 0.03 0.53 -0.43 -0.13 -0.35 0.66 -0.19
Emerin Complex 25 5613 0.18 -0.49 -0.38 0.36 0.43 -0.01 -0.23
Emerin Complex 24 5611 0.16 -0.51 -0.35 0.54 0.30 -0.01 -0.14
742 0.00 -0.44 0.14 -0.29 0.15 -0.15 -0.17
Dystrophin Glycoprote…C Skeletal Muscle 340 -0.42 -0.24 -0.40 0.09 0.57 0.37 -0.64
Cdc5l Complex 1183 -0.08 -0.75 0.07 0.22 0.30 -0.29 -0.11
C Src Immune Complex 6938 -0.76 -0.24 0.50 -0.23 -0.12 -0.39 -0.22
C Complex Spliceosome 1181 0.27 -0.50 -0.16 0.42 0.10 -0.22 -0.09
Bbs Chaperonin Complex 6247 -0.66 -0.09 -0.03 -0.21 0.29 0.30 -0.42
60s Ribosomal Subunit Cytoplasmic 308 -0.64 -0.59 0.25 -0.14 -0.10 -0.17 -0.22
40s Ribosomal Subunit Cytoplasmic 305 -0.46 -0.28 0.05 -0.15 -0.13 -0.05 -0.13
Er Localized Multipro…f Ig Heavy Chains 414 -0.08 0.40 0.24 -0.14 -0.40 -0.13 0.16
Er Localized Multipro…Chains Associated 413 0.20 0.46 0.24 -0.05 -0.50 -0.16 0.30
CSA
Capillary density
Satellite cells / fibre
Soleus pyruvate
Plantaris pyruvate
Soleus octanoyl
PAS vs vehicle
r
1.0
0.5
0.0
-0.5
-1.0


Male: Complexes, treatment effect and phenotype associations
Pearson r across this sex's ten mice (treatment: PAS = 1); box: FDR ≤ 0.05 (limma, BH within level)
Phenotype associations
Treatment effect
V Atpase Ragulator Ax…kb1 Ampk Complex 6888
5380
Spermatoproteasome 8532
Sdh Mabc1 Pic Ant Atpase Complex 139
Sarcoglycan Sarcospan…trobrevin Complex 349
Sarcoglycan Sarcospan…troglycan Complex 346
Respiratory Chain Com…eta Mitochondrial 402
Respiratory Chain Com…pha Mitochondrial 400
Respiratory Chain Complex I Mitochondrial 382
Respiratory Chain Complex I Mitochondrial 381
Respiratory Chain Com…te Mitochondrial 2905
Respiratory Chain Com…yme Mitochondrial 178
Respiratory Chain Com…it Mitochondrial 2919
Respiratory Chain Com…ly Mitochondrial 2884
Respiratory Chain Com…it Mitochondrial 2929
Respiratory Chain Com…it Mitochondrial 2914
Pyruvate Dehydrogenase Complex 1375
Profilin 1 Complex 2836
Pa28 20s Proteasome 192
Oligosaccharyltransferase Ostc Iii 1209
Micos Complex Mitochondrial 6255
Lebercilin Complex 5899
H2ax Complex II 1227
7961
7960
Emerin Complex 52 5615
Cct Prefoldin Complex Cct6b Variant 3074
Cct Complex Chaperoni…t Testis Specific 150
55s Ribosome Mitochondrial 320
39s Ribosomal Subunit Mitochondrial 324
26s Proteasome 181
20s Proteasome 38
-0.08 -0.24 -0.52 -0.14 0.40 0.41 -0.28
-0.55 -0.31 0.24 -0.05 -0.02 -0.32 -0.20
-0.22 -0.10 0.42 -0.30 0.08 -0.14 0.38
0.02 0.38 0.20 -0.44 -0.61 0.25 0.31
-0.33 -0.25 -0.53 0.27 0.58 0.35 -0.65
-0.42 -0.18 -0.42 0.06 0.52 0.44 -0.62
-0.05 0.47 0.21 -0.55 -0.01 0.07 0.26
-0.38 0.33 0.39 -0.72 -0.10 -0.01 0.18
-0.34 0.35 0.39 -0.71 -0.07 -0.01 0.19
-0.30 0.38 0.37 -0.70 -0.06 -0.01 0.20
-0.42 0.33 0.41 -0.76 -0.02 0.01 0.10
-0.34 0.35 0.39 -0.71 -0.06 -0.00 0.20
-0.37 0.33 0.40 -0.71 -0.08 0.03 0.17
-0.30 0.38 0.40 -0.69 -0.14 -0.06 0.23
-0.00 0.51 0.19 -0.53 -0.03 0.10 0.29
-0.06 0.45 0.22 -0.53 -0.02 0.08 0.28
-0.26 0.30 0.24 -0.55 -0.42 0.10 0.04
0.81 0.10 -0.18 0.30 -0.26 -0.05 0.45
0.06 -0.35 0.16 -0.09 0.13 0.06 0.56
0.23 0.36 0.49 -0.18 -0.40 -0.18 0.63
-0.23 0.47 0.26 -0.53 -0.30 0.07 0.25
-0.42 -0.19 -0.01 -0.43 0.62 -0.04 -0.51
0.16 -0.35 -0.11 0.11 0.05 0.01 -0.19
-0.05 0.25 0.46 -0.14 -0.62 -0.35 0.47
-0.05 0.26 0.47 -0.15 -0.61 -0.35 0.47
0.52 -0.36 -0.15 0.30 0.27 0.02 0.09
-0.09 -0.11 -0.24 0.04 0.14 0.23 -0.38
-0.45 -0.03 -0.21 -0.27 0.14 0.37 -0.46
-0.20 0.09 0.22 -0.11 -0.27 -0.13 -0.02
-0.22 0.11 0.23 -0.09 -0.28 -0.20 -0.02
0.10 -0.35 0.03 -0.20 -0.01 0.17 0.51
0.04 -0.30 0.19 -0.12 0.09 -0.00 0.55
CSA
Capillary density
Satellite cells / fibre
Soleus pyruvate
Plantaris pyruvate
Soleus octanoyl
PAS vs vehicle
r
1.0
0.5
0.0
-0.5
-1.0

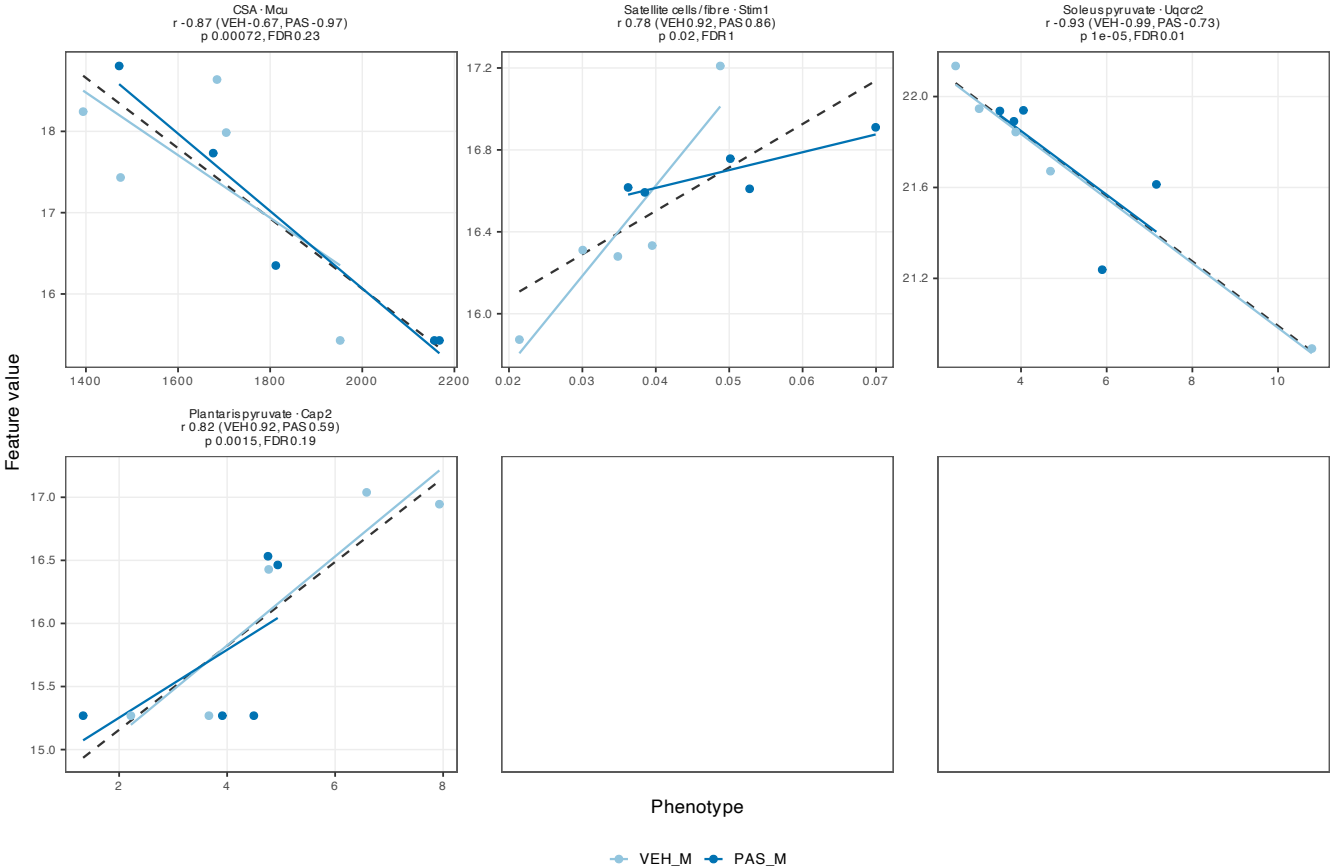

Male: Complexes, treatment effect and phenotype associations
Pearson r across this sex's ten mice (treatment: PAS = 1); box: FDR ≤ 0.05 (limma, BH within level)
Phenotype associations
Treatment effect
Vacuolar Atpase 442 0.11 0.10 -0.29 -0.02 0.26 0.20 0.01
Tle1 Corepressor Complex 1729 0.43 -0.30 -0.13 0.30 0.45 -0.17 -0.05
Tim23 Sort Complex Ro…50 Mitochondrial 8723 -0.28 0.26 0.43 -0.48 -0.52 -0.14 0.20
Tim22 Holo Complex Ag…29 Mitochondrial 8557 0.41 -0.07 -0.20 0.07 -0.56 0.12 0.32
Thymoproteasome 8522 -0.09 -0.10 0.40 -0.25 0.00 -0.15 0.52
Sarcoglycan Sarcospan Complex Sg Spn 343 -0.33 -0.08 -0.45 0.15 0.47 0.45 -0.62
Respiratory Chain Sup…tochondrial Heart 589 -0.61 0.29 0.33 -0.72 -0.08 0.00 -0.07
Respiratory Chain Com…bda Mitochondrial 401 -0.20 0.38 0.26 -0.65 -0.10 -0.03 0.23
Respiratory Chain Com…it Mitochondrial 2920 -0.38 0.31 0.42 -0.74 -0.05 -0.08 0.19
Respiratory Chain Com…kd Mitochondrial 2904 -0.44 0.30 0.42 -0.77 -0.05 0.02 0.11
Respiratory Chain Com…te Mitochondrial 2948 -0.48 0.27 0.41 -0.77 -0.06 0.00 0.09
Pa700 20s Pa28 Complex 193 0.55 -0.03 -0.34 0.57 -0.24 -0.05 0.30
P2x7 Receptor Signalling Complex 725 0.22 -0.04 -0.05 -0.01 0.17 0.14 0.17
8918 -0.67 0.13 0.43 -0.35 0.17 -0.07 0.03
Mitrac Complex Mitochondrial 8569 0.47 -0.35 -0.08 -0.12 0.01 0.23 0.43
Mitofilin Complex Mitochondrial 8587 -0.53 0.28 0.23 -0.66 -0.37 0.12 -0.19
Kinase Maturation Complex 1 5199 0.65 -0.21 0.18 -0.03 0.10 0.22 0.48
Immunoproteasome 39 0.21 -0.22 -0.01 0.20 0.11 0.07 0.50
Dystrophin Sarcoglyca…x Skeletal Muscle 337 -0.32 -0.15 -0.58 0.20 0.51 0.41 -0.71
Dynactin 6130 -0.43 0.16 -0.13 -0.47 0.59 0.05 -0.55
Ddb2 Complex 726 0.58 -0.26 -0.32 0.53 0.06 0.28 0.34
Cytochrome C Oxidase Liver Mitochondrial 546 -0.81 0.08 0.14 -0.46 -0.05 0.01 -0.44
Cytochrome C Oxidase Liver Mitochondrial 538 -0.81 0.08 0.14 -0.46 -0.05 0.01 -0.44
7305 -0.15 -0.14 -0.21 0.15 -0.18 -0.14 -0.36
Cytochrome C Oxidase Heart Mitochondrial 542 -0.61 0.23 -0.05 -0.60 -0.07 0.31 -0.44
Cytochrome Bc1 Comple…2 2 Mitochondrial 403 -0.36 0.38 0.27 -0.68 -0.24 0.06 0.06
Coa3 Hspa9 Timm44 Tim…ex Mitochondrial 8563 -0.25 0.20 0.37 -0.57 -0.49 -0.07 0.15
Cct Prefoldin Complex Cct6a Variant 145 -0.12 0.21 -0.15 -0.07 0.18 0.15 -0.49
51 -0.43 0.26 -0.13 -0.33 0.18 0.27 -0.54
Alix Actomyosin Tight Junction Complex 8172 0.30 -0.12 -0.02 -0.01 0.31 -0.26 0.16
Actb Anp32a C1qbp Psm…ma Psma1 Complex 7298 0.74 -0.04 -0.36 0.33 0.08 0.23 0.36
28s Ribosomal Subunit Mitochondrial 315 -0.18 0.05 0.20 -0.14 -0.25 -0.06 -0.01
CSA
Capillary density
Satellite cells / fibre
Soleus pyruvate
Plantaris pyruvate
Soleus octanoyl
PAS vs vehicle
r
1.0
0.5
0.0
-0.5
-1.0


Male: Complexes, treatment effect and phenotype associations
Pearson r across this sex's ten mice (treatment: PAS = 1); box: FDR ≤ 0.05 (limma, BH within level)
Phenotype associations
Treatment effect
Respiratory Chain Com…it Mitochondrial 2928
Aqp2 Binding Multipro…Generator Complex 437
-0.37
0.16
0.48
-0.85
-0.03
0.06
0.27
-0.26
0.14
-0.06
-0.29
0.35
0.17
-0.07
CSA
Capillary density
Satellite cells / fibre
Soleus pyruvate
Plantaris pyruvate
Soleus octanoyl
PAS vs vehicle
r
1.0
0.5
0.0
-0.5
-1.0


Male: Clusters, treatment effect and phenotype associations
Pearson r across this sex's ten mice (treatment: PAS = 1); box: FDR ≤ 0.05 (limma, BH within level)
Phenotype associations
Treatment effect
String 07 Unannotated
String 06 Myogenesis
String 05 Extracellular Matrix Organization
String 04 Translation
String 03 Hemostasis
String 02 Protein Secretion
String 01 Oxidative Phosphorylation
0.05 0.26 0.41 -0.07 -0.61 -0.34 0.51
-0.52 0.41 -0.02 -0.16 0.15 -0.09 -0.45
-0.40 -0.17 -0.29 0.38 0.63 -0.13 -0.65
0.19 -0.39 0.00 0.30 -0.21 -0.24 0.06
0.12 -0.30 -0.22 0.24 0.56 0.15 -0.09
0.17 0.18 -0.28 0.00 -0.06 0.41 0.04
-0.16 0.54 0.32 -0.60 -0.49 0.03 0.21
CSA
Capillary density
Satellite cells / fibre
Soleus pyruvate
Plantaris pyruvate
Soleus octanoyl
PAS vs vehicle
r
1.0
0.5
0.0
-0.5
-1.0

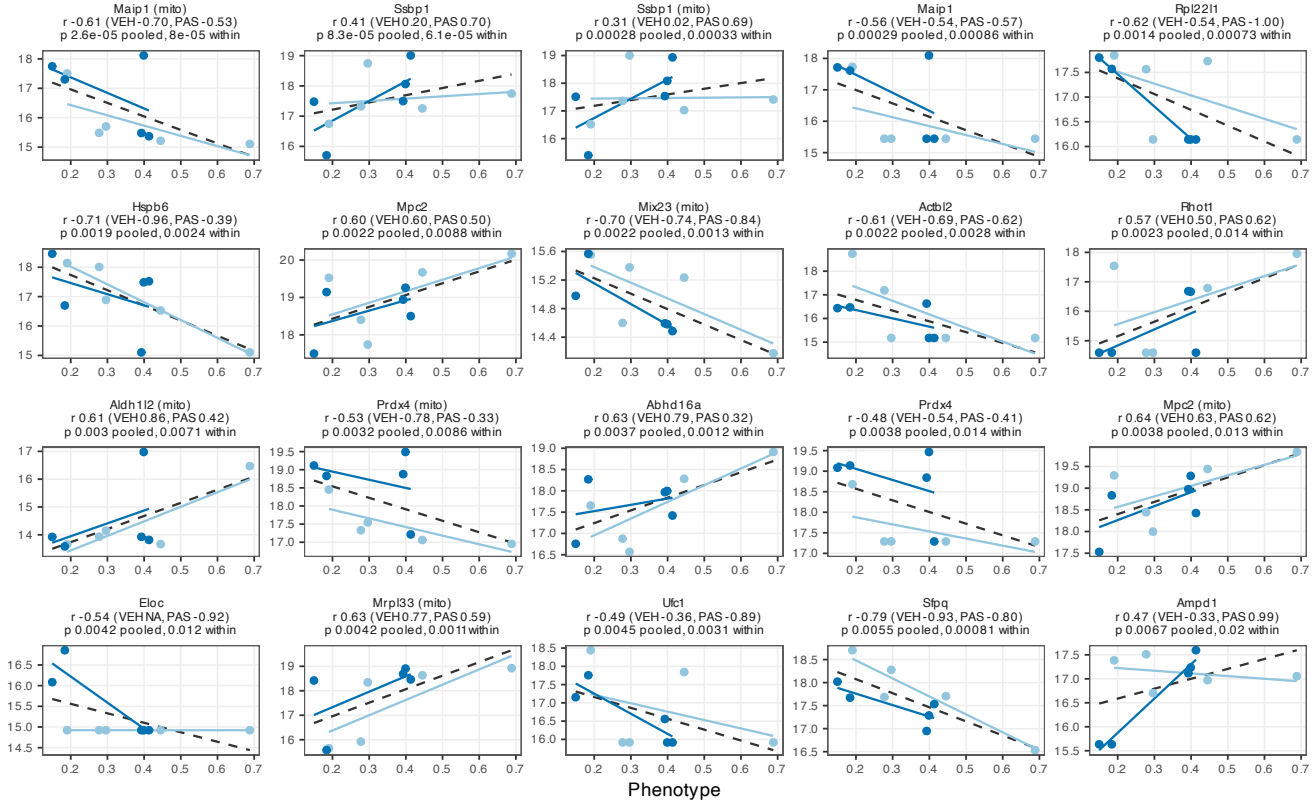
**Male: strongest robust protein per phenotype**

*Smallest association p that keeps its sign and p ≤ 0.05 with any one mouse left out*



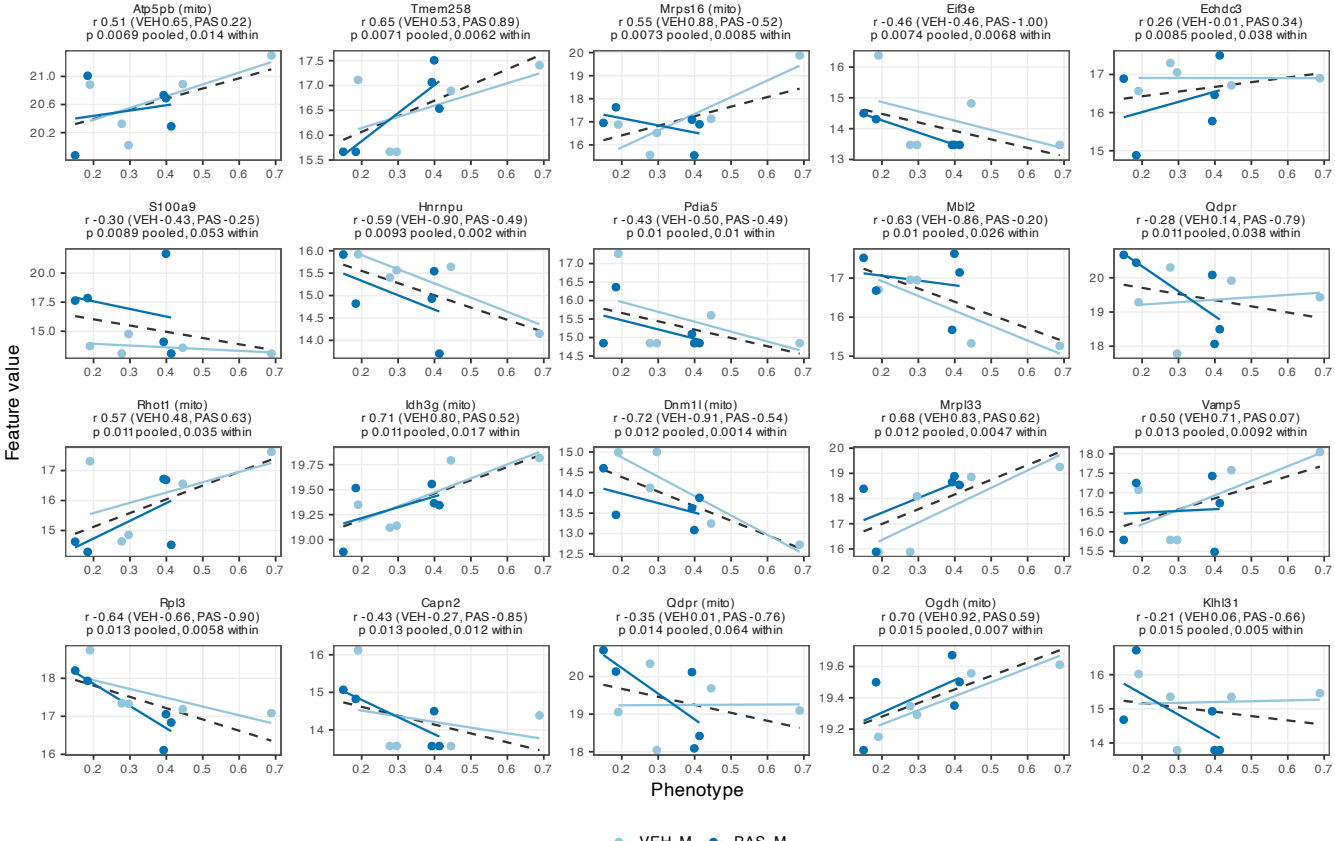
**Male: Proteins associated with Capillary density, p ≤ 0.05 in either model (n = 205)**

*Phenotype effect of PAS in this sex: -18% (p 0.54), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



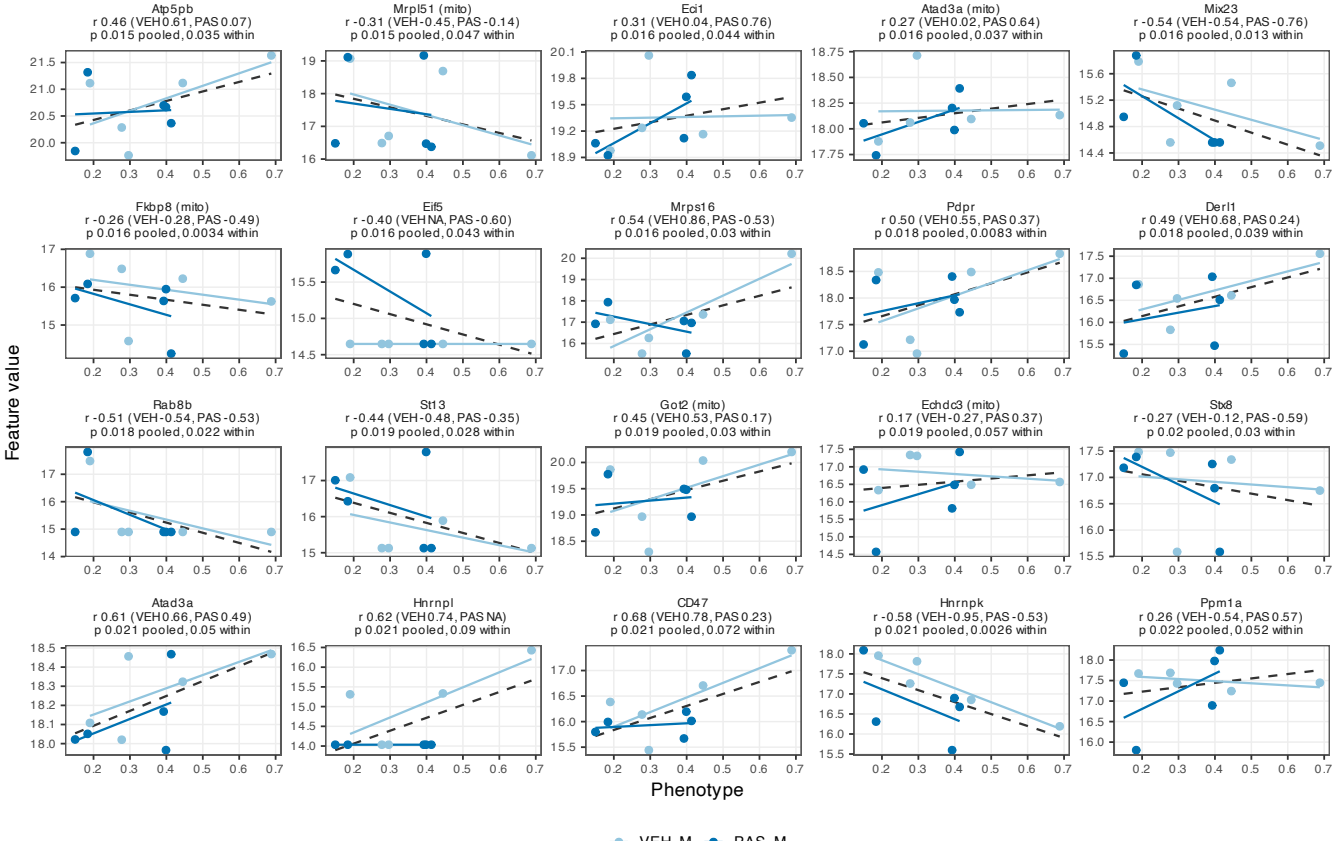
**Male: Proteins associated with Capillary density, p ≤ 0.05 in either model (n = 205)**

*Phenotype effect of PAS in this sex: -18% (p 0.54), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



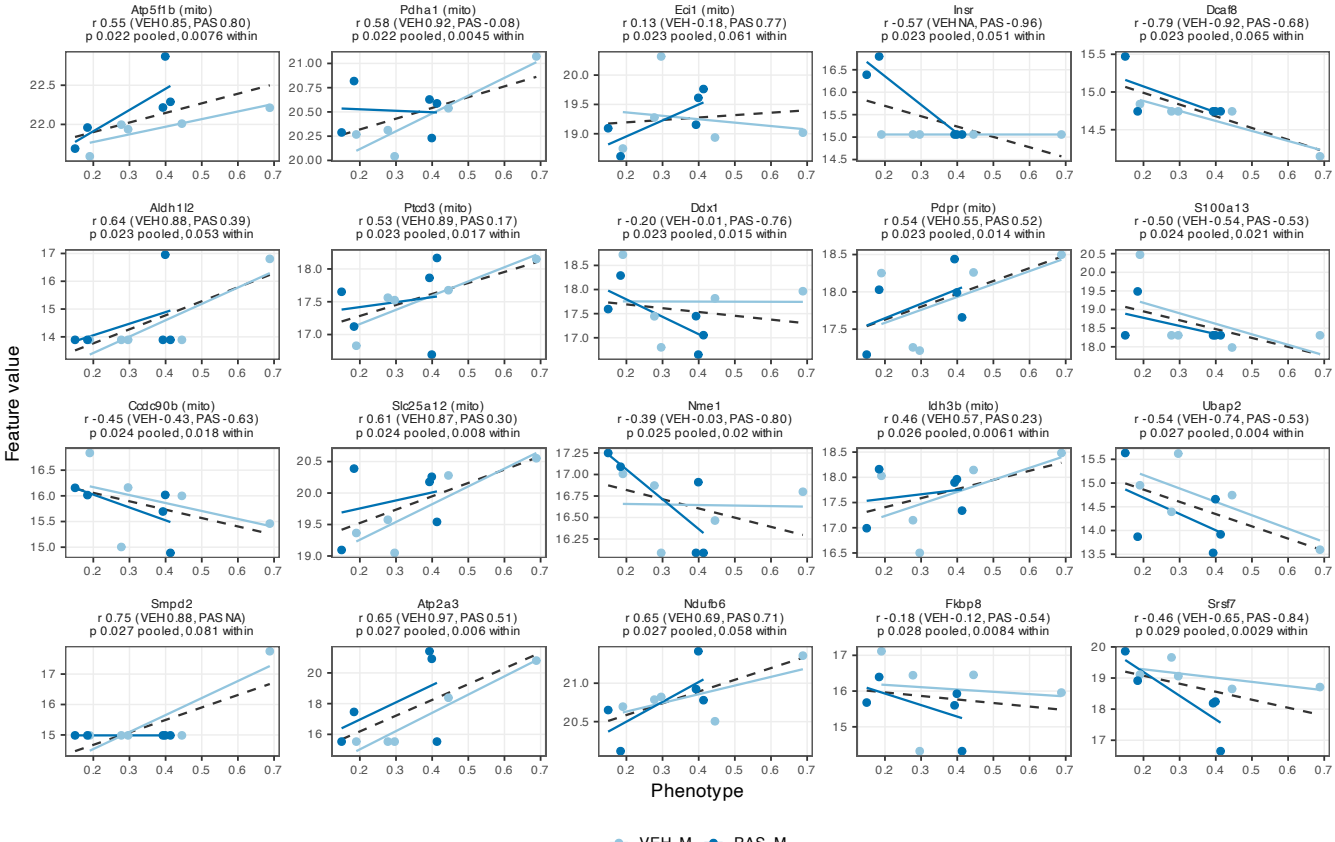
**Male: Proteins associated with Capillary density, p ≤ 0.05 in either model (n = 205)**

*Phenotype effect of PAS in this sex: -18% (p 0.54), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



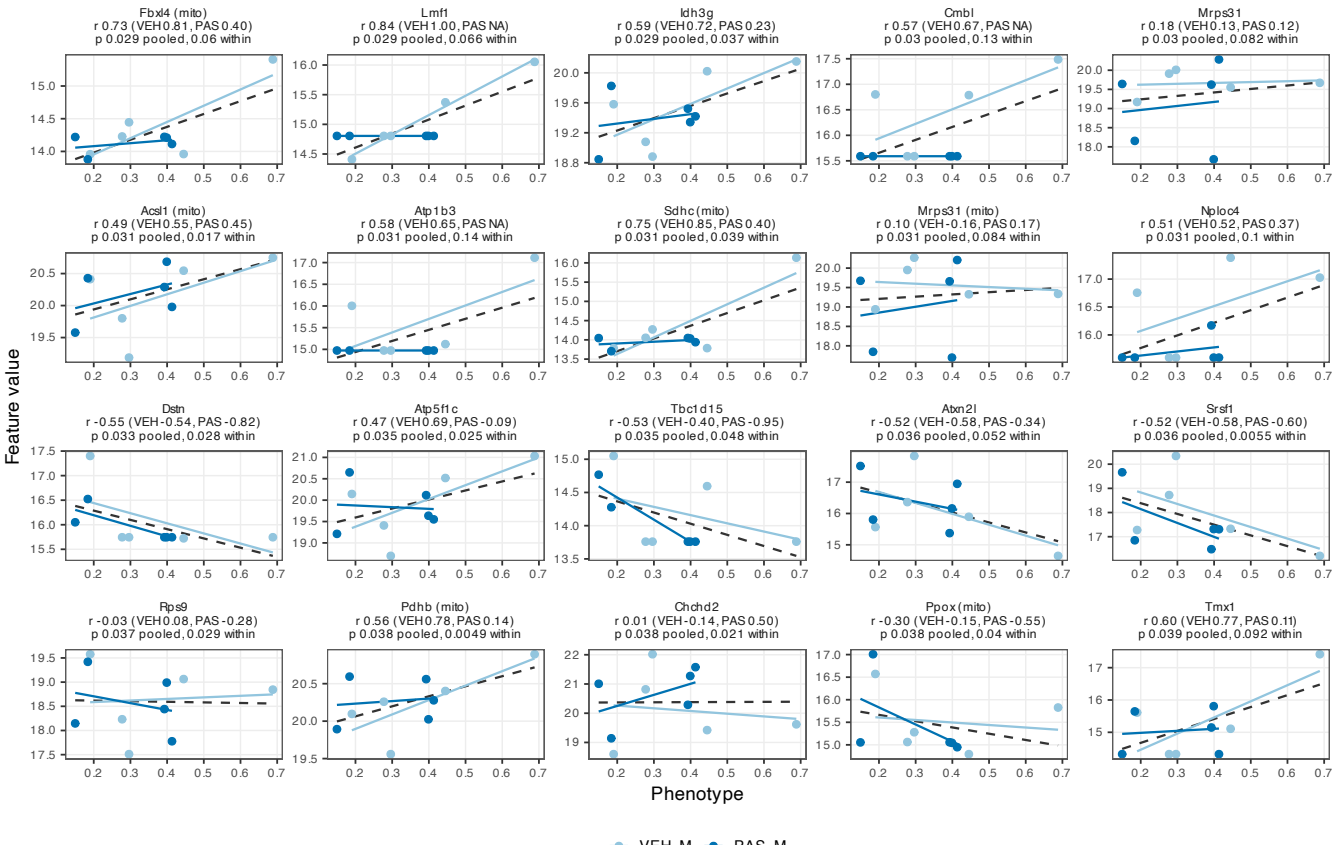
**Male: Proteins associated with Capillary density, p ≤ 0.05 in either model (n = 205)**

*Phenotype effect of PAS in this sex: -18% (p 0.54), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



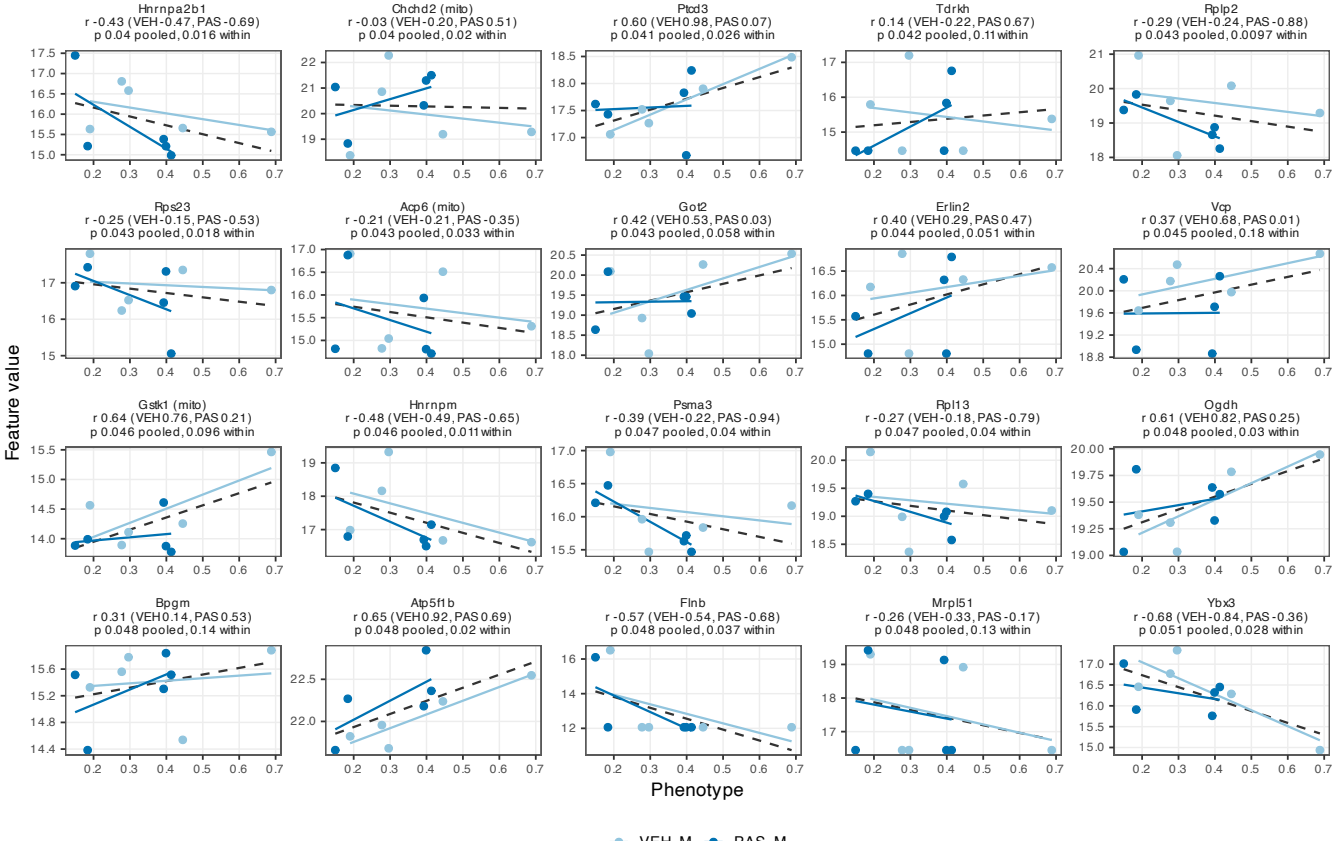
**Male: Proteins associated with Capillary density, p ≤ 0.05 in either model (n = 205)**

*Phenotype effect of PAS in this sex: -18% (p 0.54), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



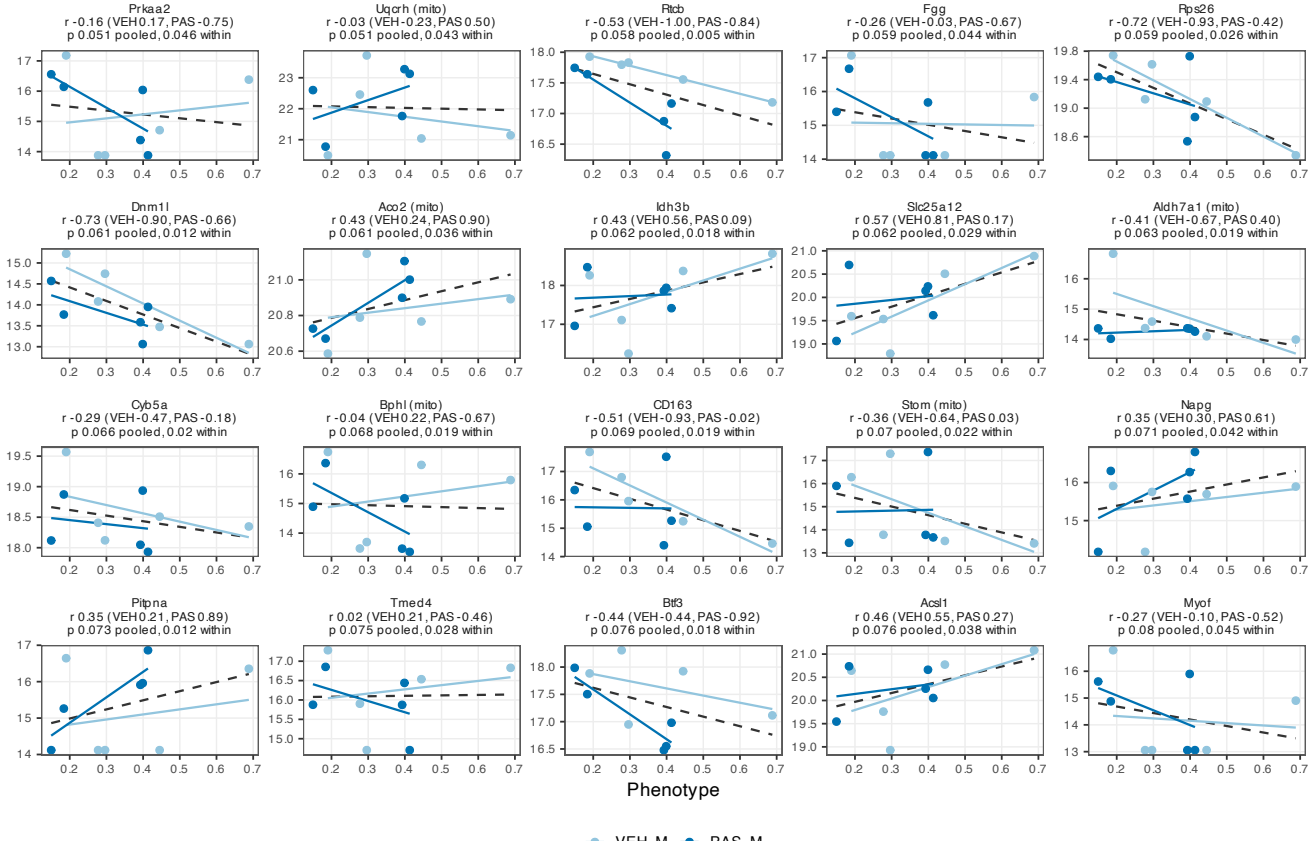
**Male: Proteins associated with Capillary density, p ≤ 0.05 in either model (n = 205)**

*Phenotype effect of PAS in this sex: -18% (p 0.54), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



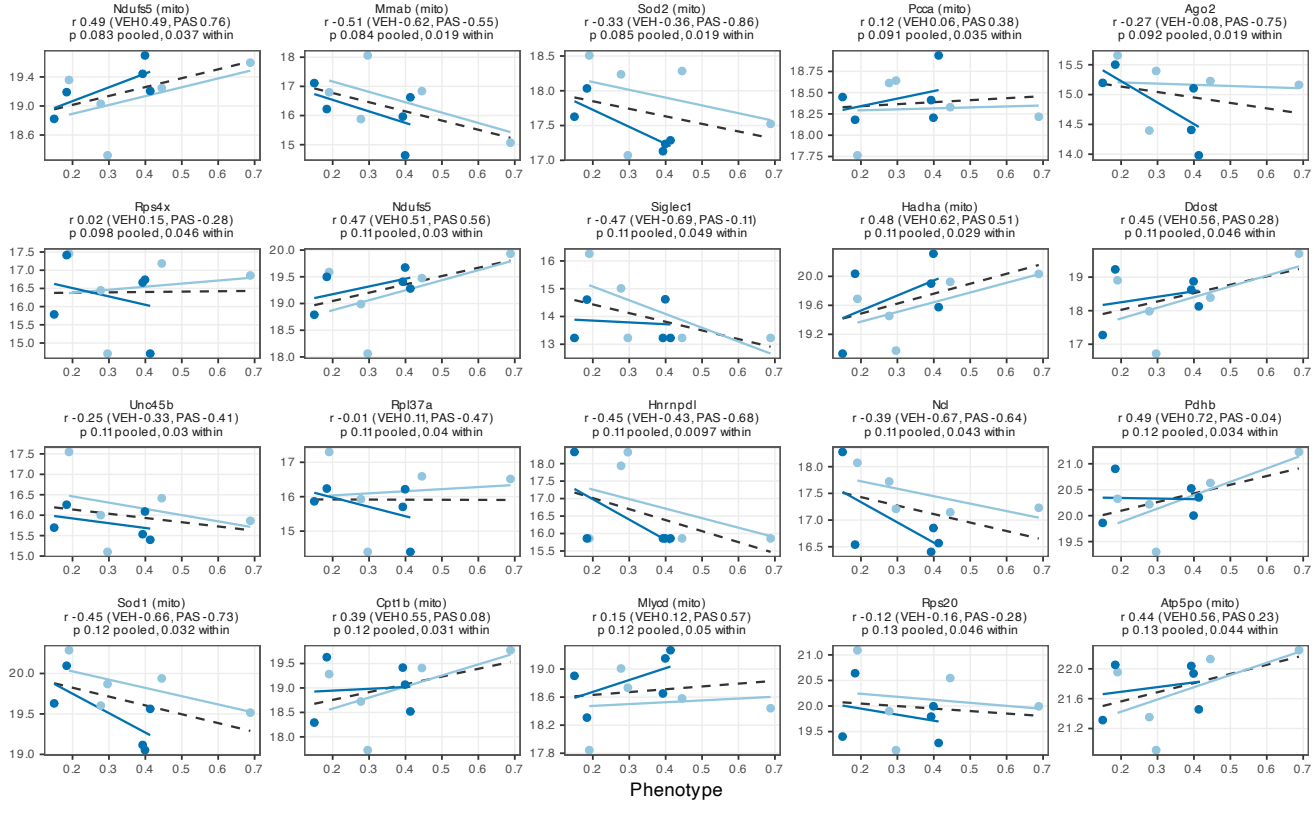
**Male: Proteins associated with Capillary density, p ≤ 0.05 in either model (n = 205)**

*Phenotype effect of PAS in this sex: -18% (p 0.54), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



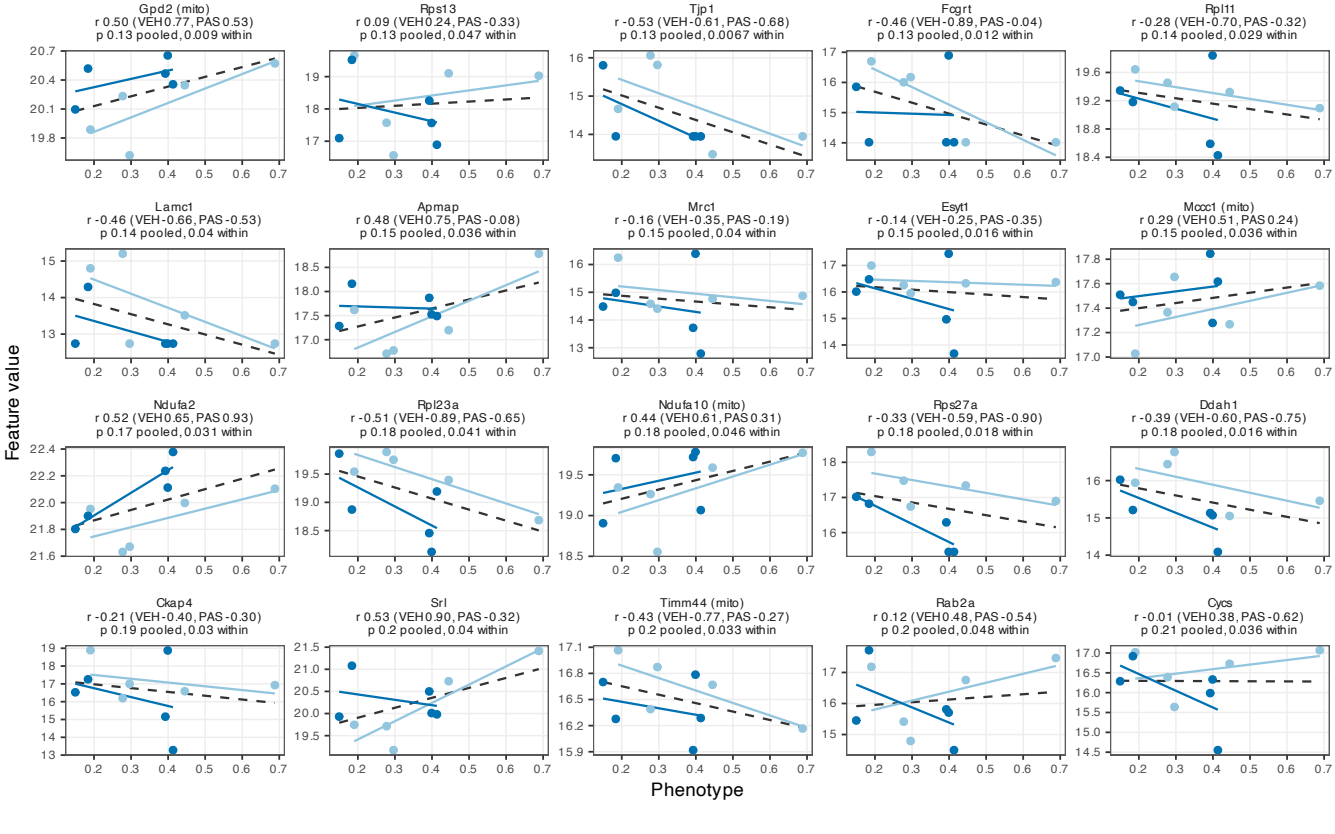
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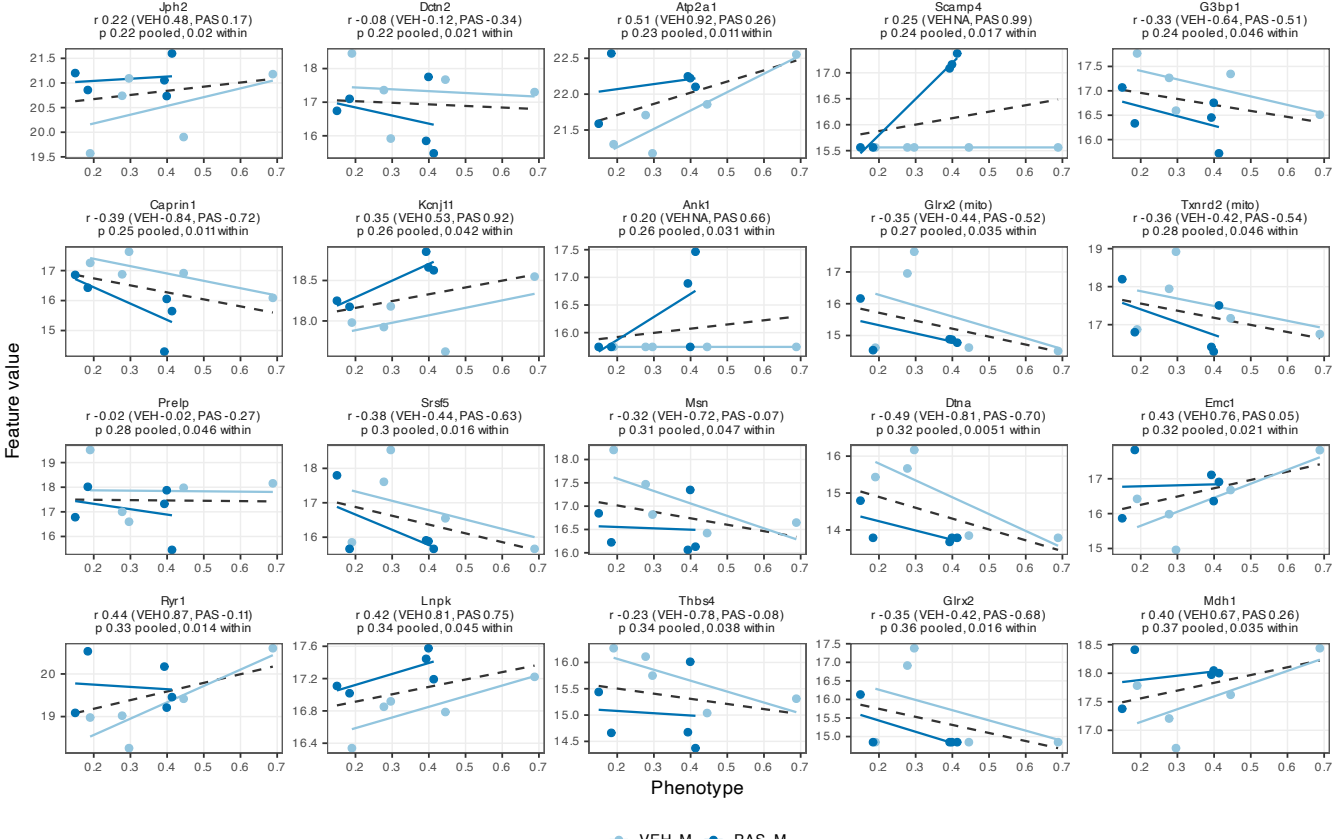
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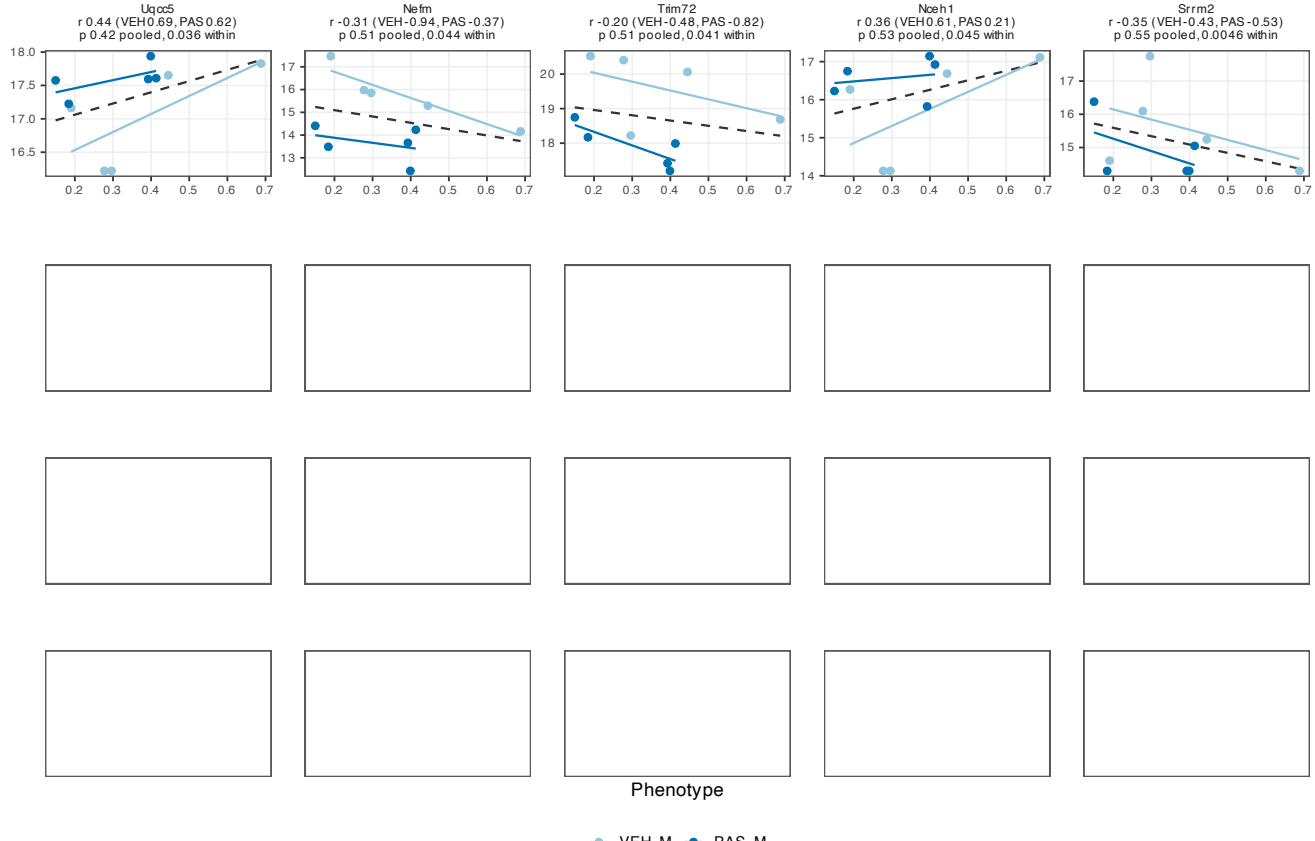
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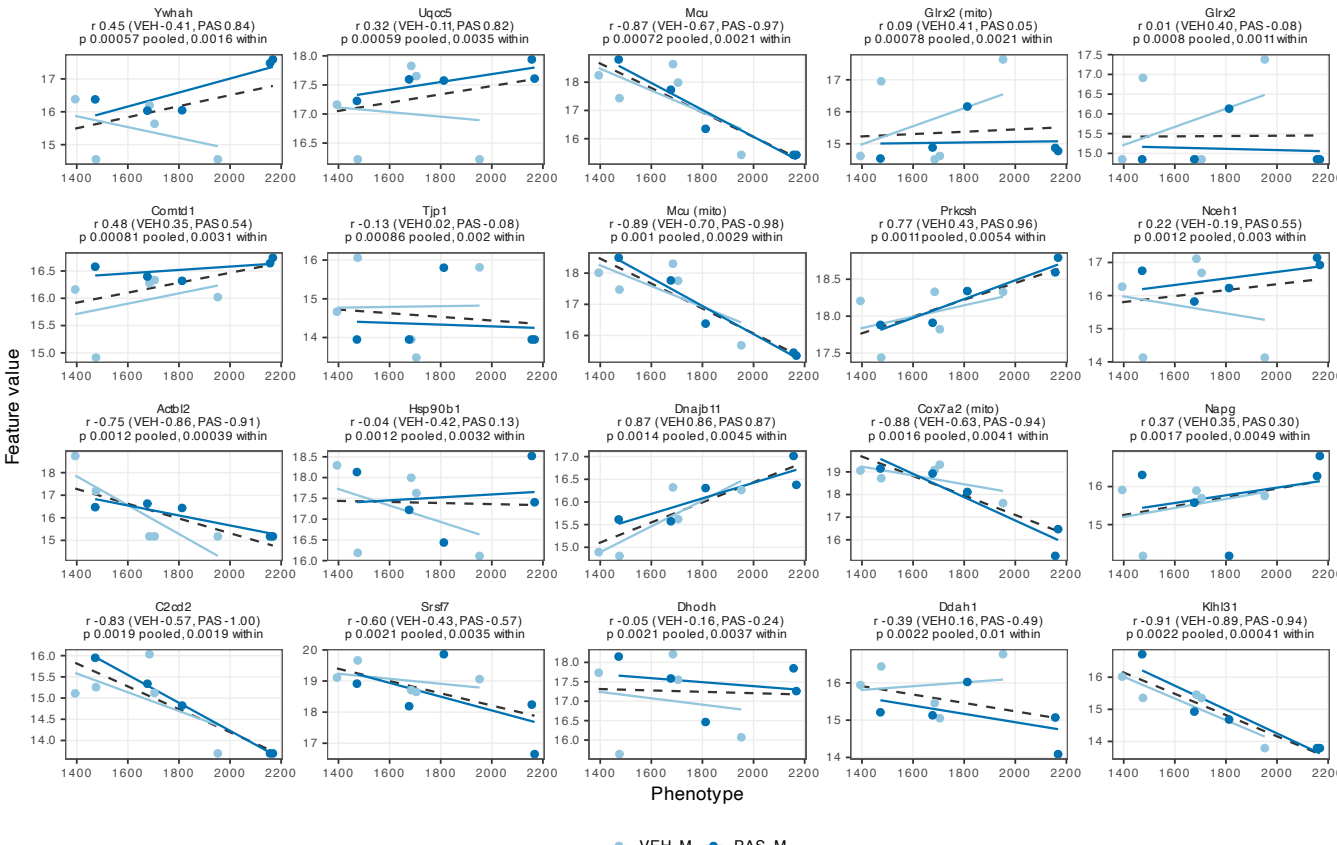
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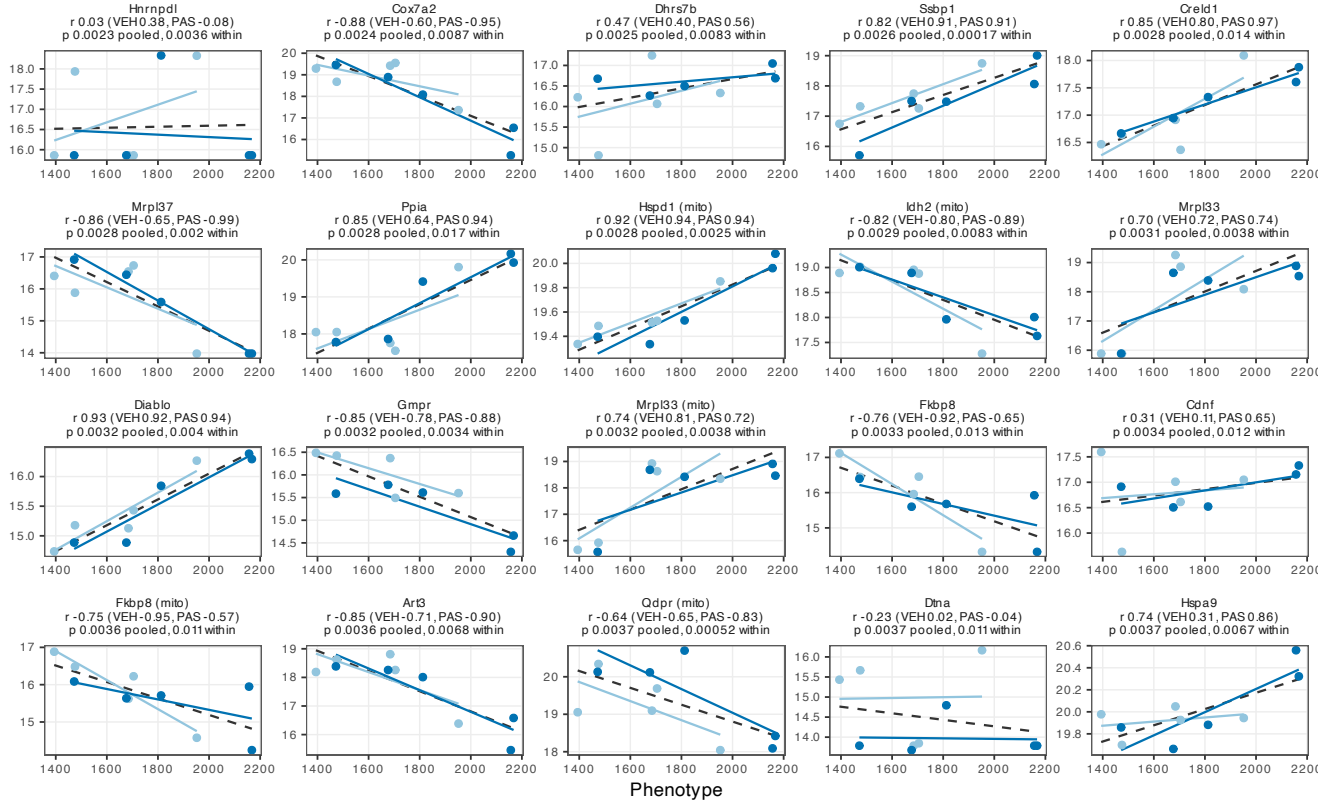
**Male: Proteins associated with CSA, p ≤ 0.05 in either model (n = 222)**

*Phenotype effect of PAS in this sex: +13% (p 0.25), so these associations follow variation between mice, not a PAS response.*
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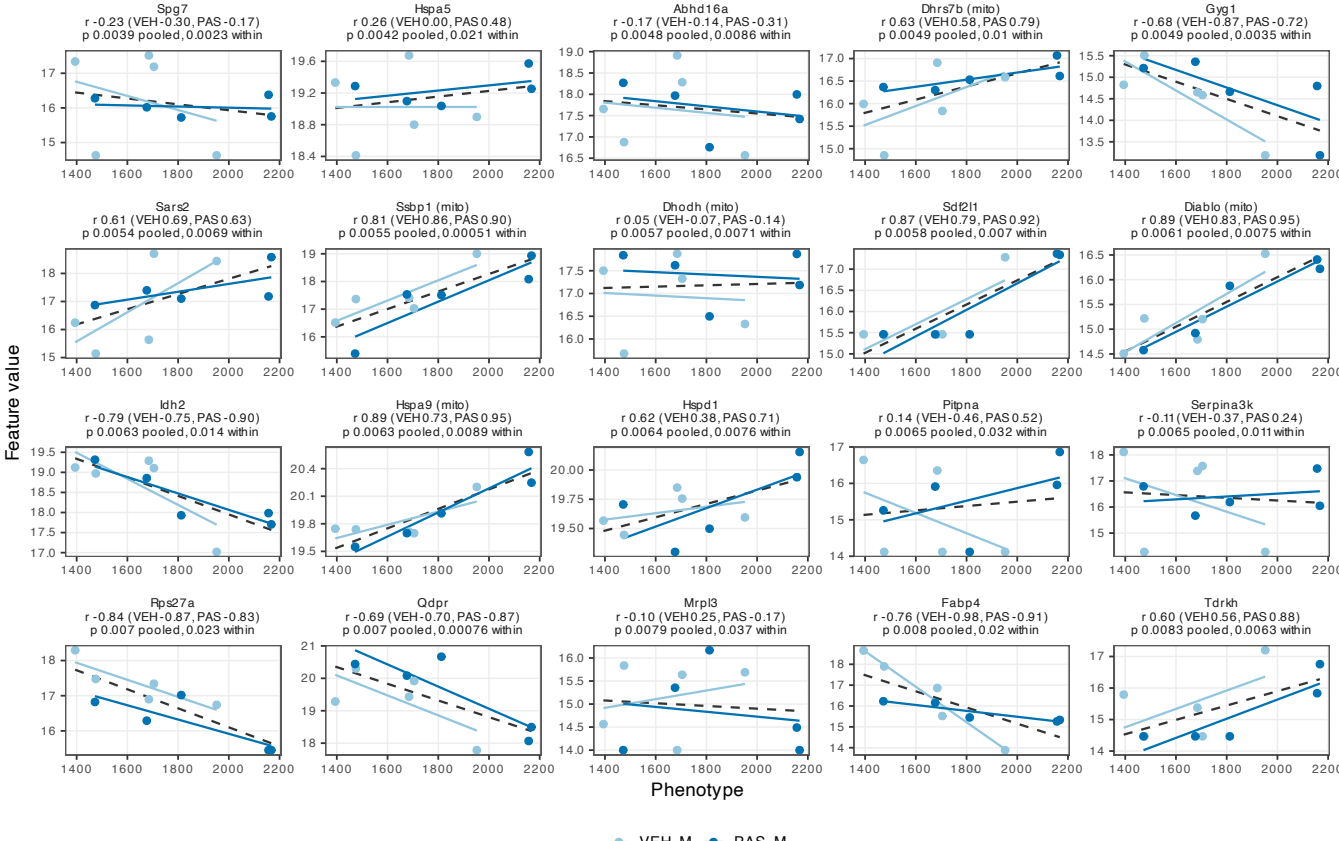
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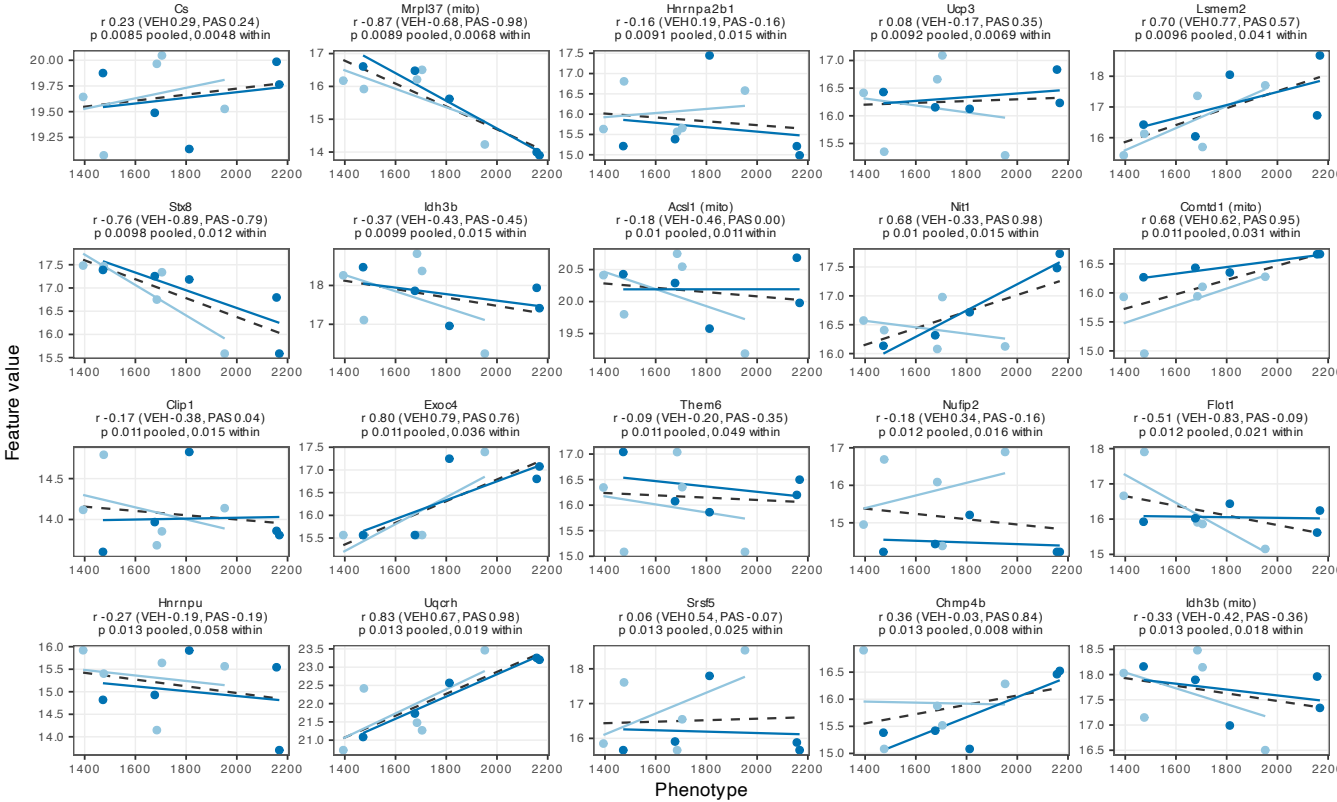
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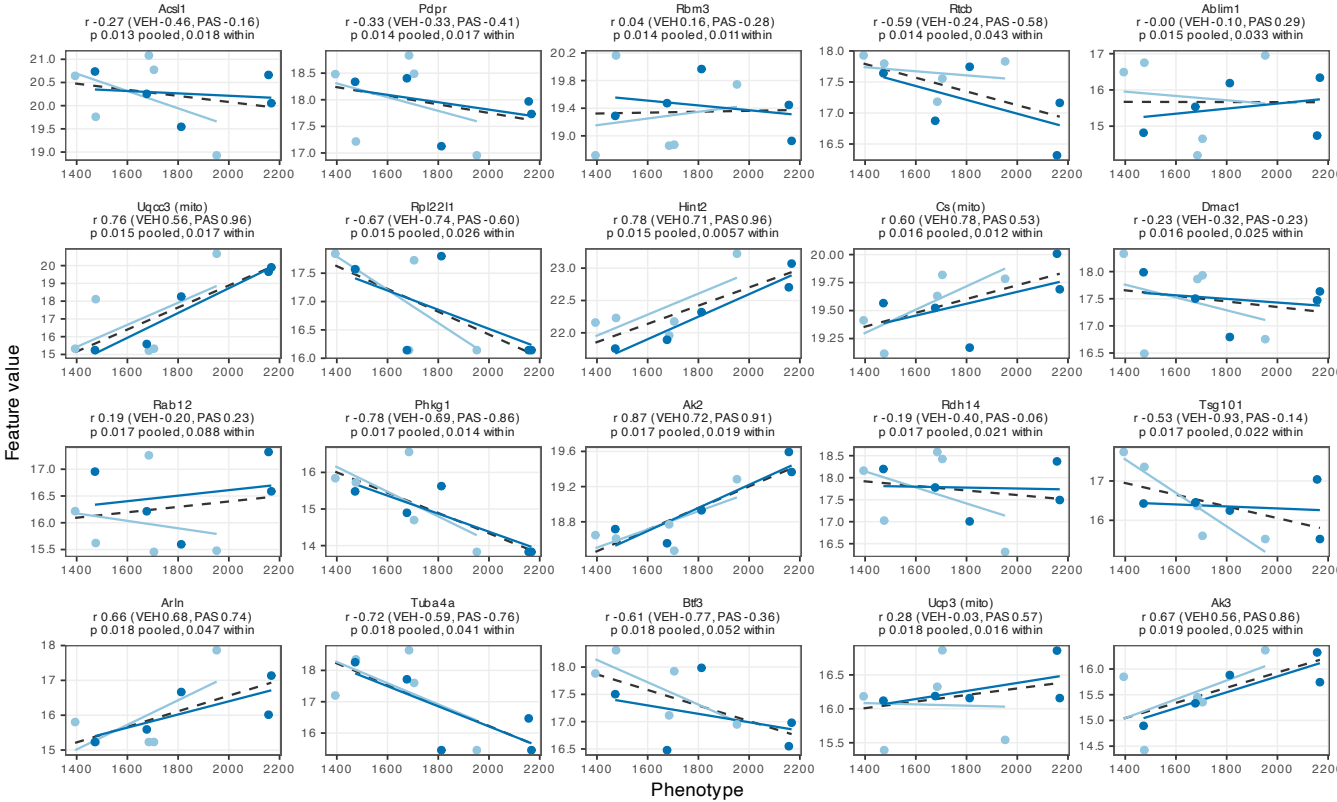
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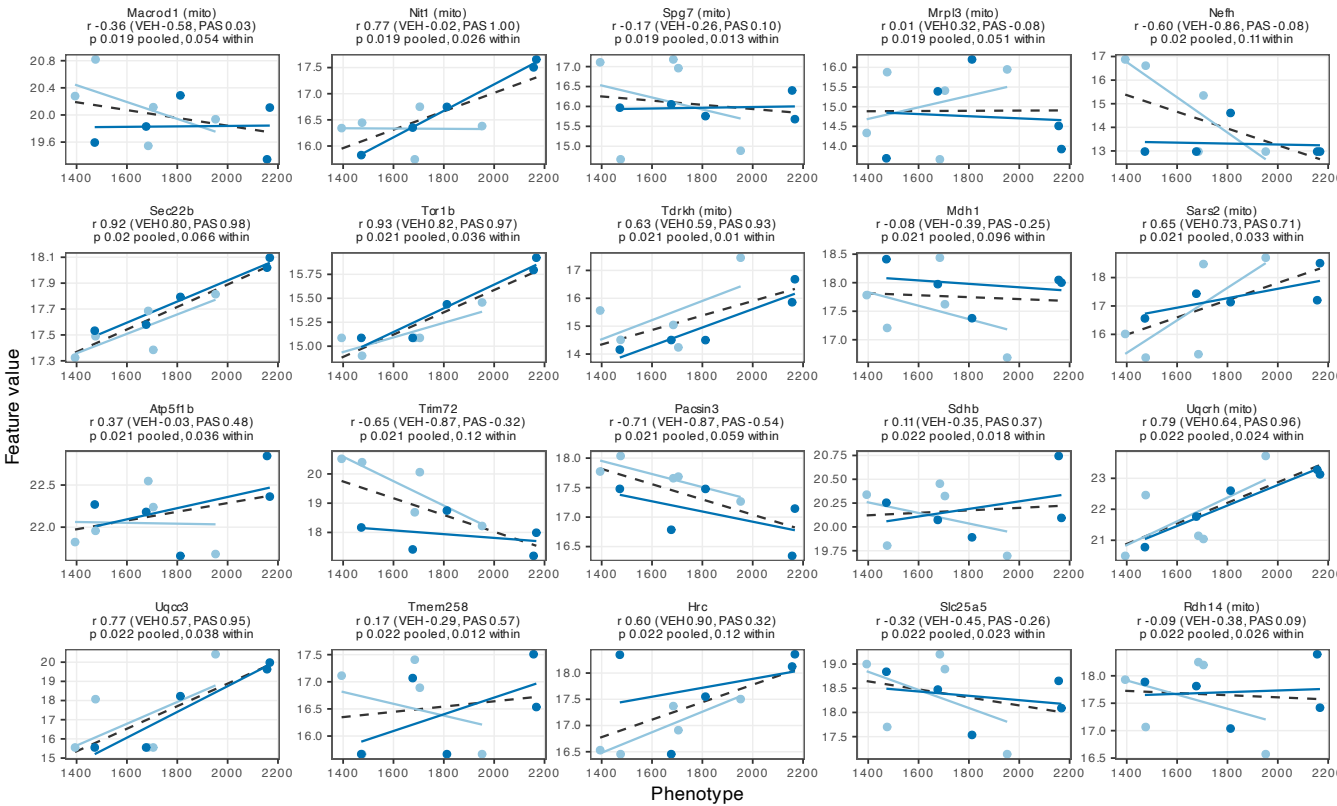
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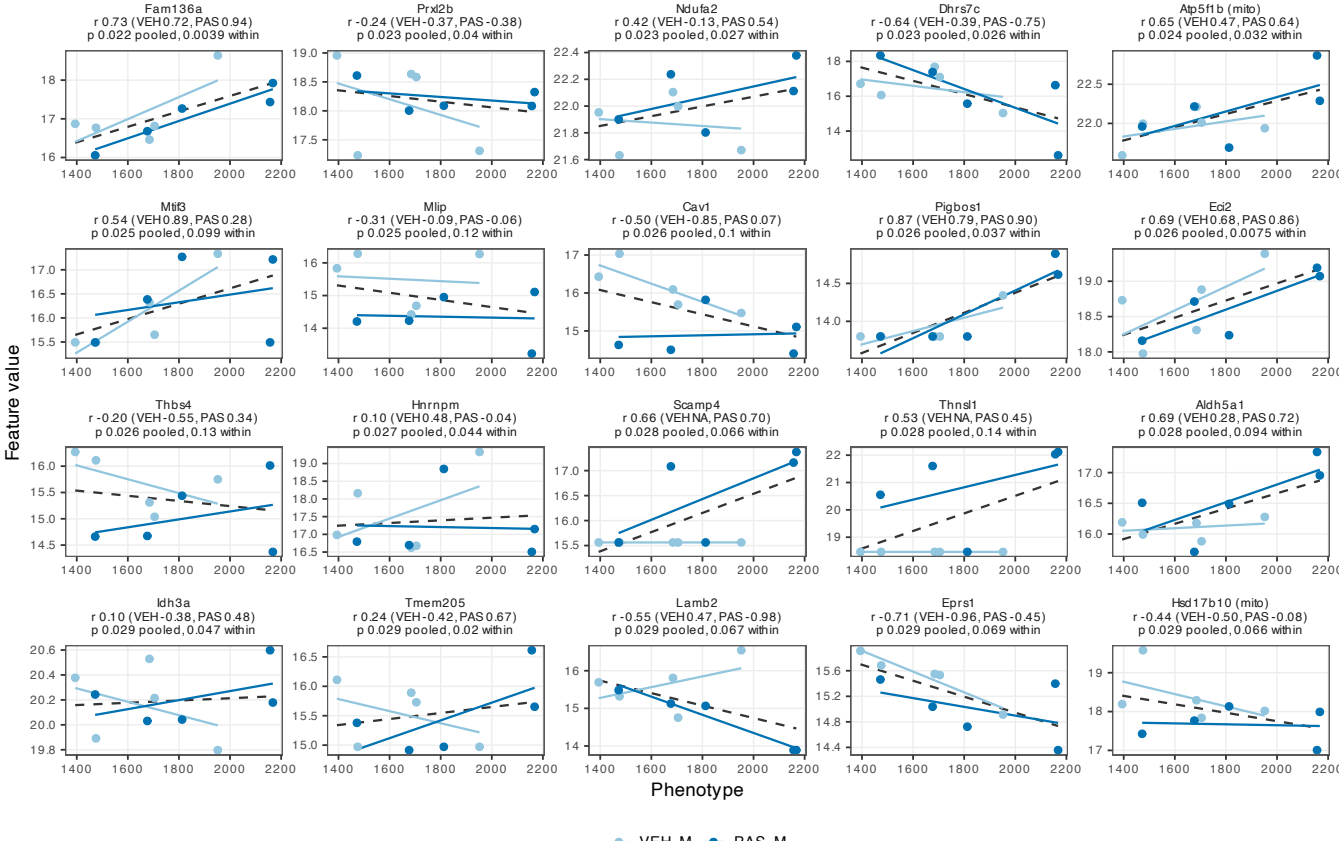
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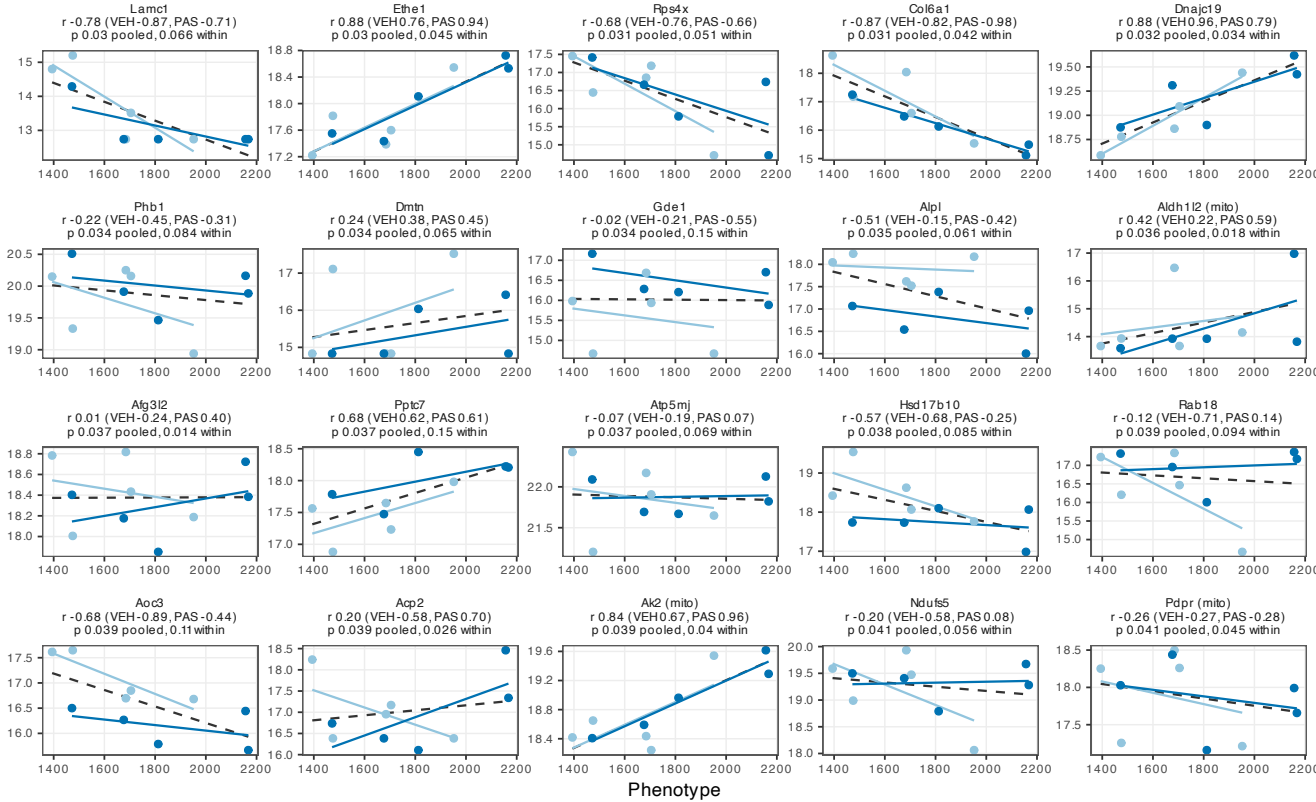
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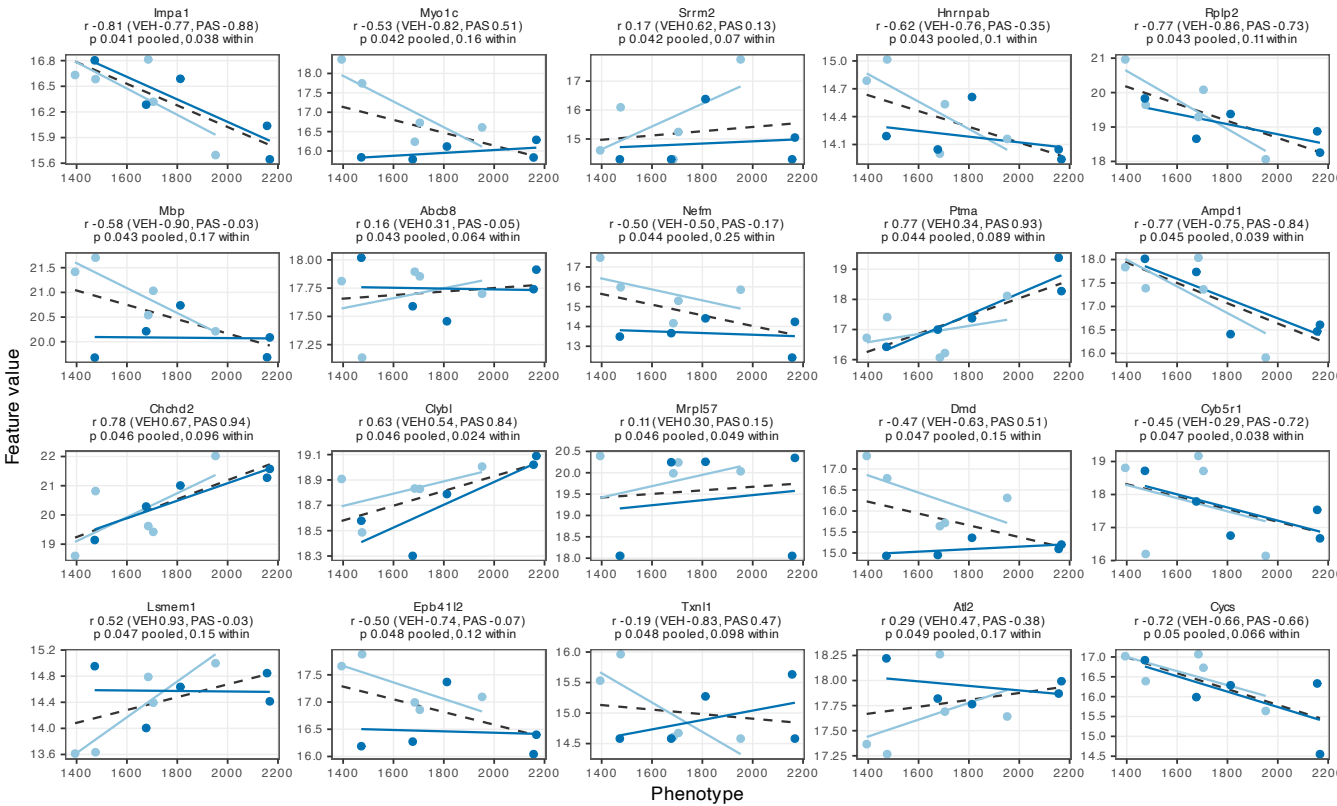
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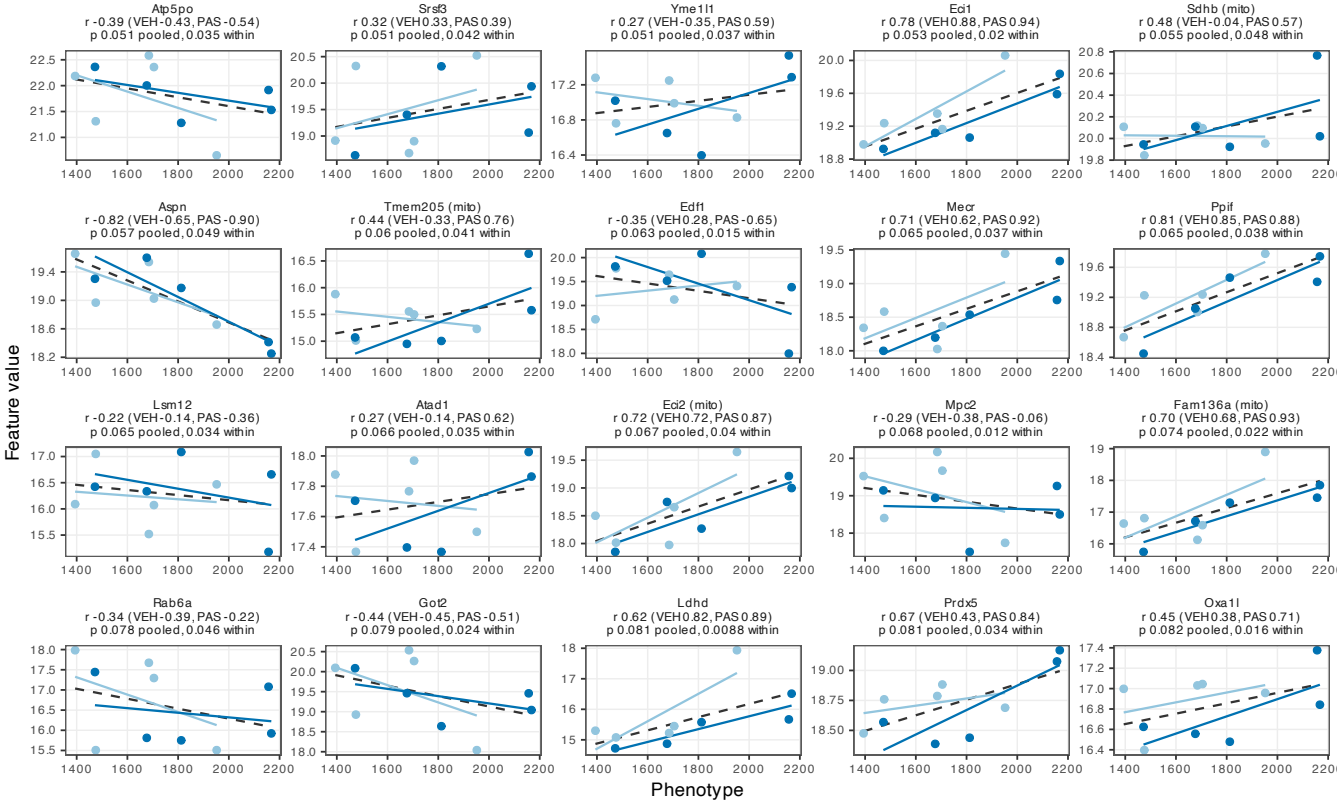
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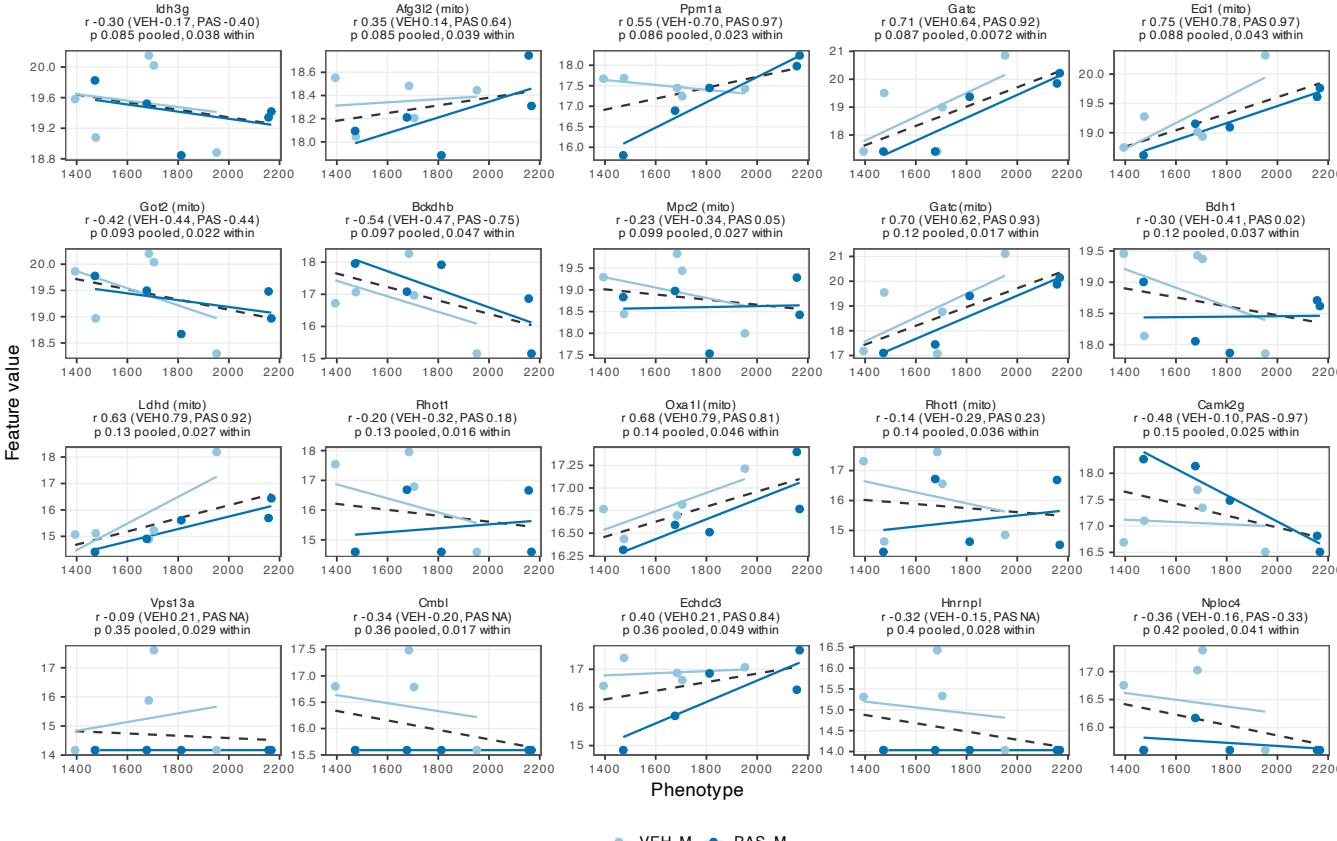
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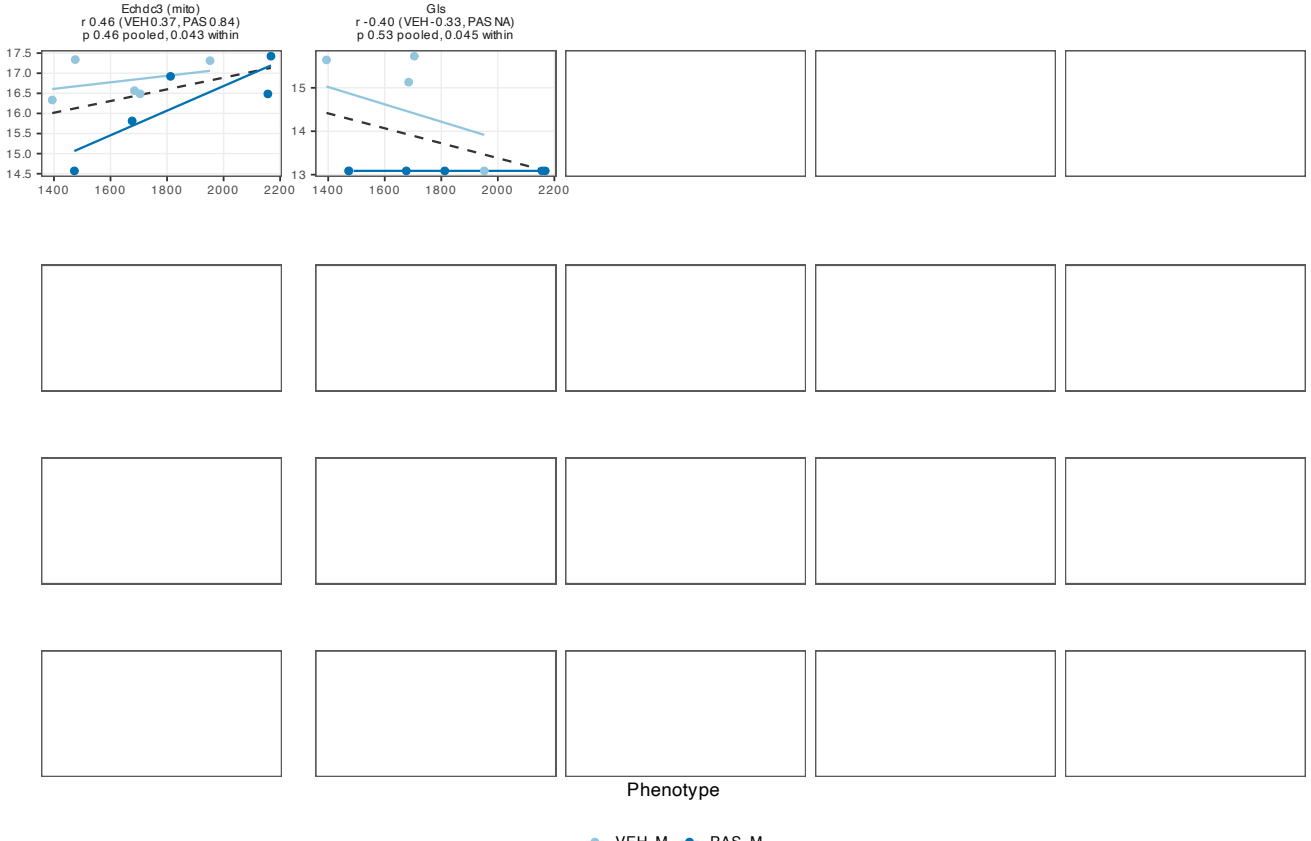
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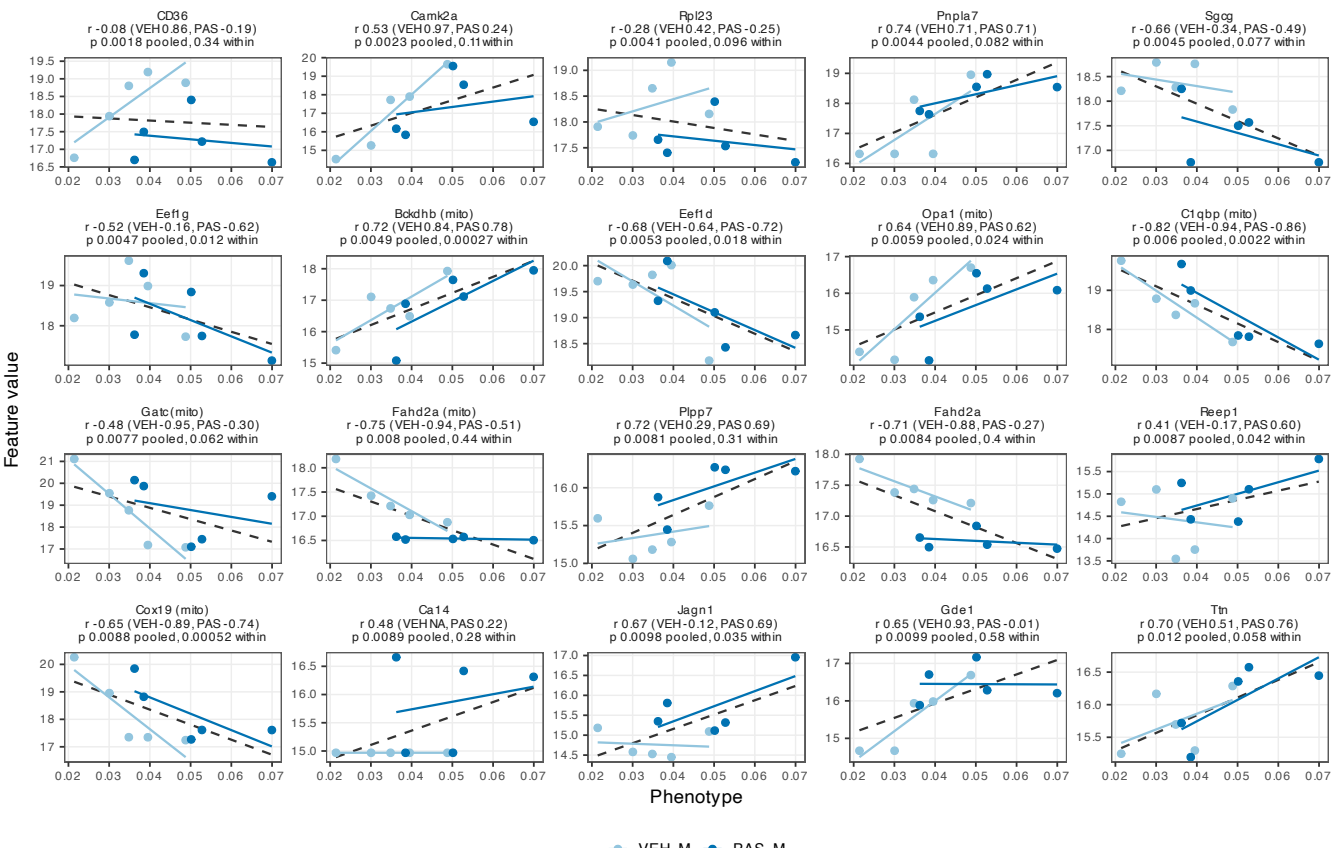
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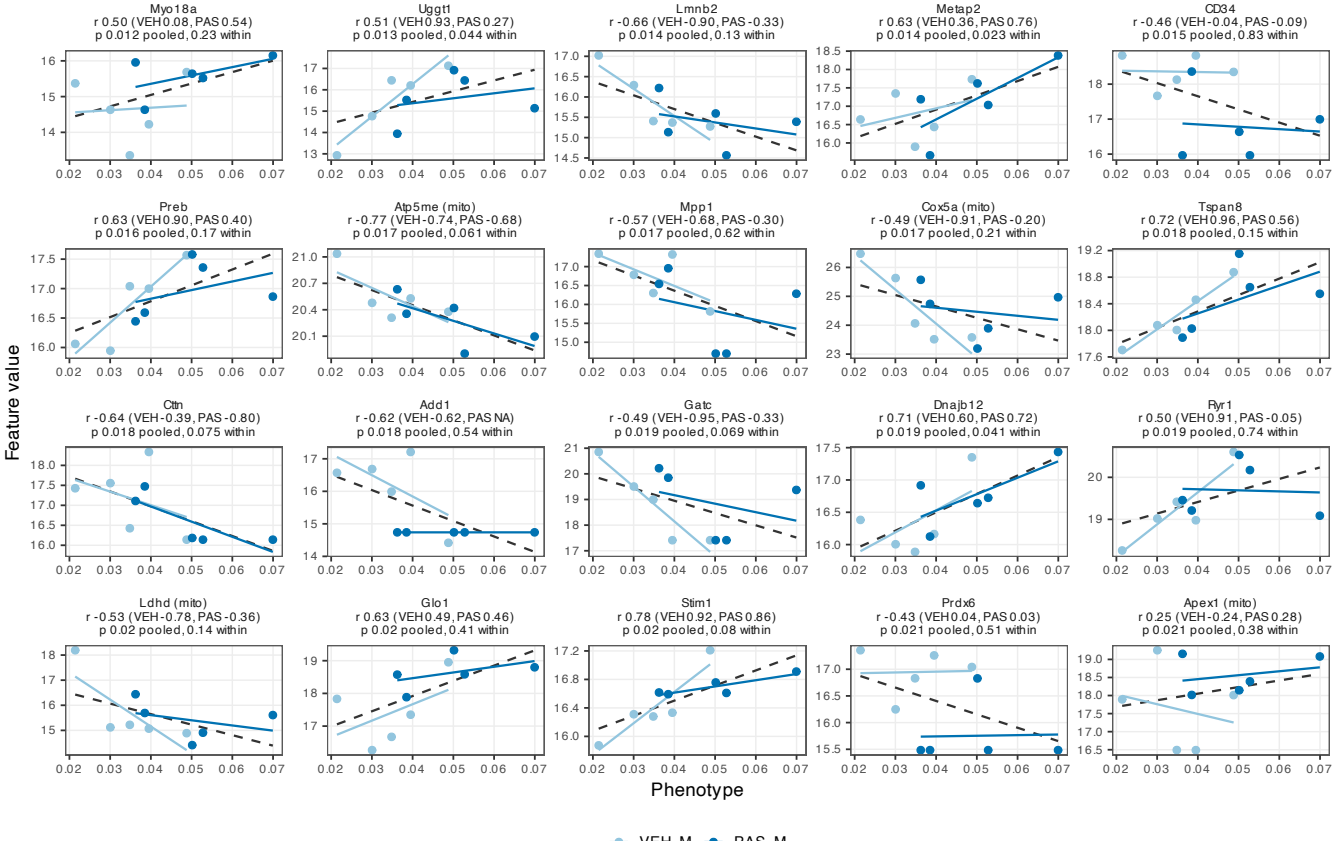
**Male: Proteins associated with Satellite cells / fibre, p ≤ 0.05 in either model (n = 157)**

*Phenotype effect of PAS in this sex: +43% (p 0.086), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



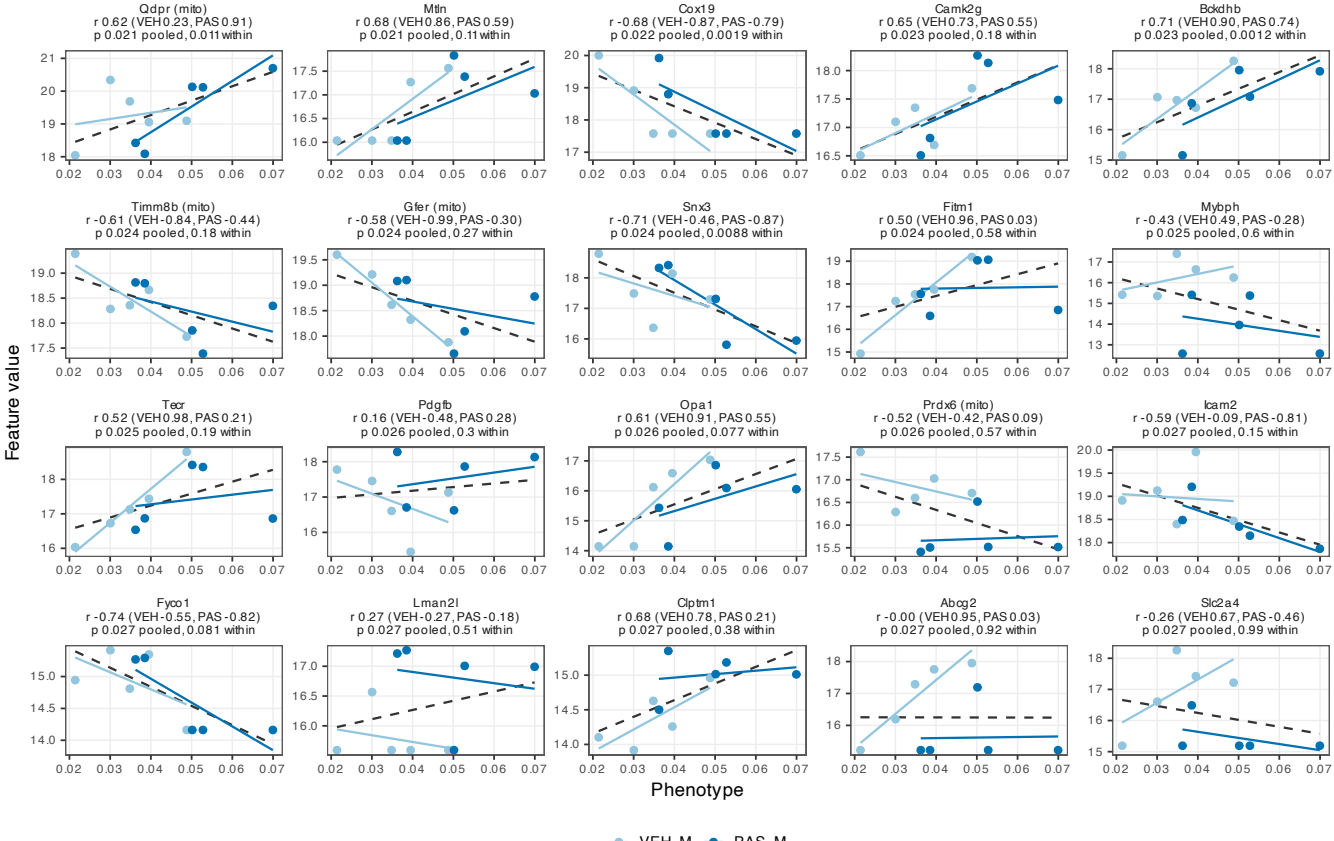
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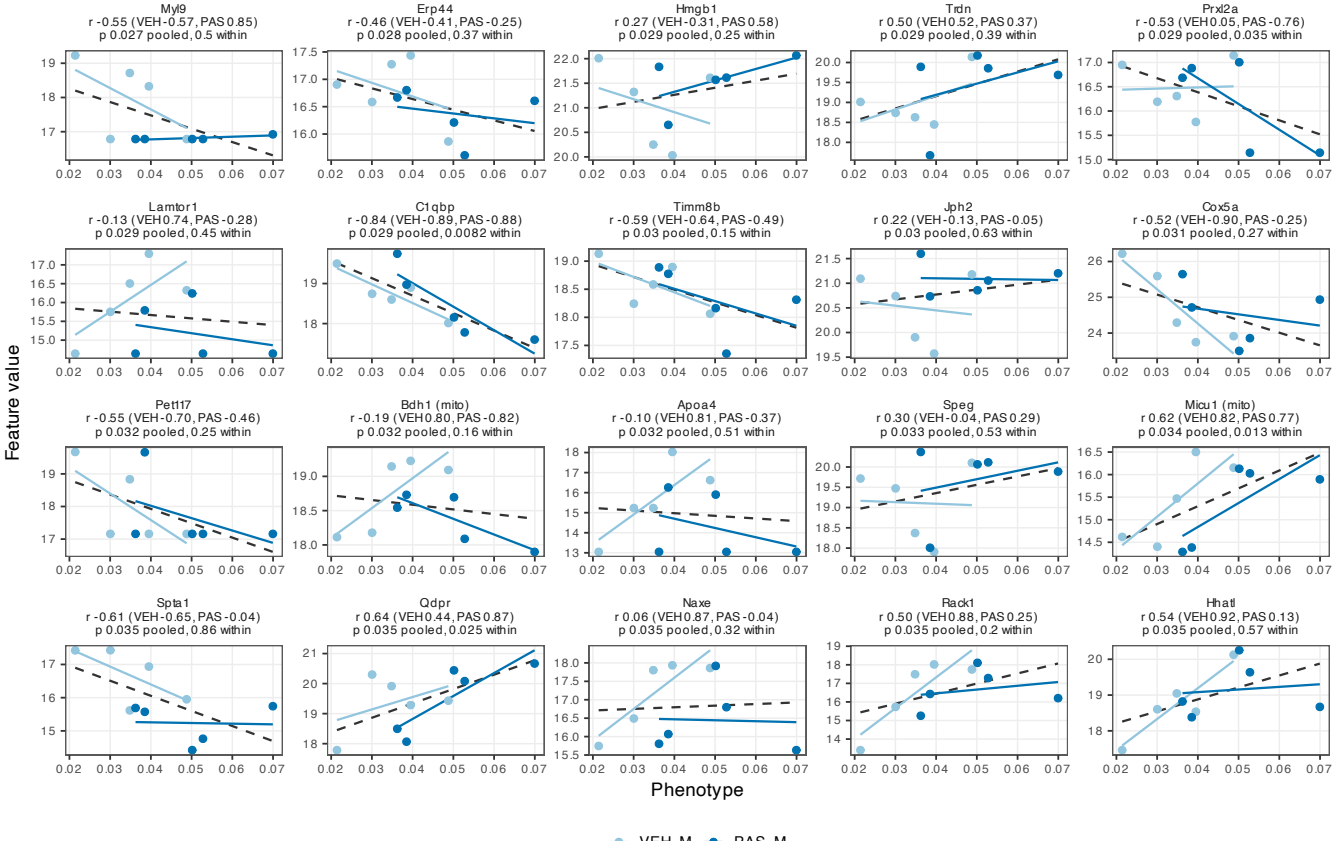
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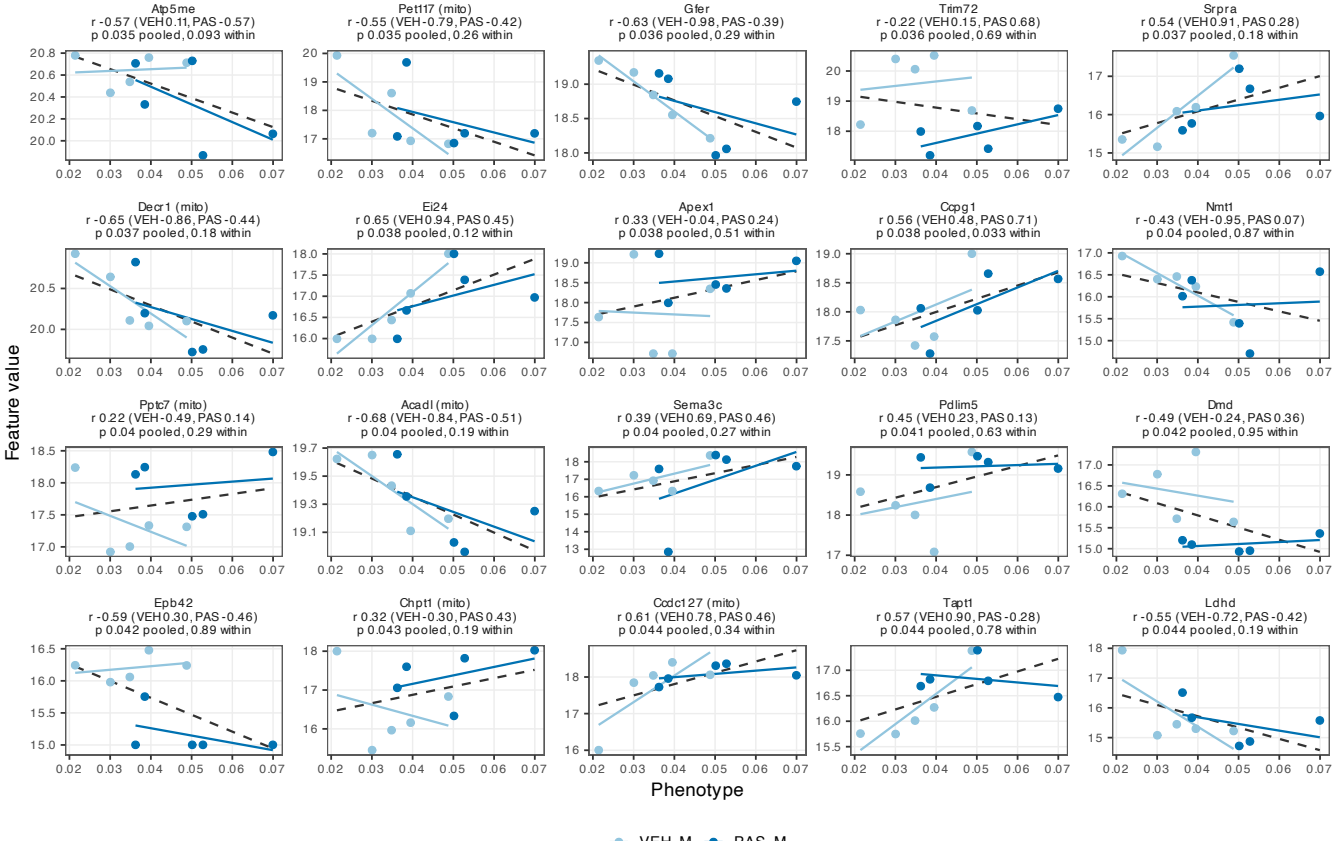
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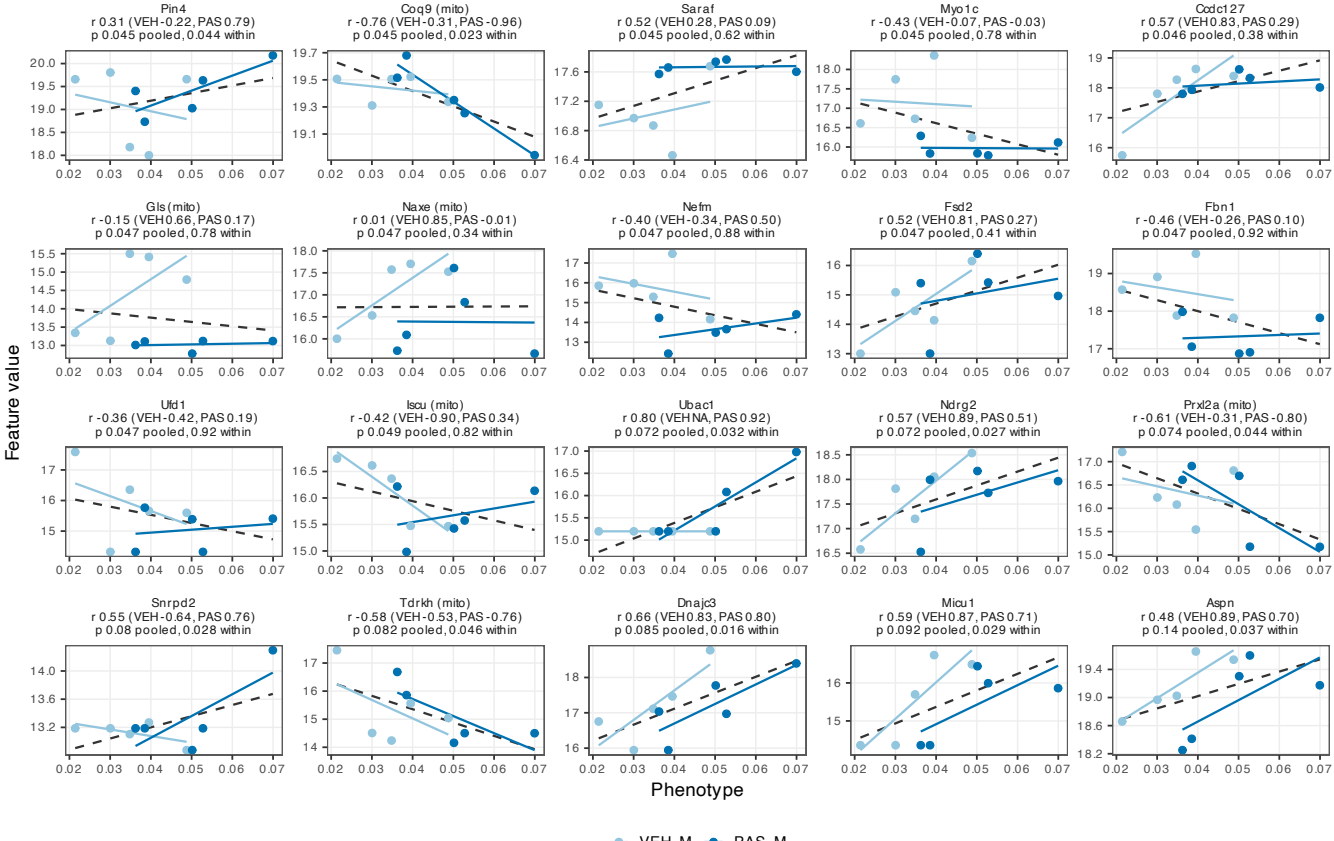
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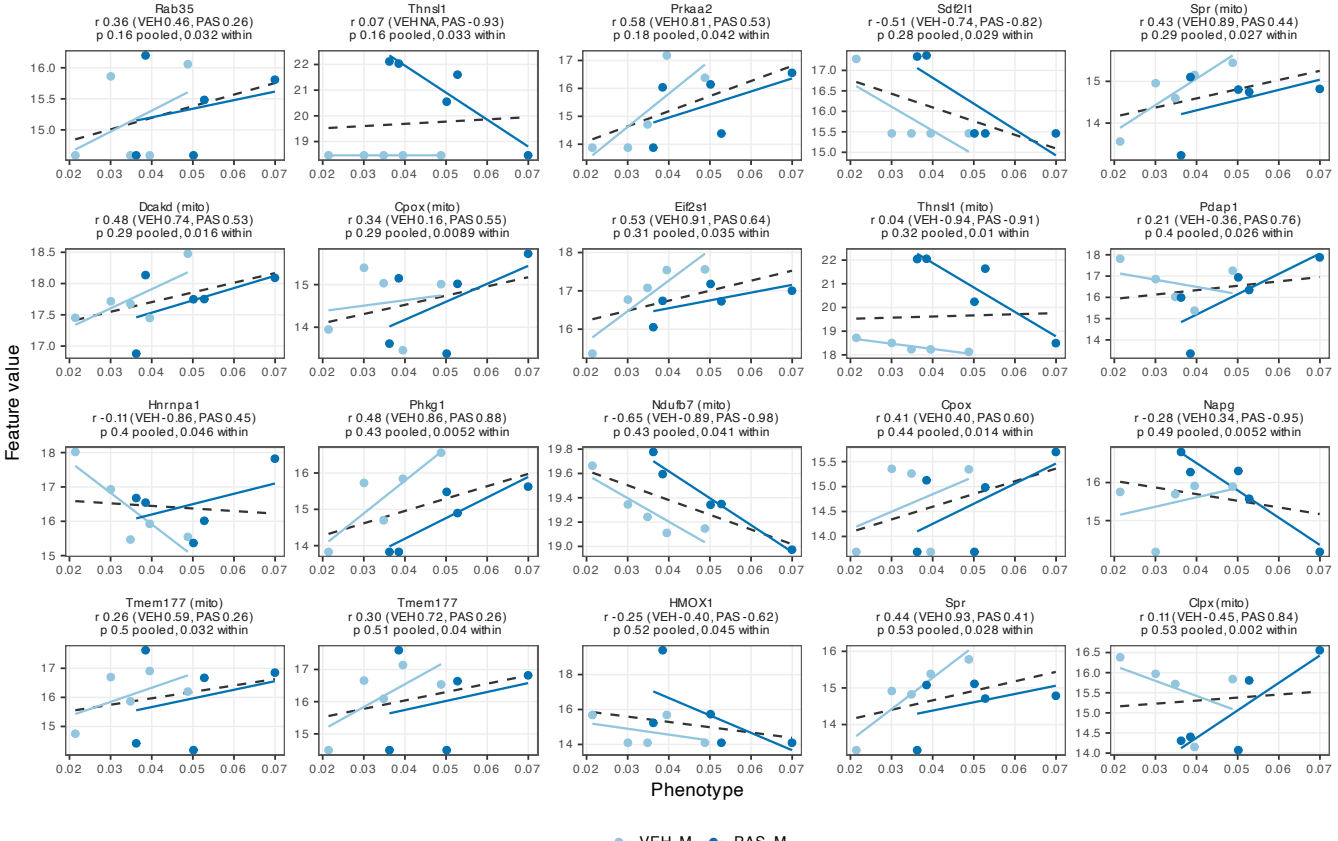
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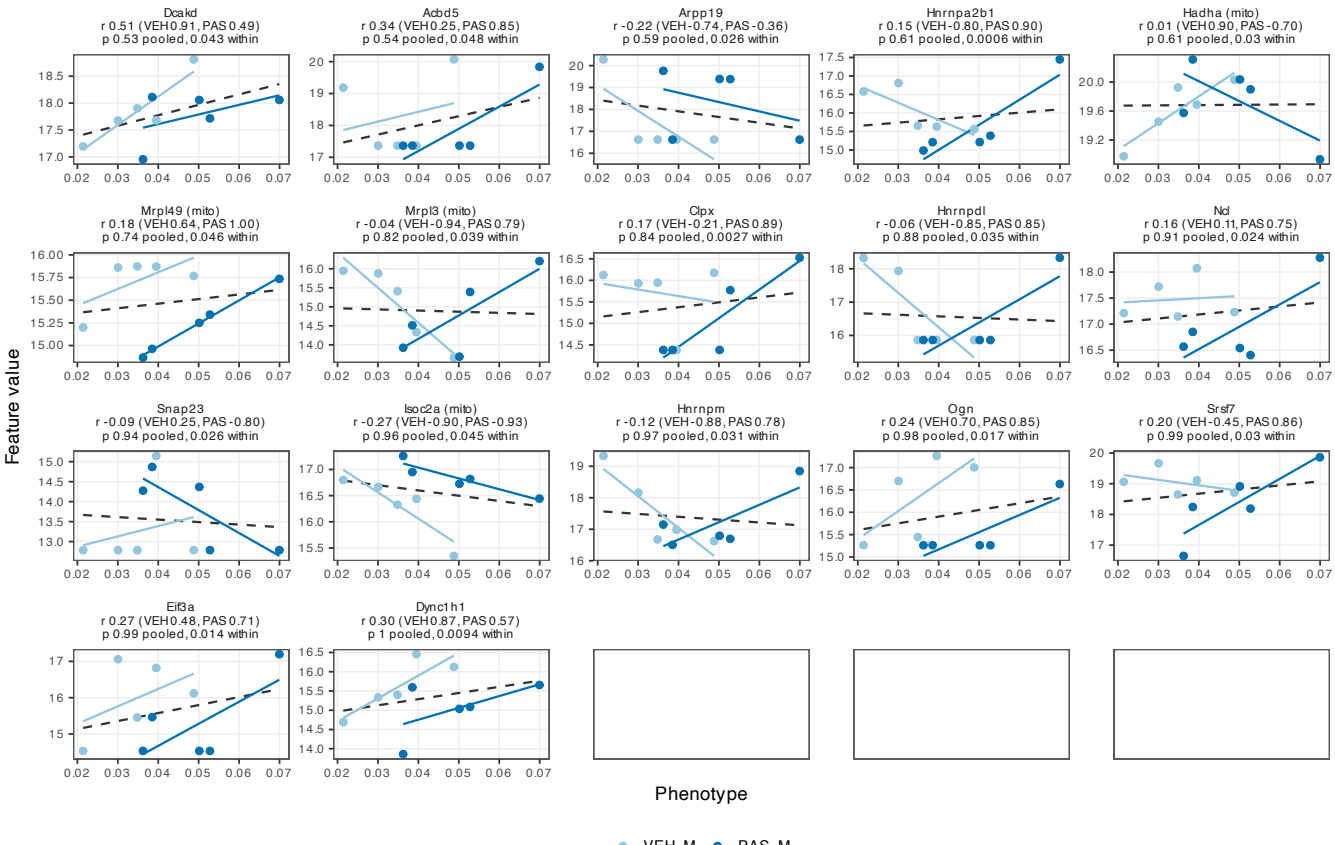
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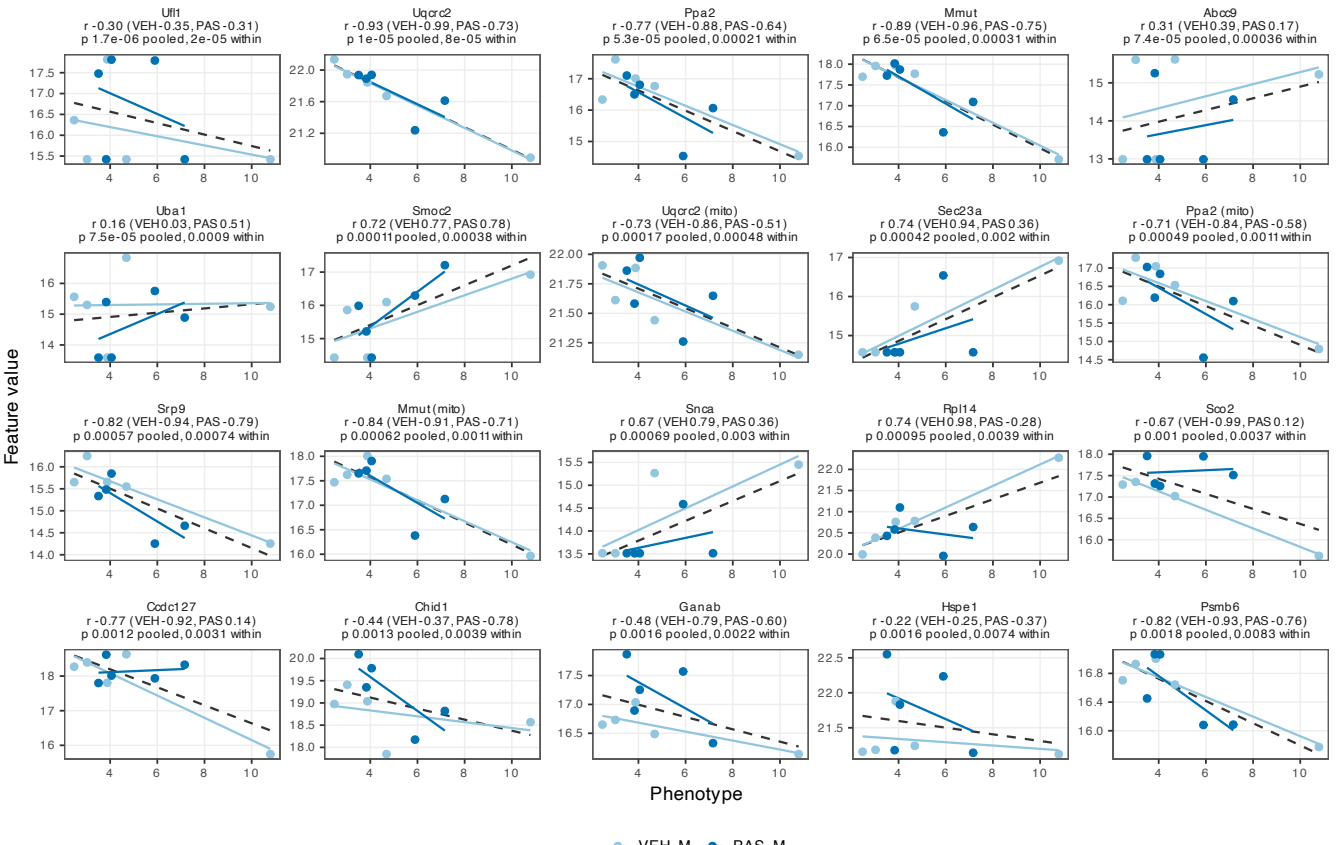
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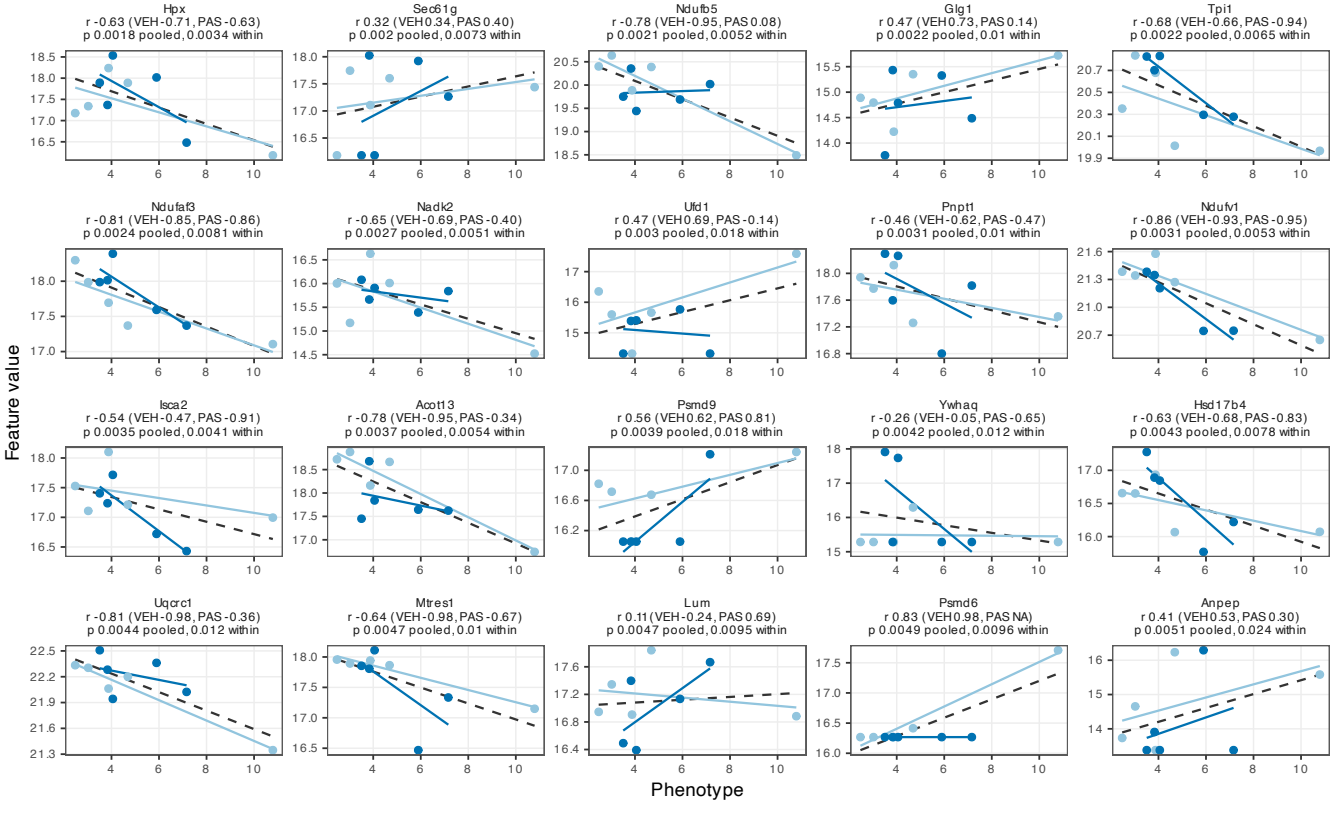
**Male: Proteins associated with Soleus pyruvate, p ≤ 0.05 in either model (n = 206)**

*Phenotype effect of PAS in this sex: +9% (p 0.77), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



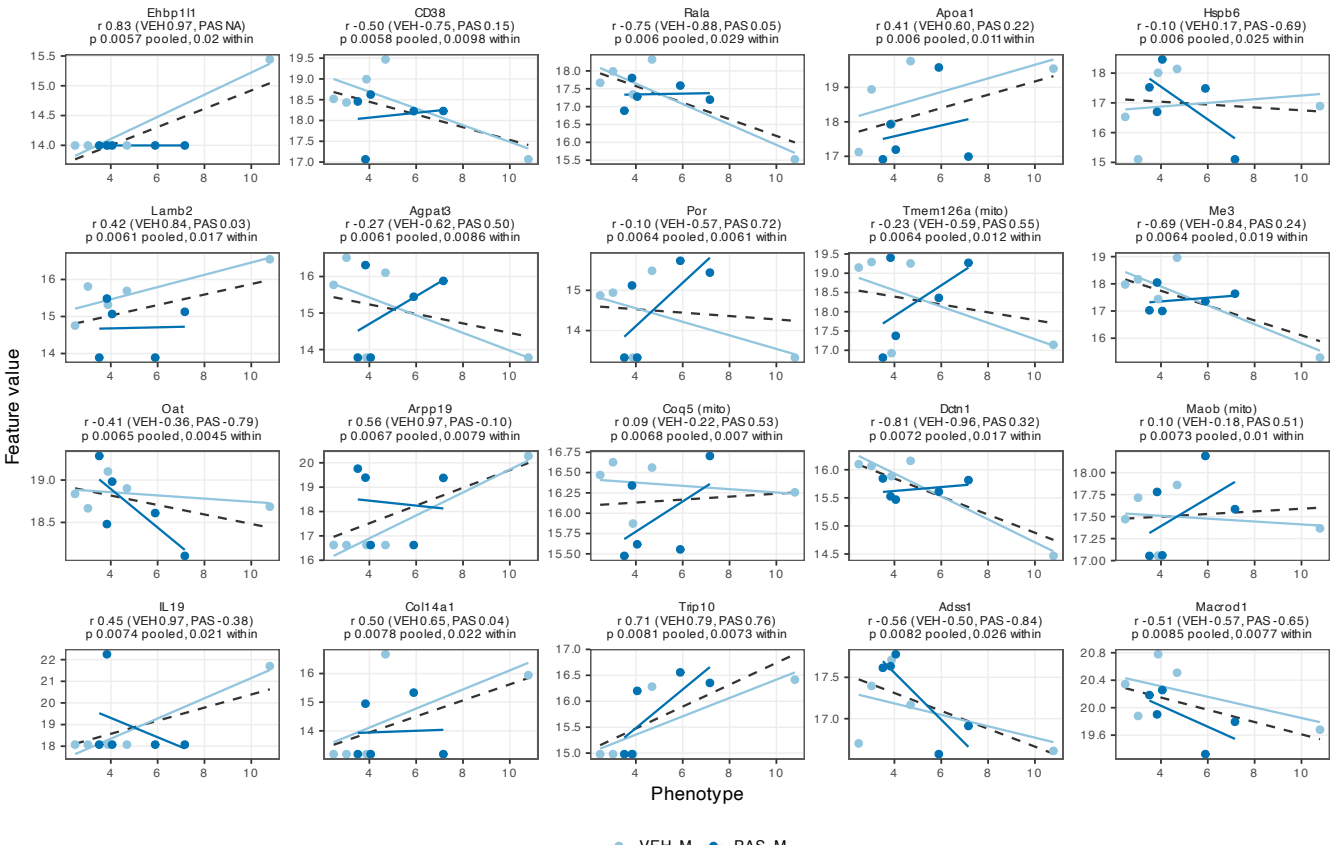
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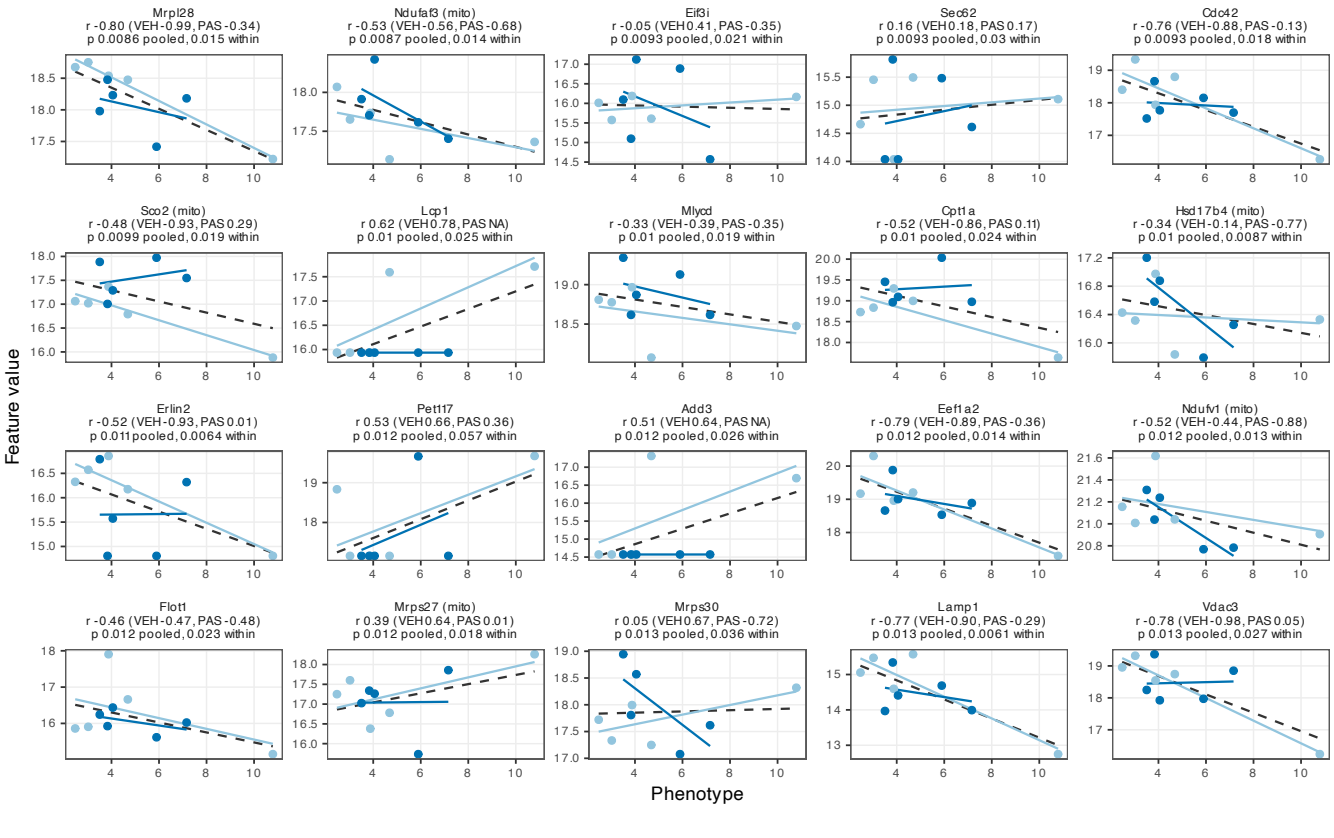
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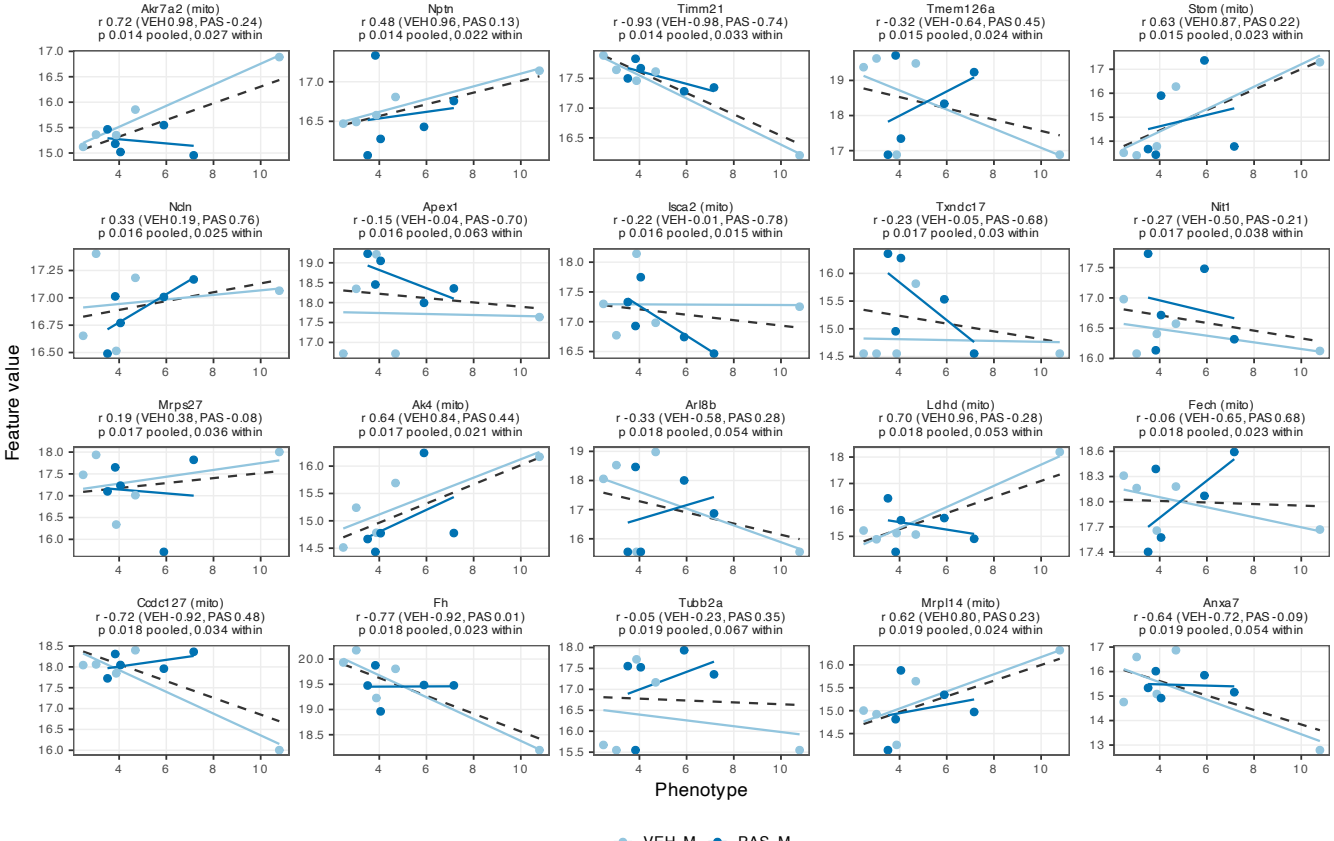
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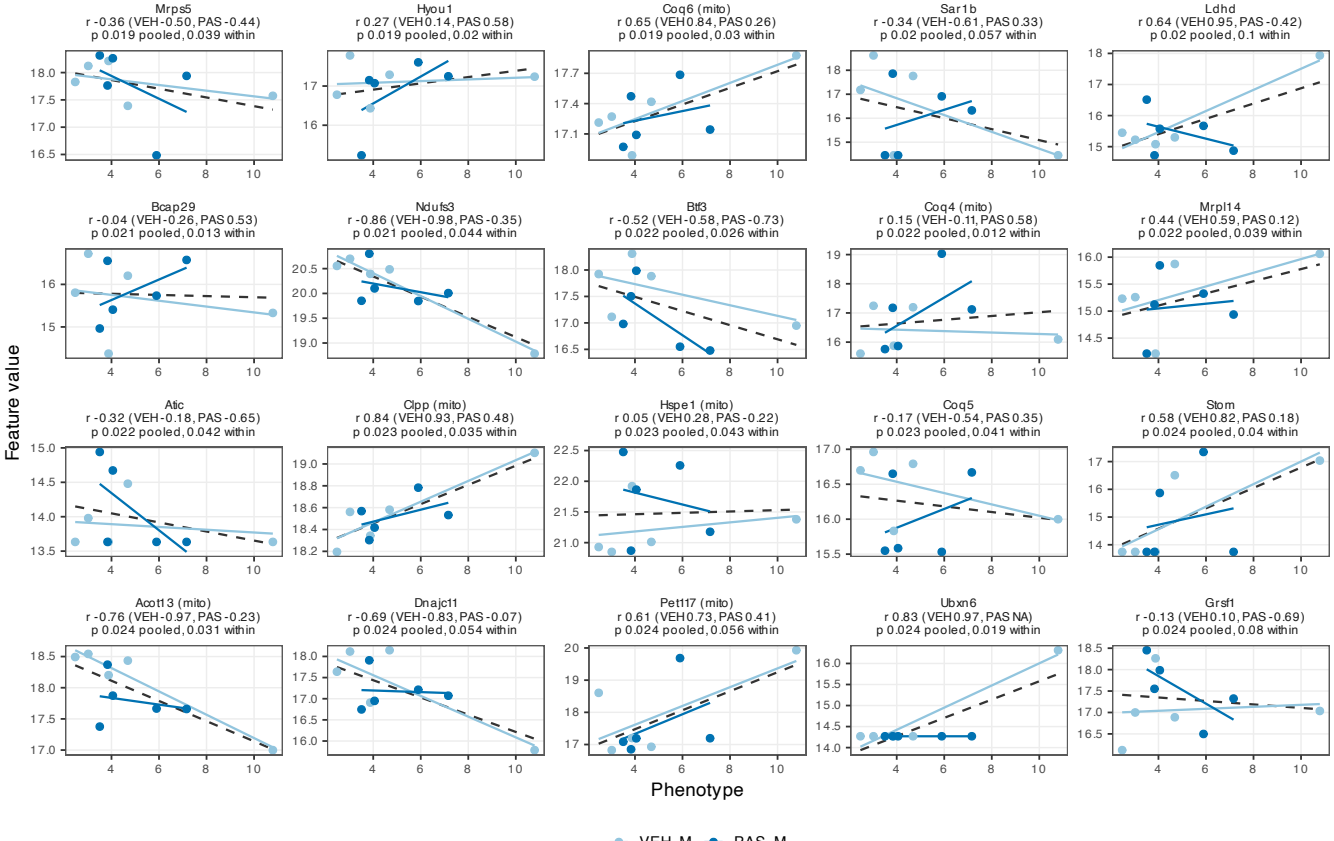
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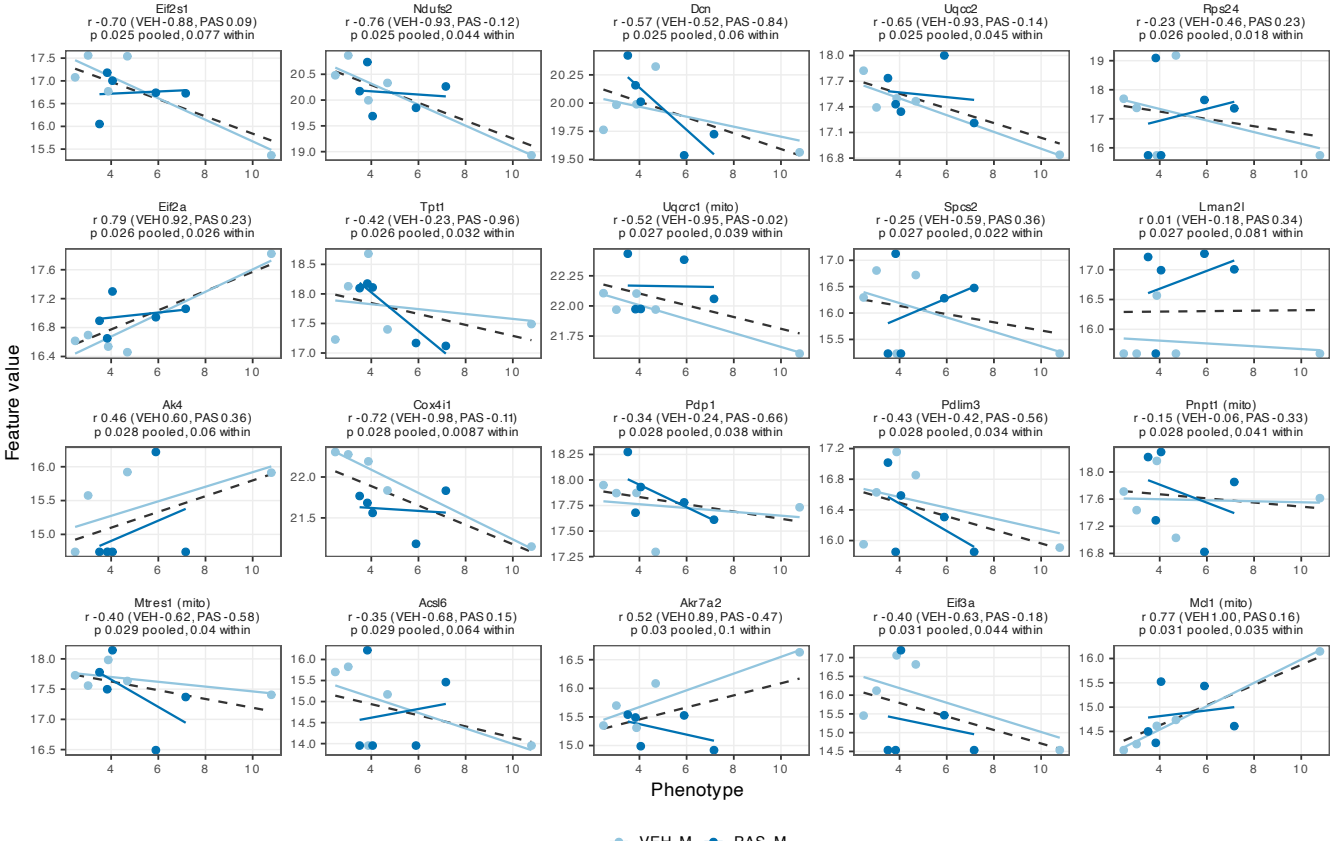
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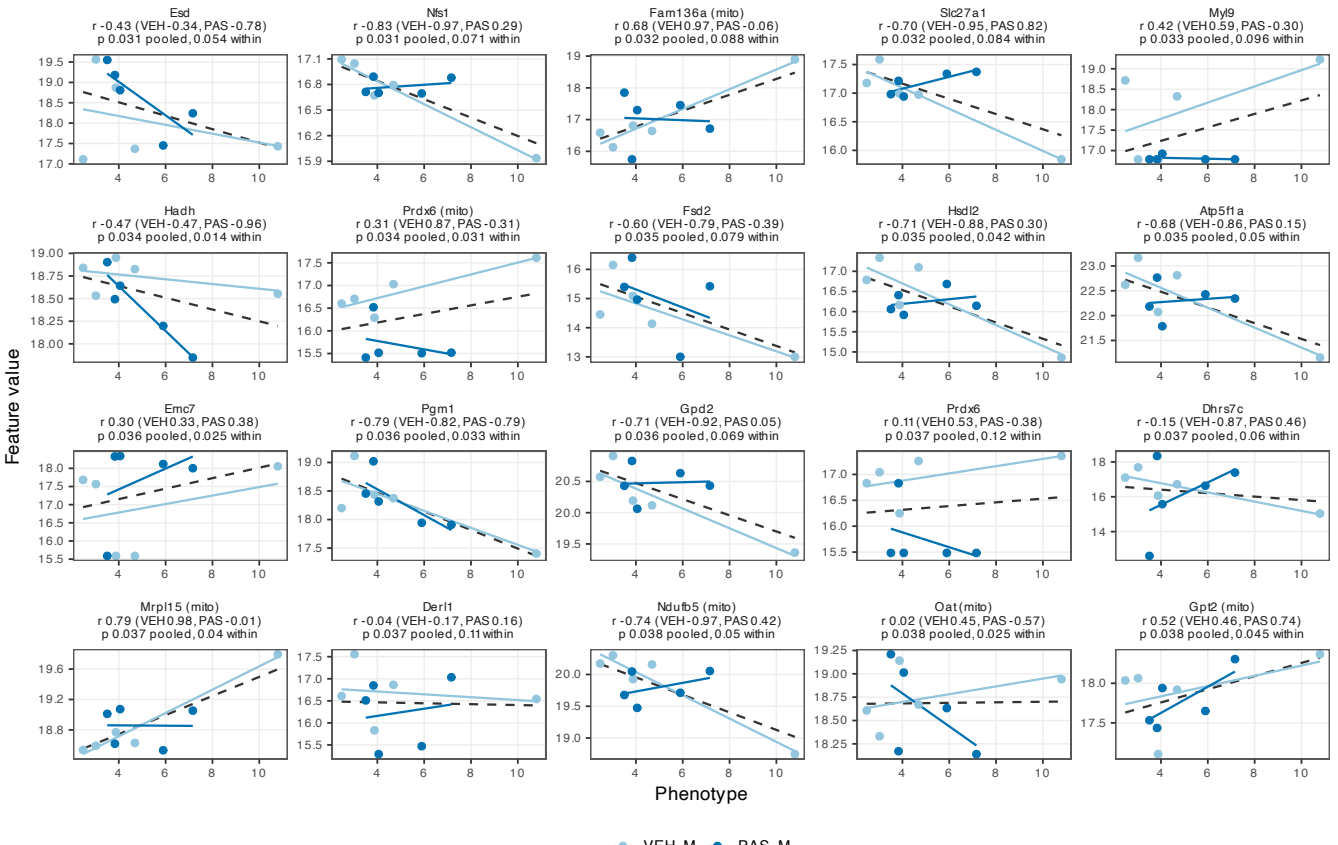
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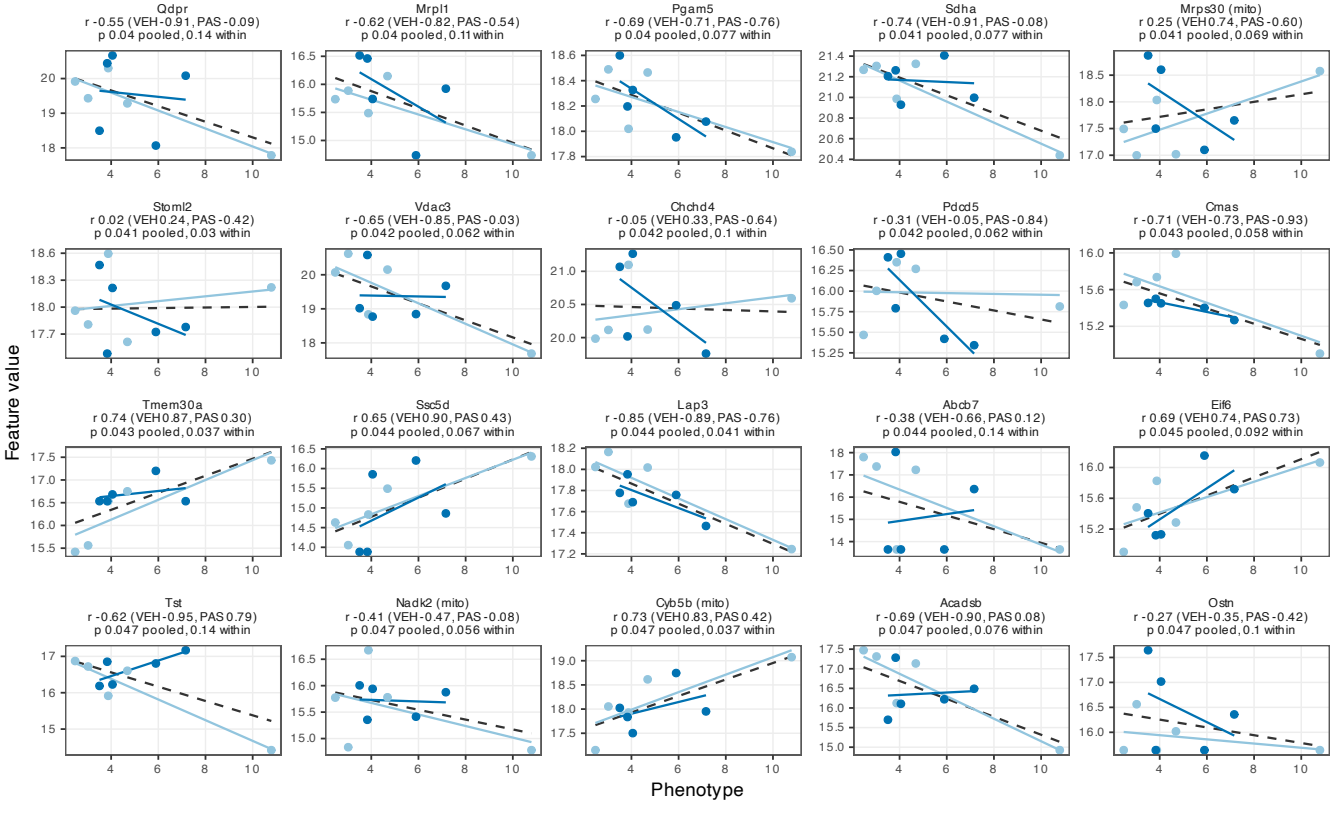
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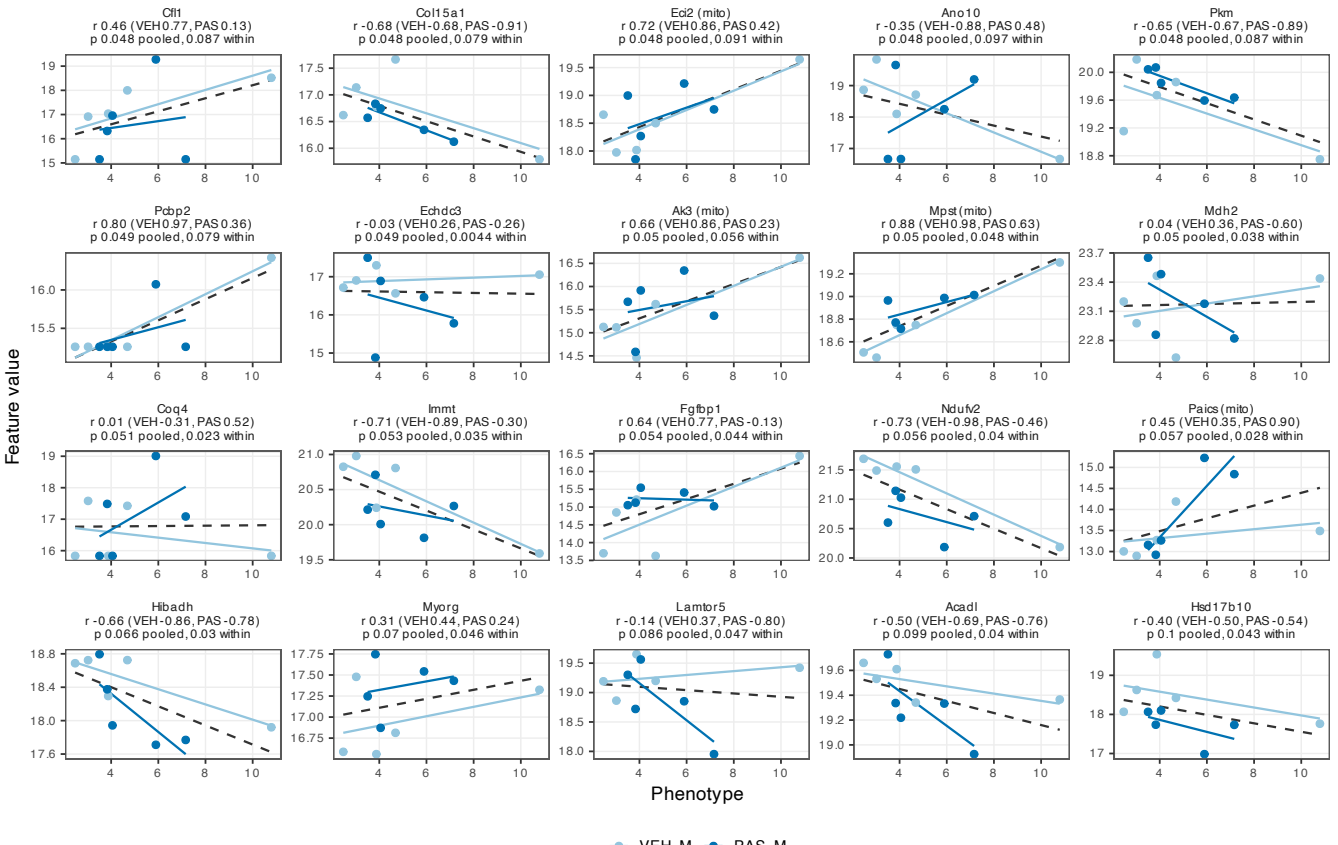
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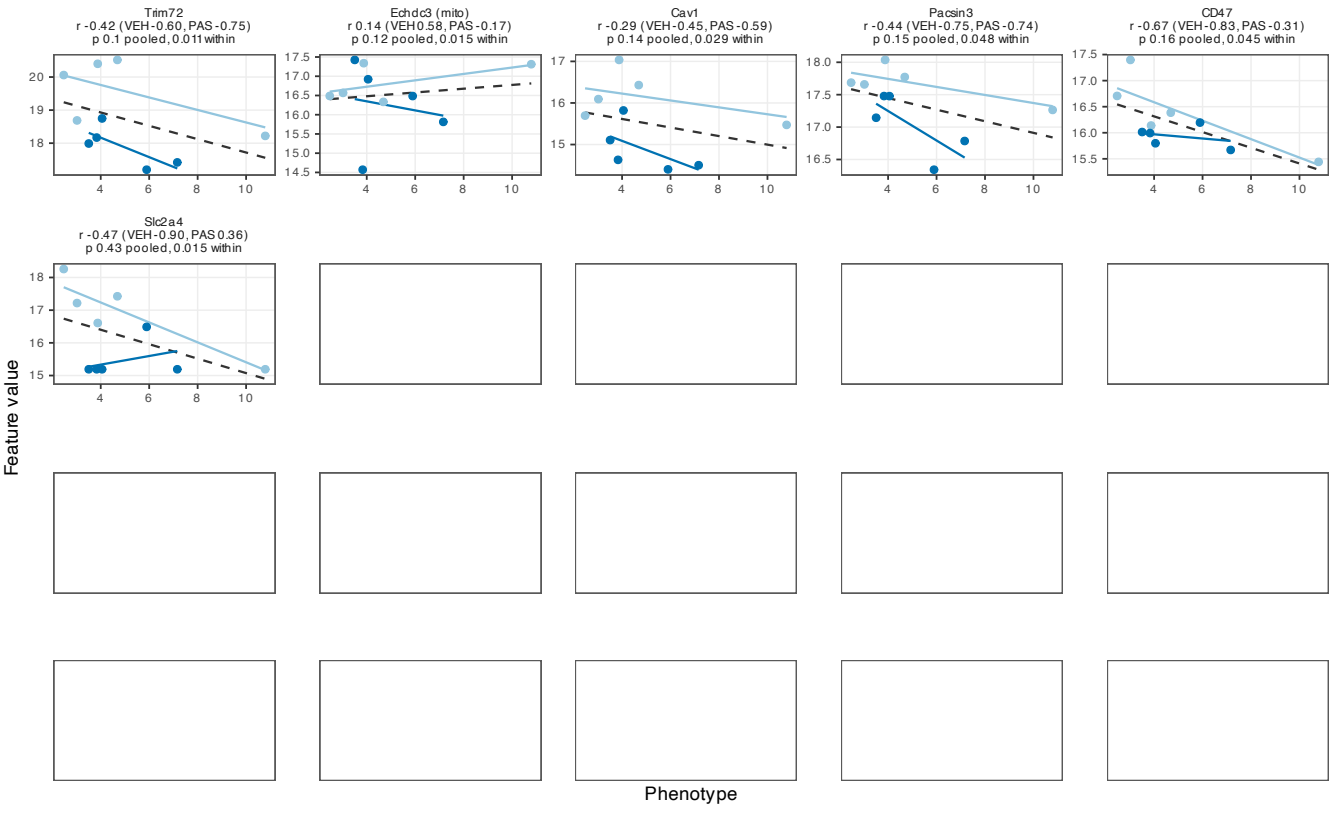
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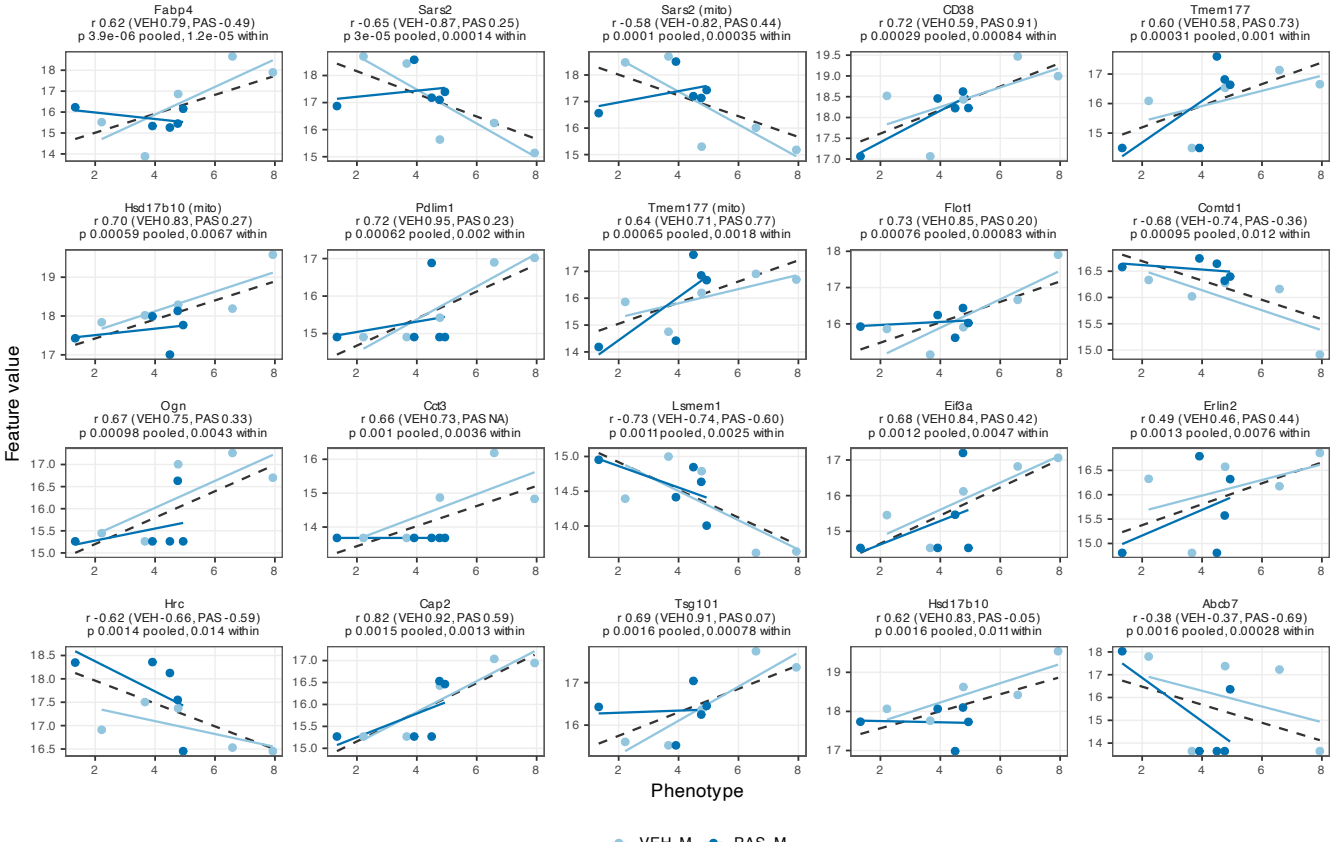
**Male: Proteins associated with Soleus pyruvate, p ≤ 0.05 in either model (n = 206)**

*Phenotype effect of PAS in this sex: +9% (p 0.77), so these associations follow variation between mice, not a PAS response. Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



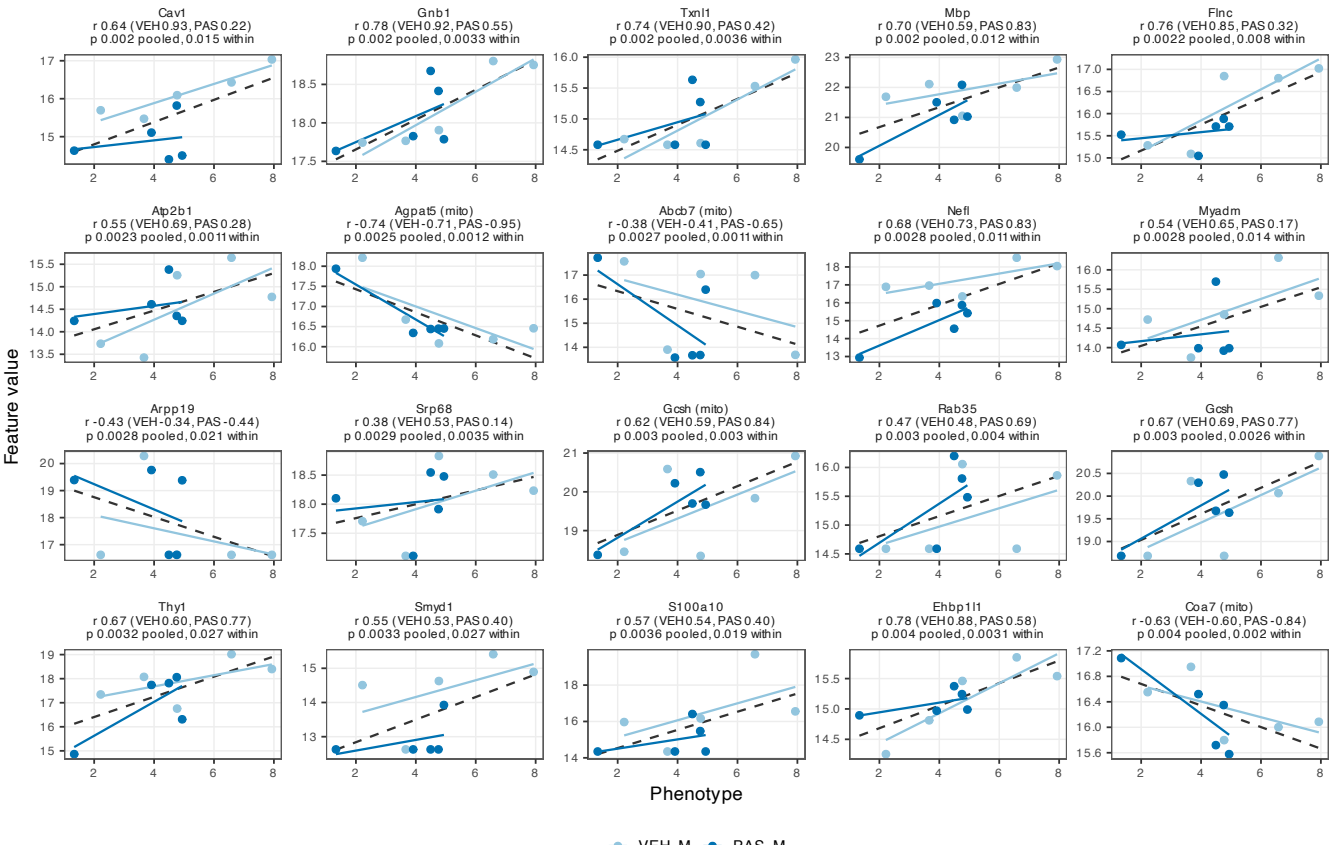
**Male: Proteins associated with Plantaris pyruvate, p ≤ 0.05 in either model (n = 283)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



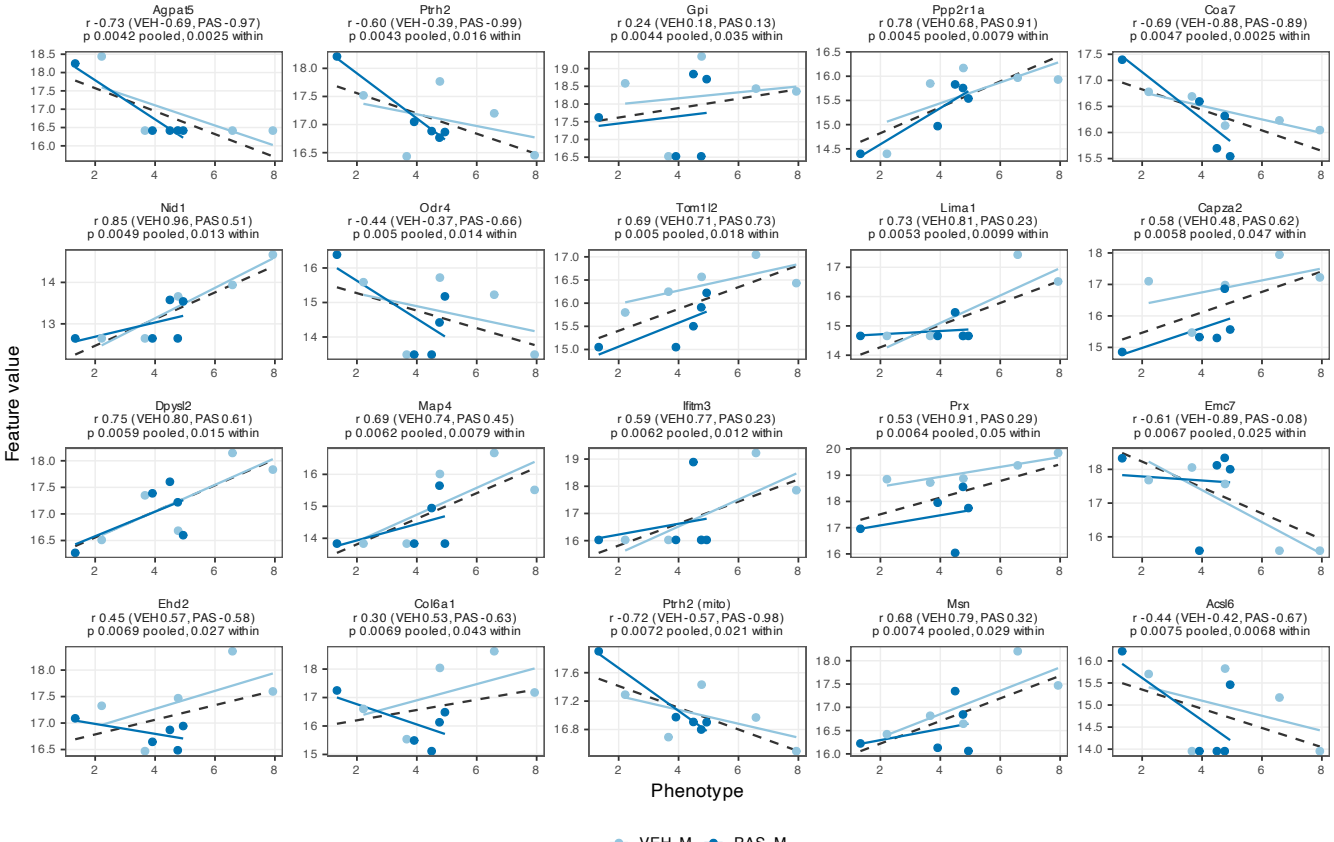
**Male: Proteins associated with Plantaris pyruvate, p ≤ 0.05 in either model (n = 283)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



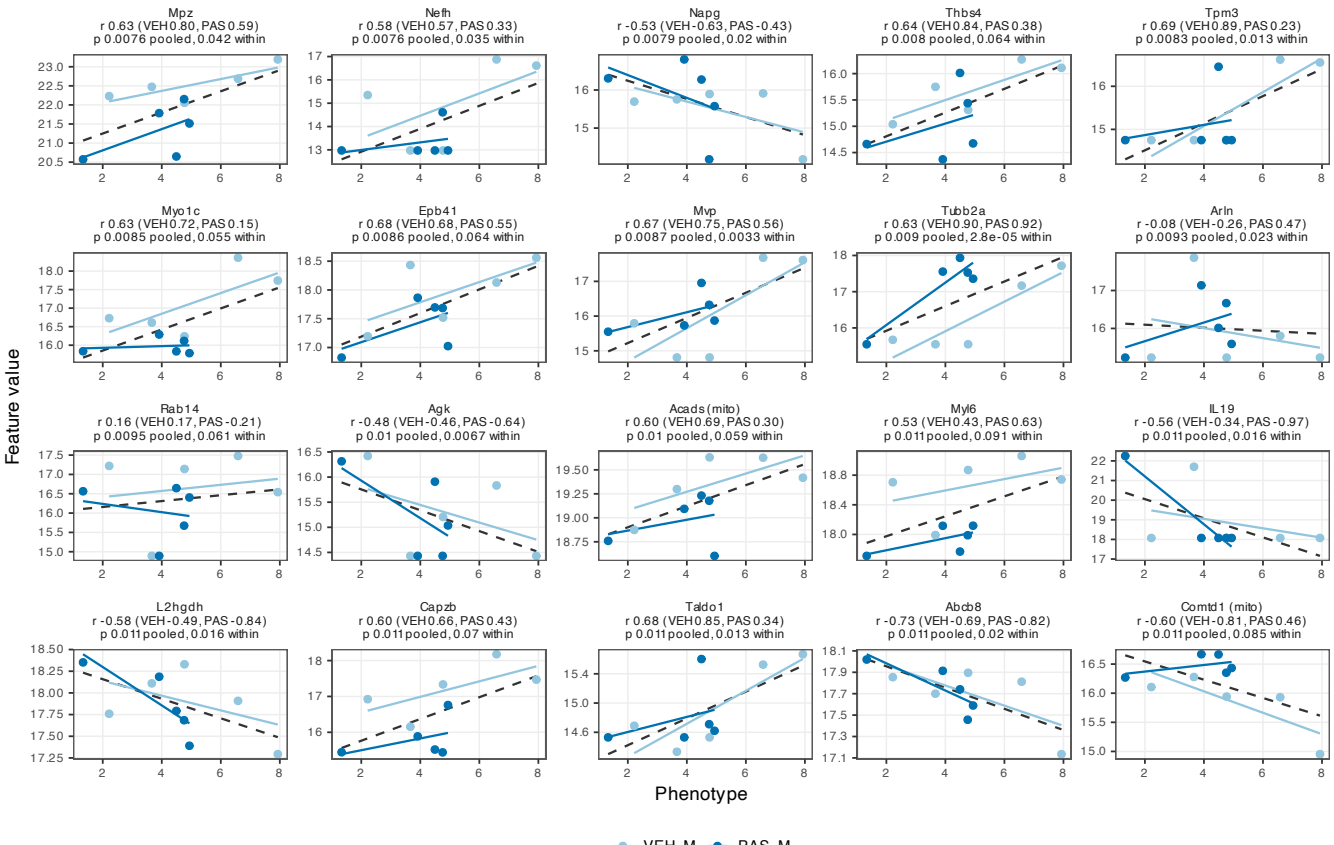
**Male: Proteins associated with Plantaris pyruvate, p ≤ 0.05 in either model (n = 283)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



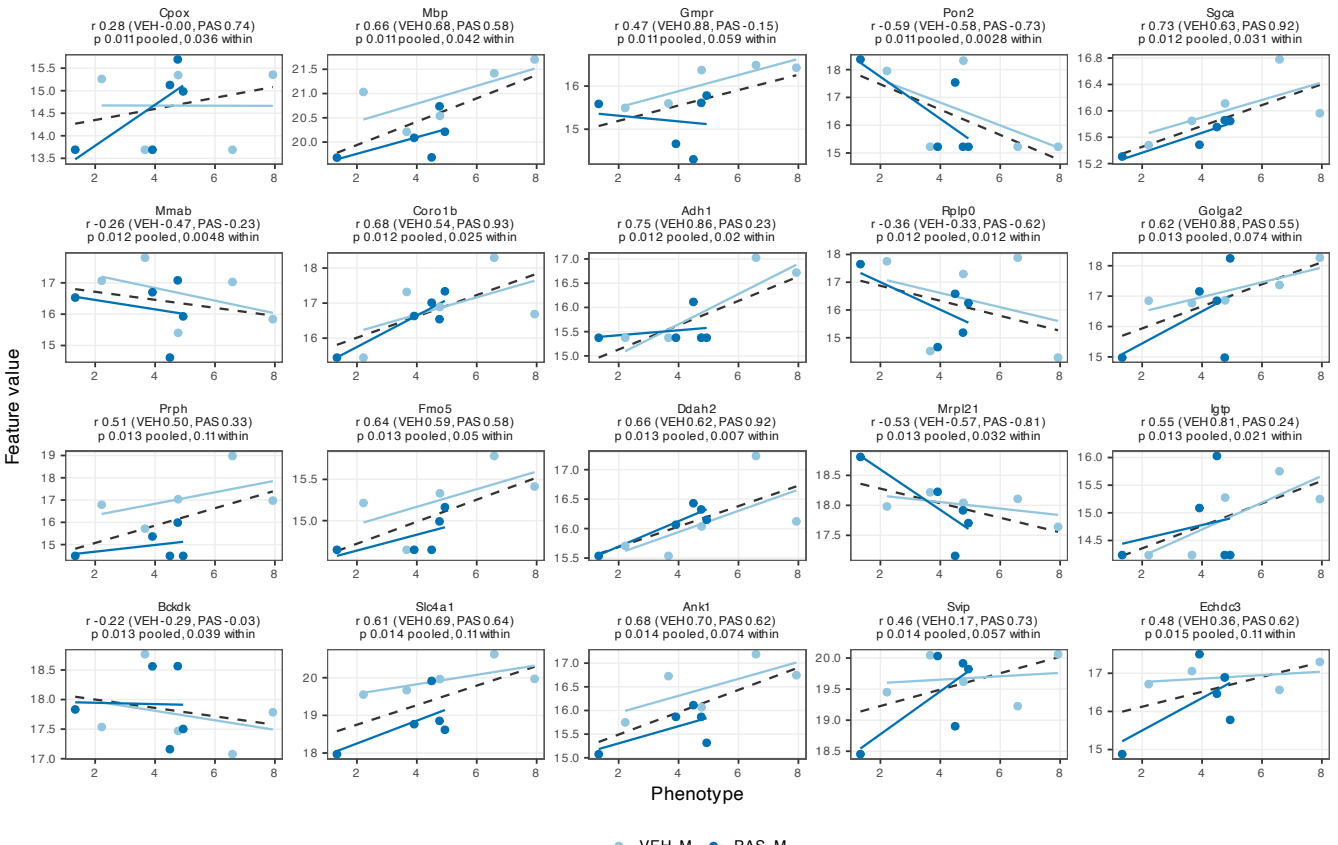
**Male: Proteins associated with Plantaris pyruvate, p ≤ 0.05 in either model (n = 283)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



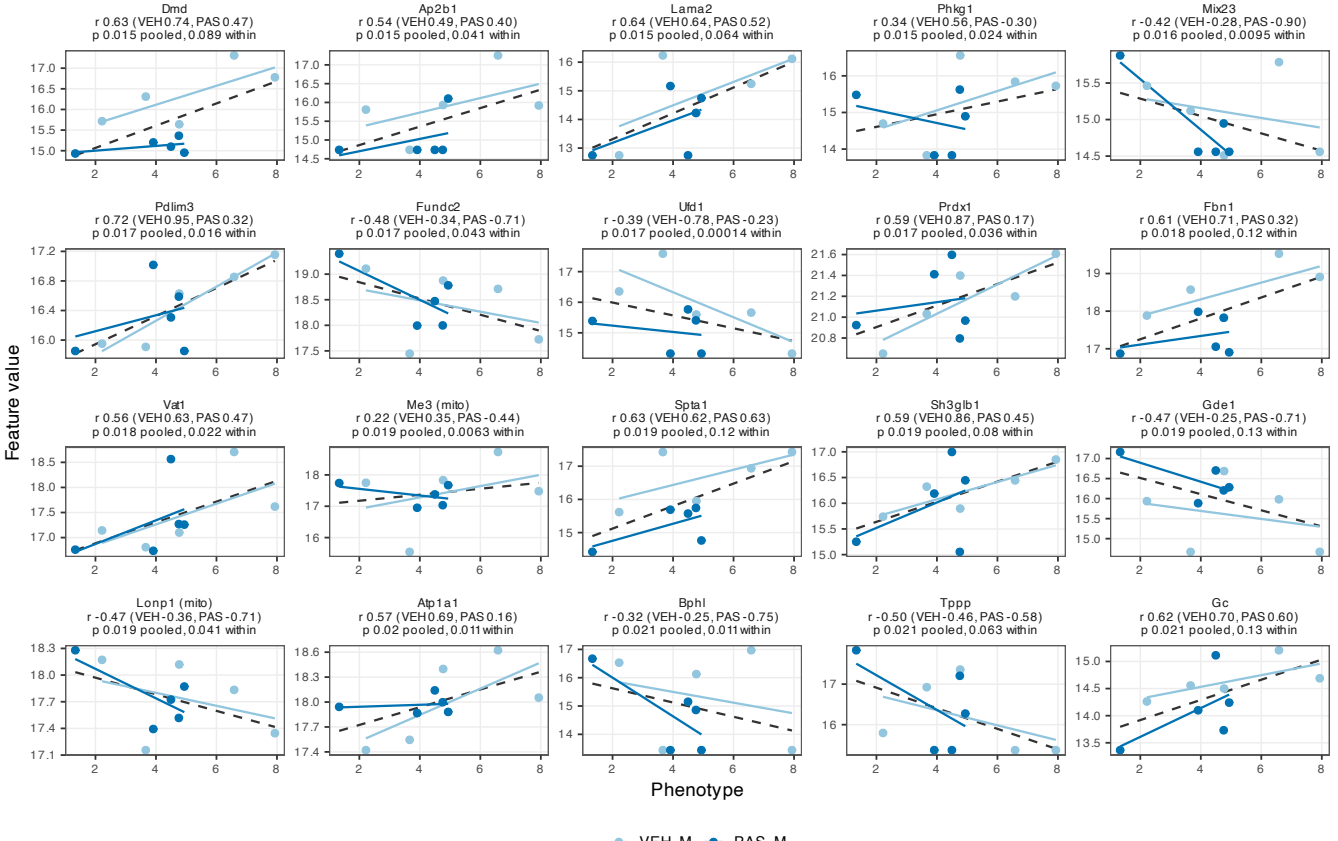
**Male: Proteins associated with Plantaris pyruvate, p ≤ 0.05 in either model (n = 283)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



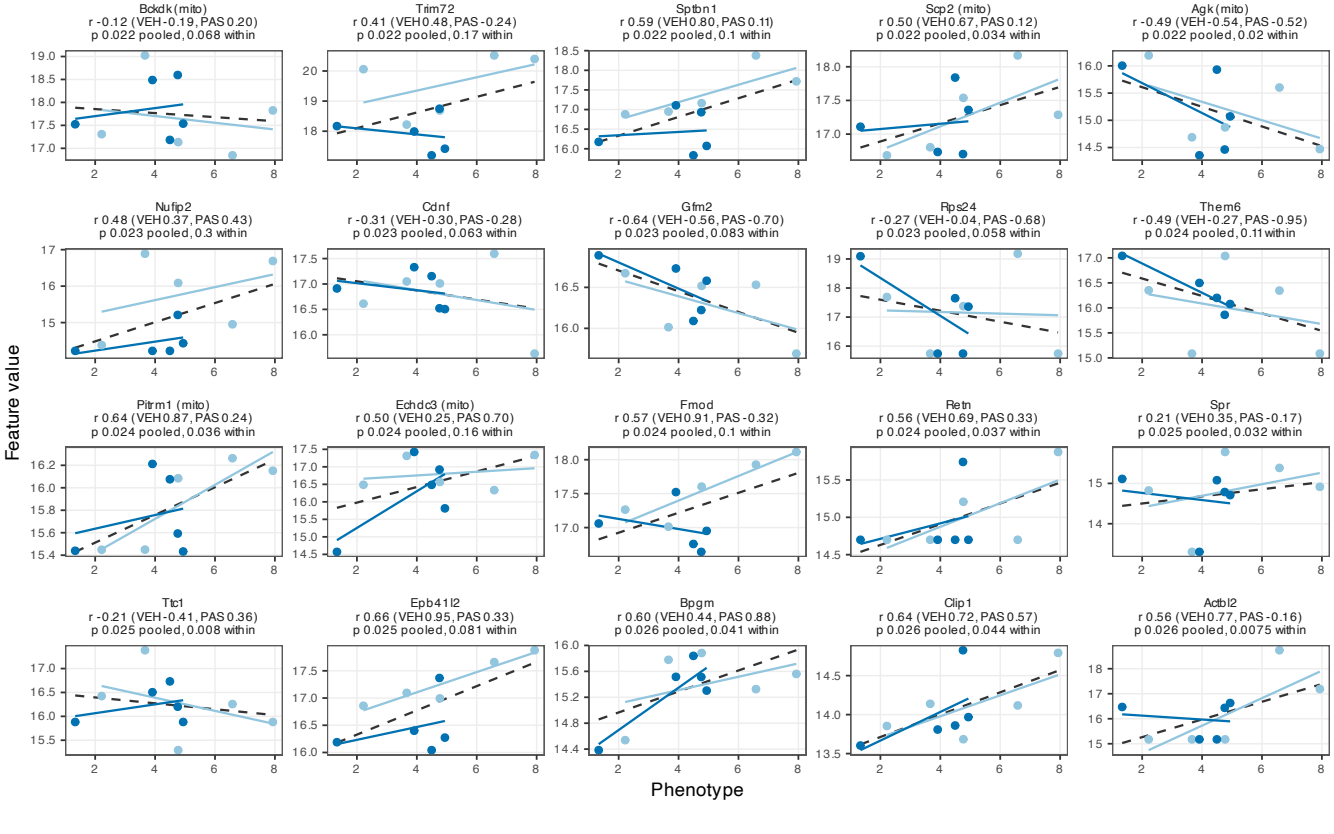
**Male: Proteins associated with Plantaris pyruvate, p ≤ 0.05 in either model (n = 283)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



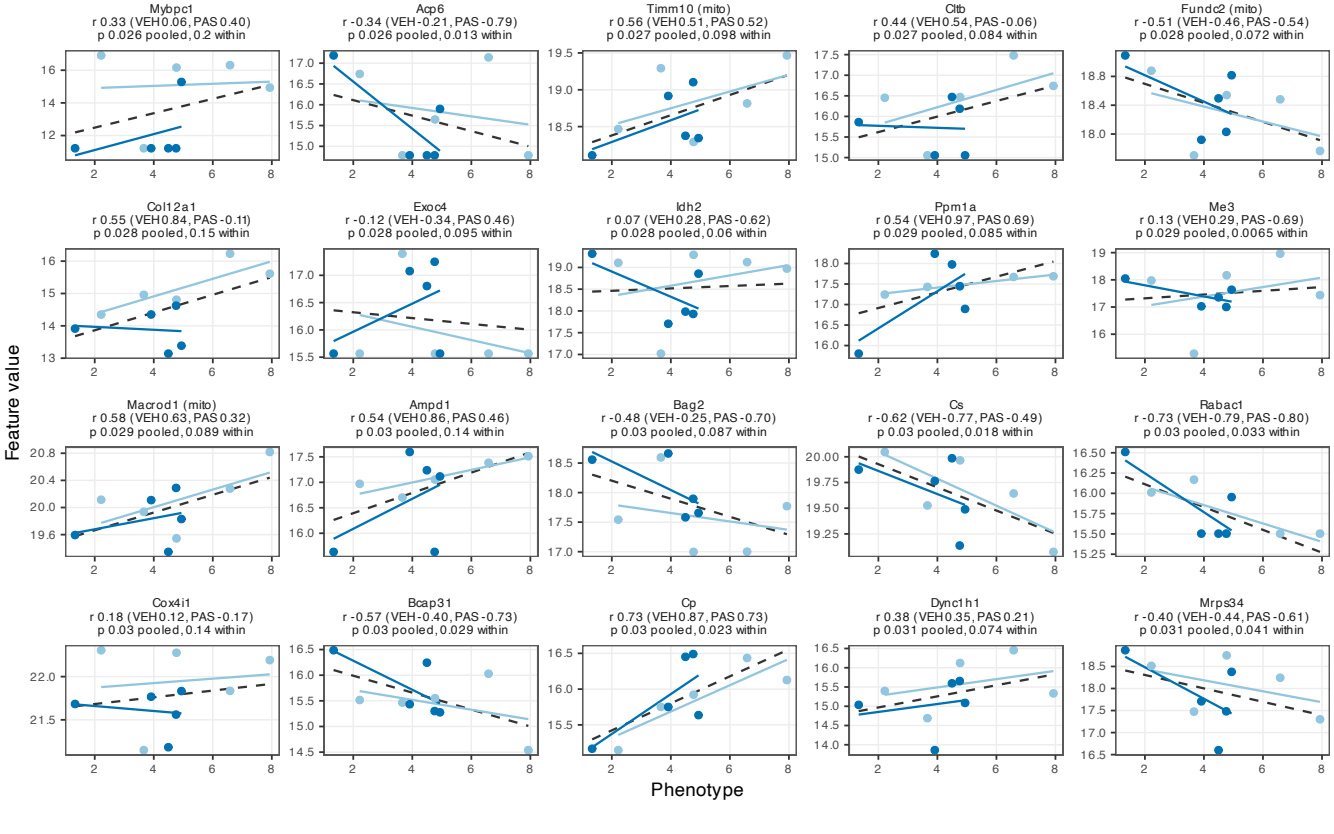
**Male: Proteins associated with Plantaris pyruvate, p ≤ 0.05 in either model (n = 283)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



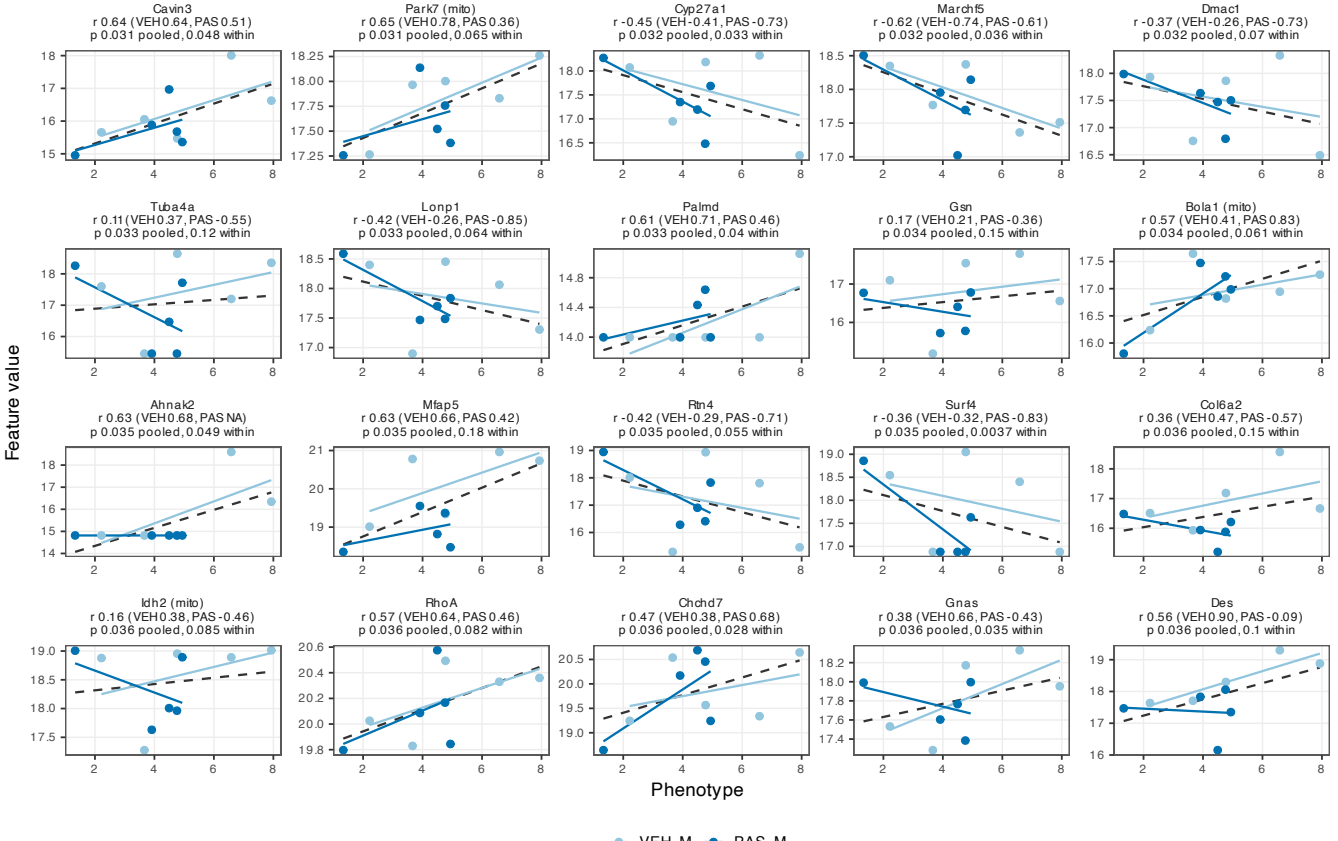
**Male: Proteins associated with Plantaris pyruvate, p ≤ 0.05 in either model (n = 283)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



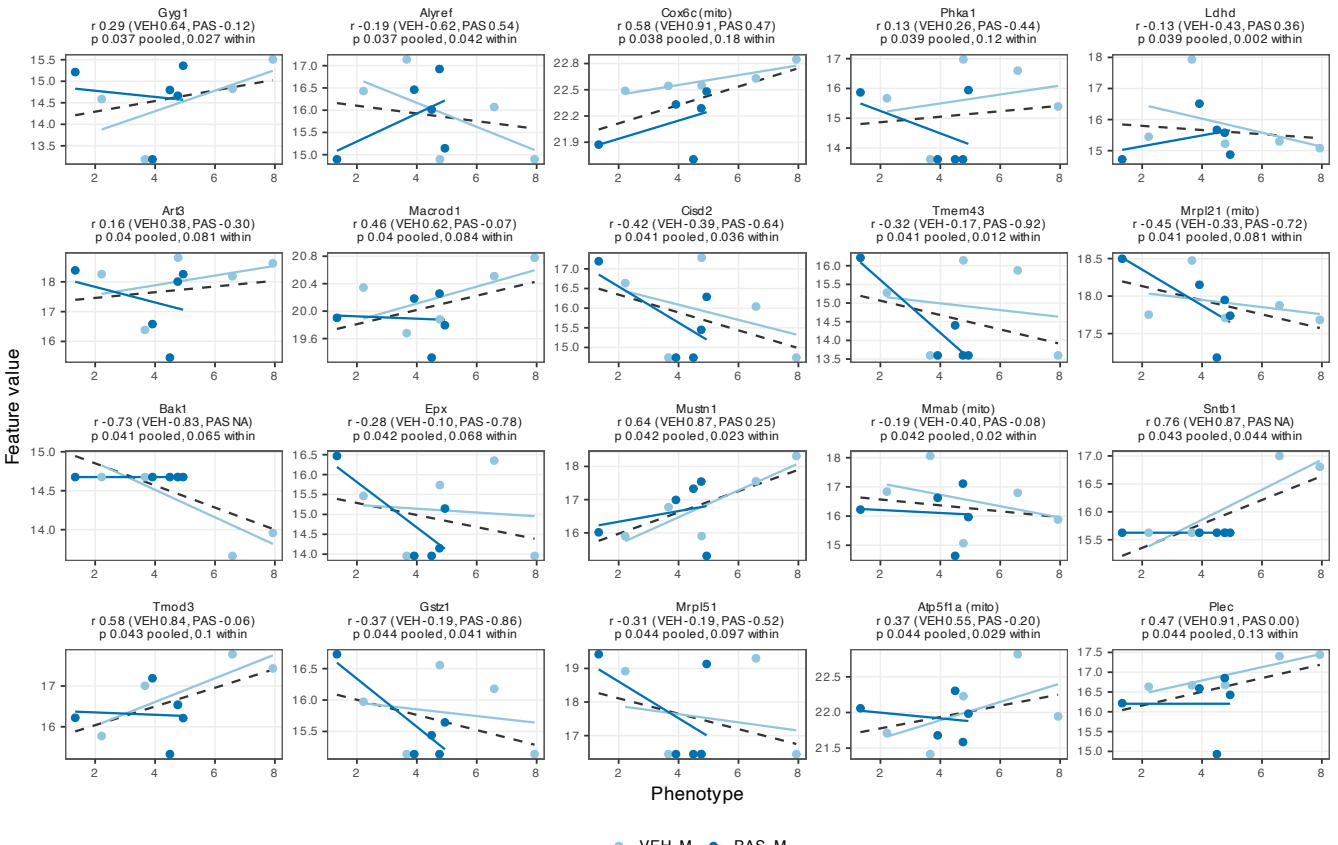
**Male: Proteins associated with Plantaris pyruvate, p ≤ 0.05 in either model (n = 283)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



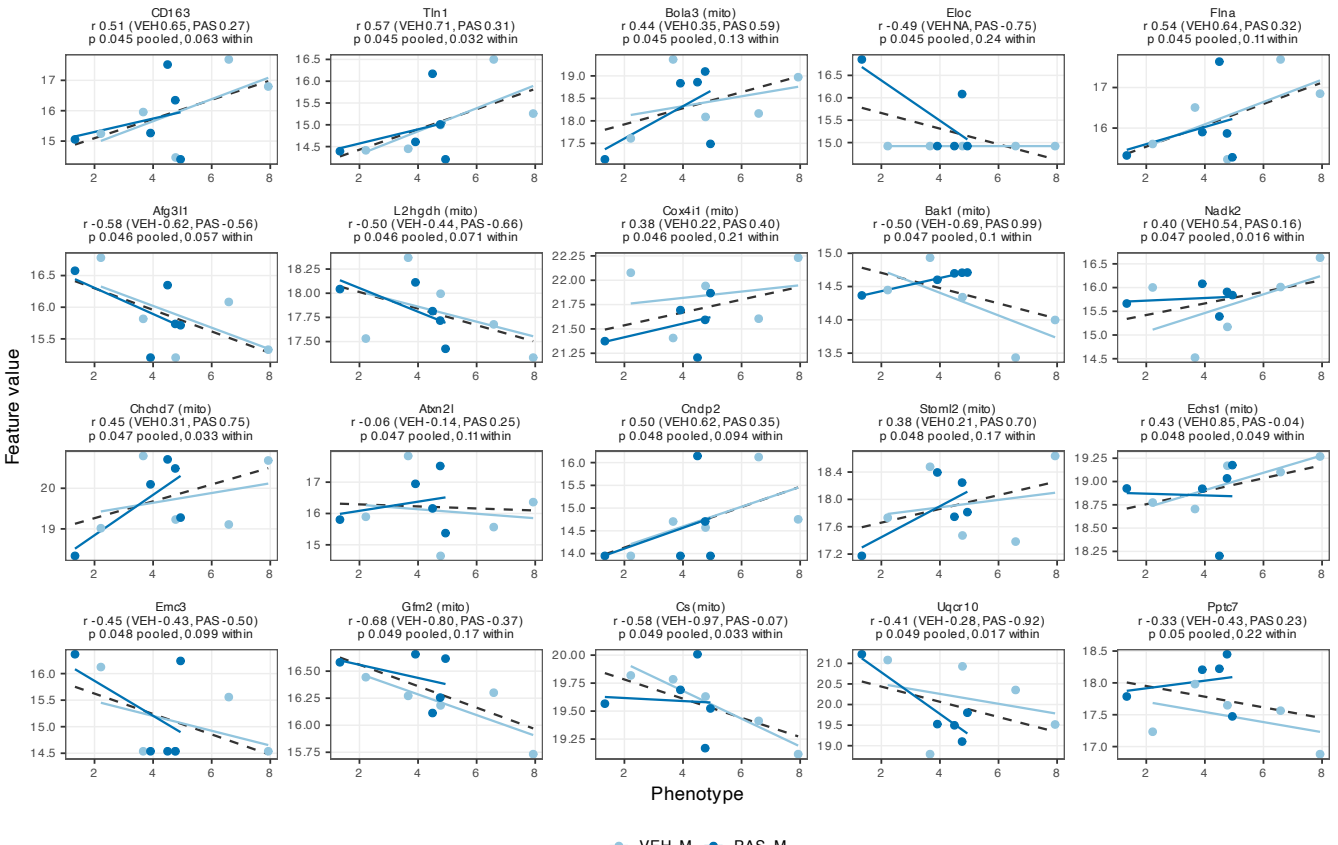
**Male: Proteins associated with Plantaris pyruvate, p ≤ 0.05 in either model (n = 283)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



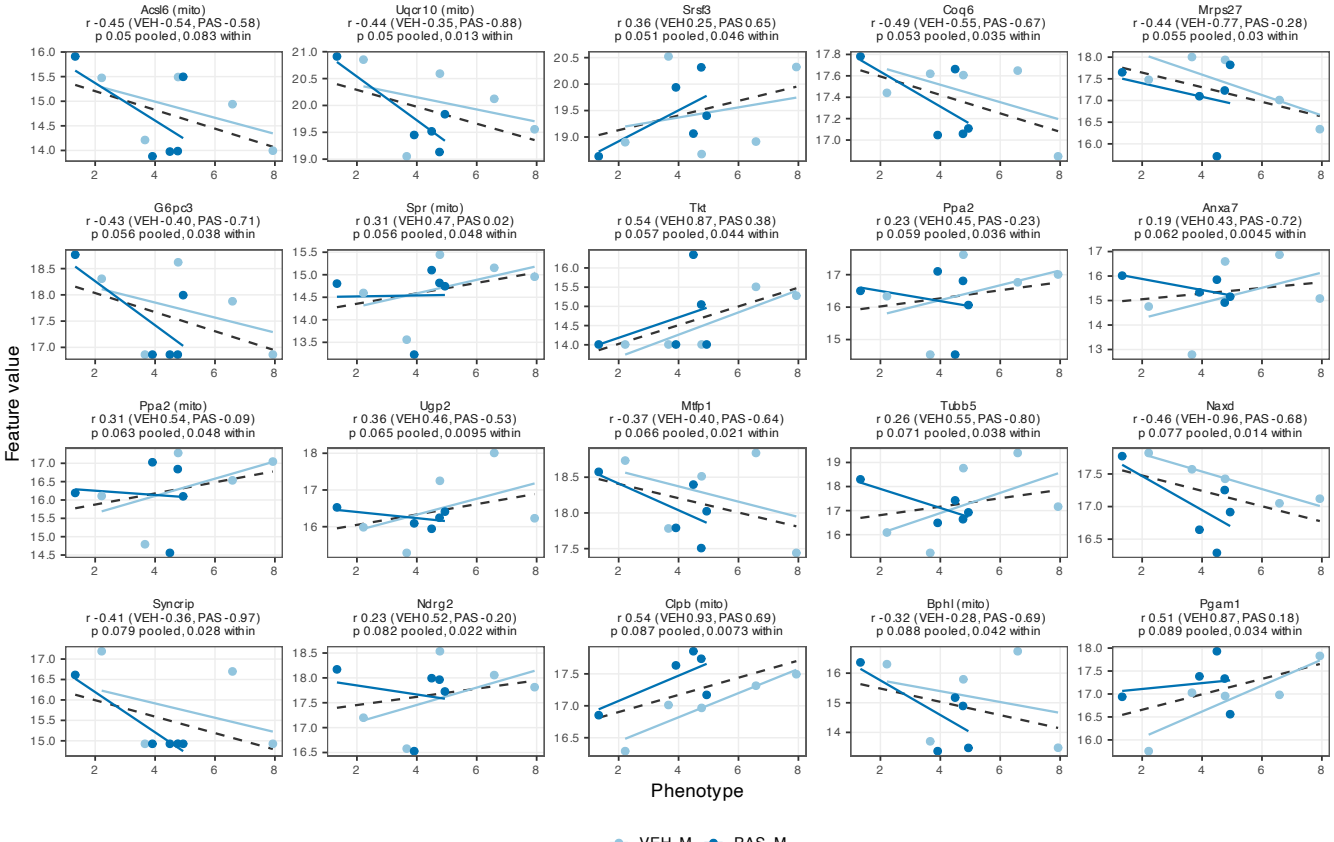
**Male: Proteins associated with Plantaris pyruvate, p ≤ 0.05 in either model (n = 283)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



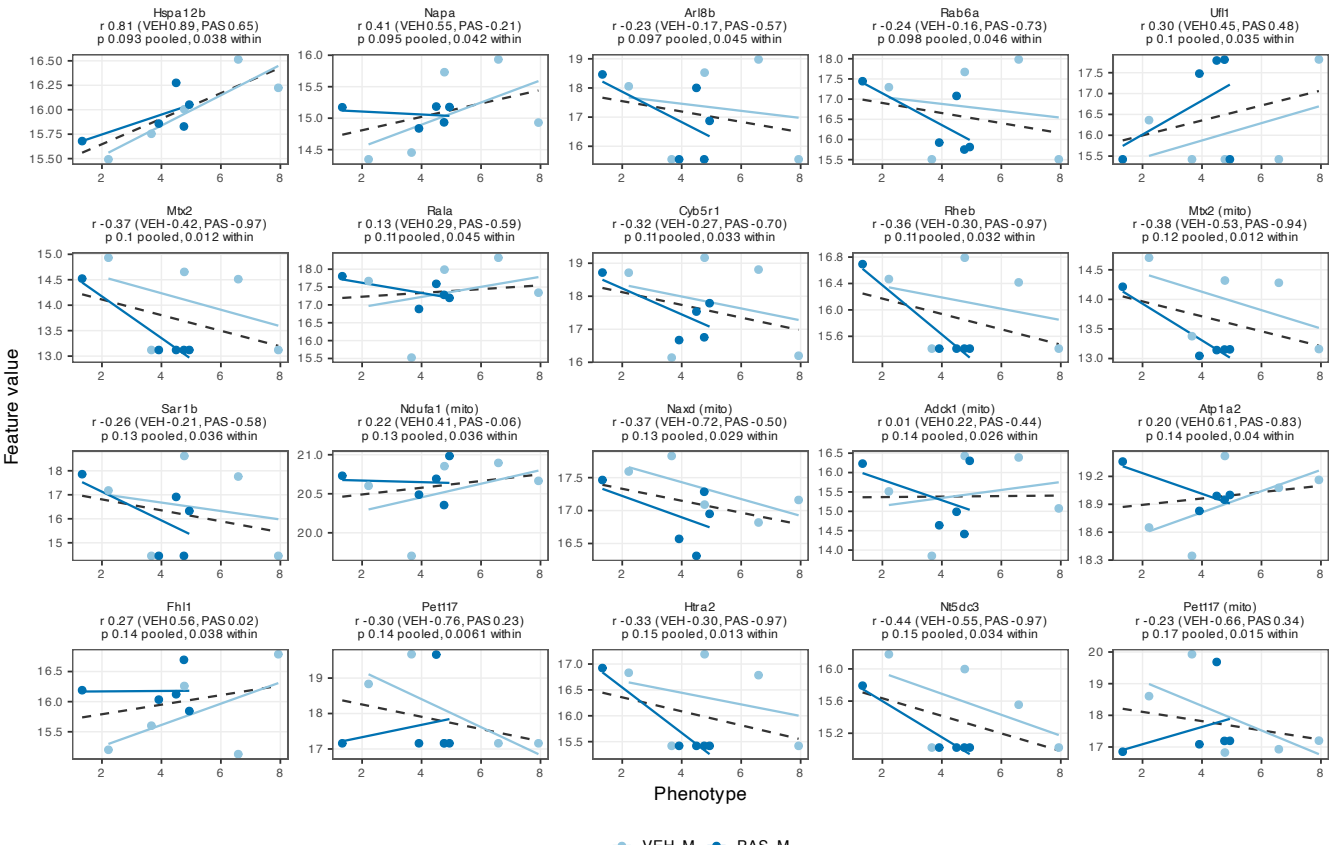
**Male: Proteins associated with Plantaris pyruvate, p ≤ 0.05 in either model (n = 283)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



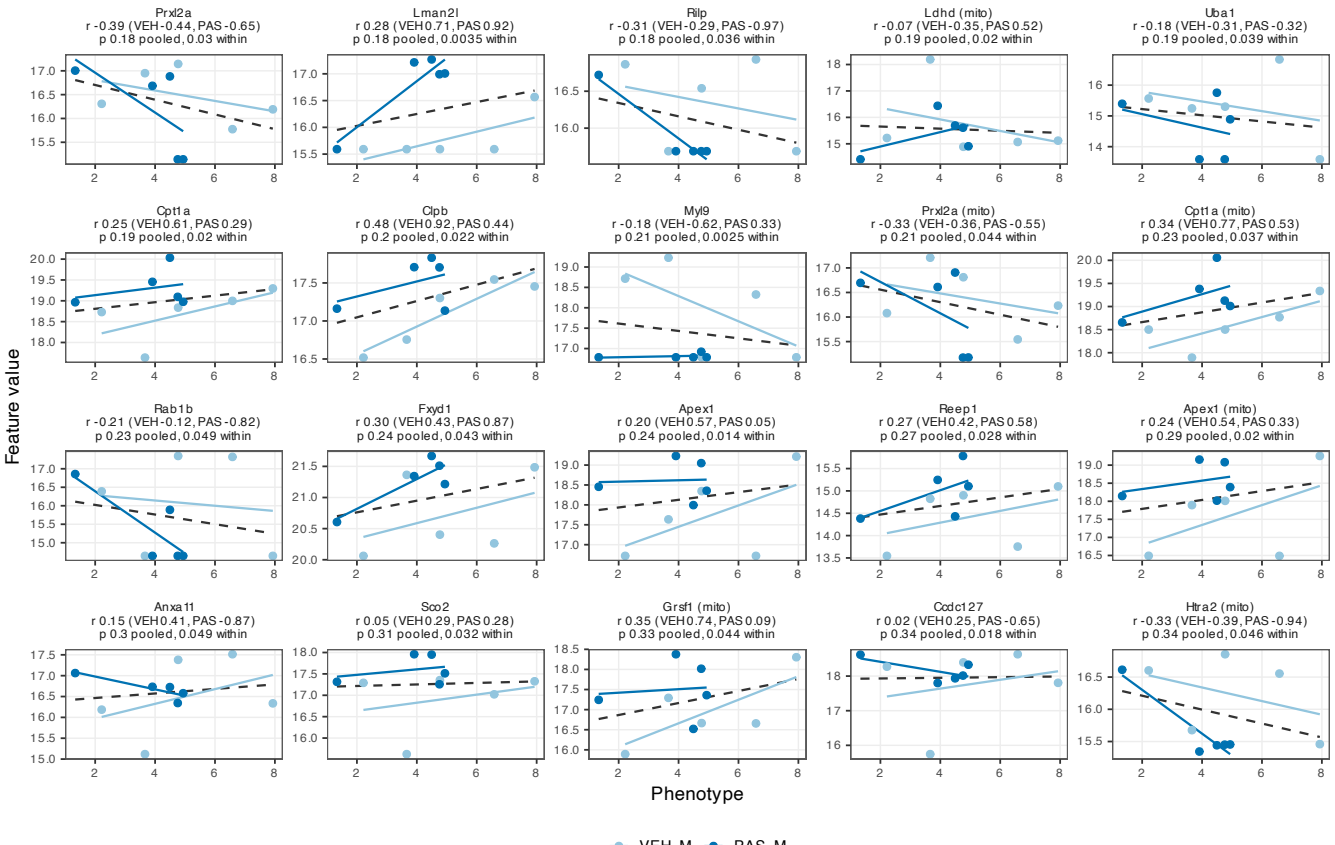
**Male: Proteins associated with Plantaris pyruvate, p ≤ 0.05 in either model (n = 283)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



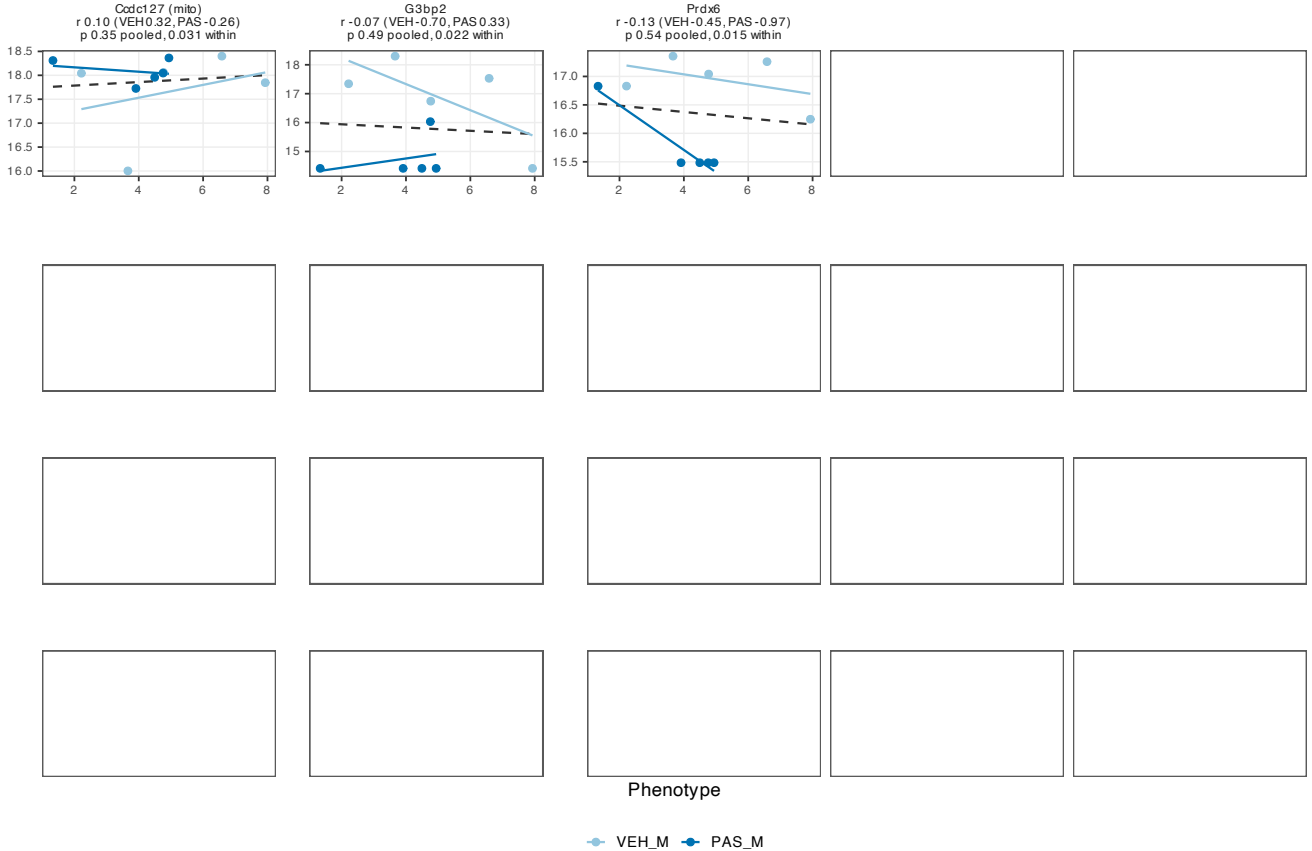
**Male: Proteins associated with Plantaris pyruvate, p ≤ 0.05 in either model (n = 283)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



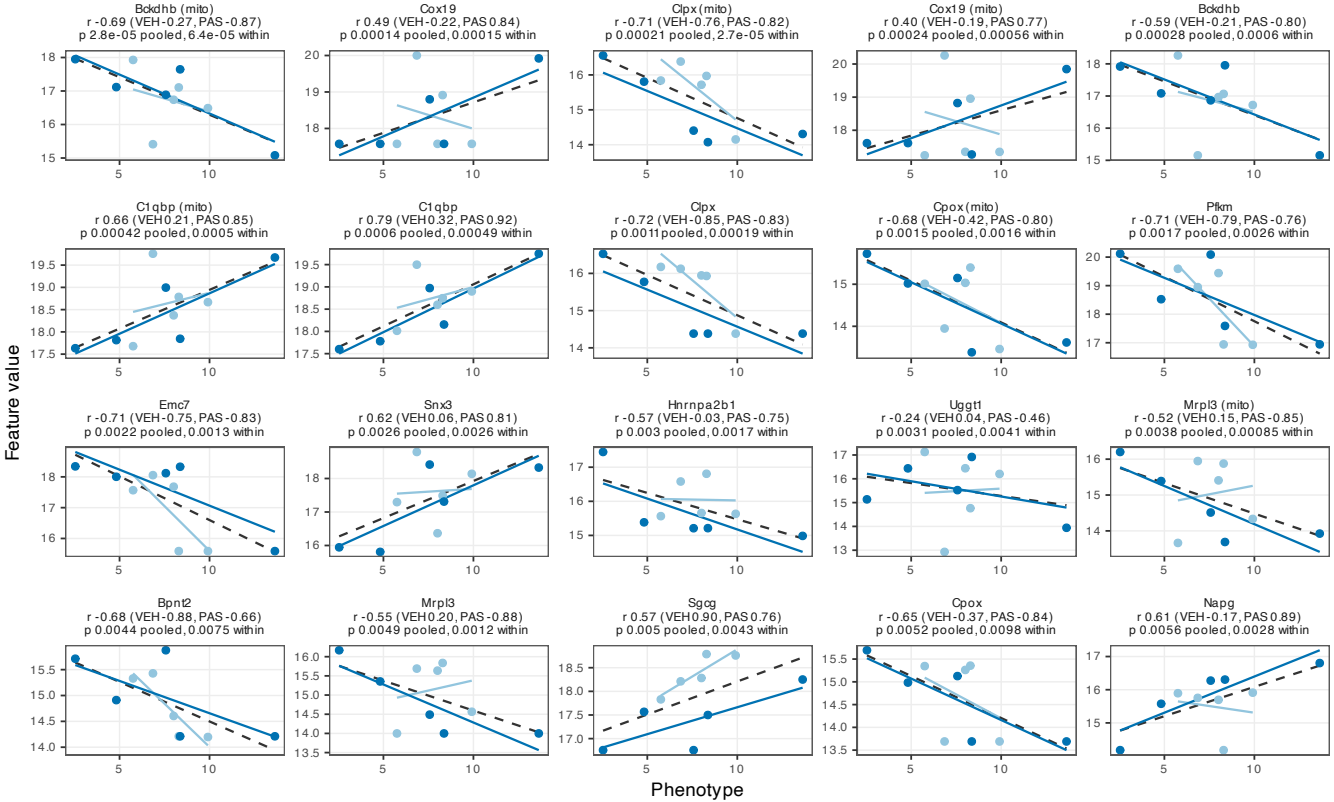
**Male: Proteins associated with Plantaris pyruvate, p ≤ 0.05 in either model (n = 283)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



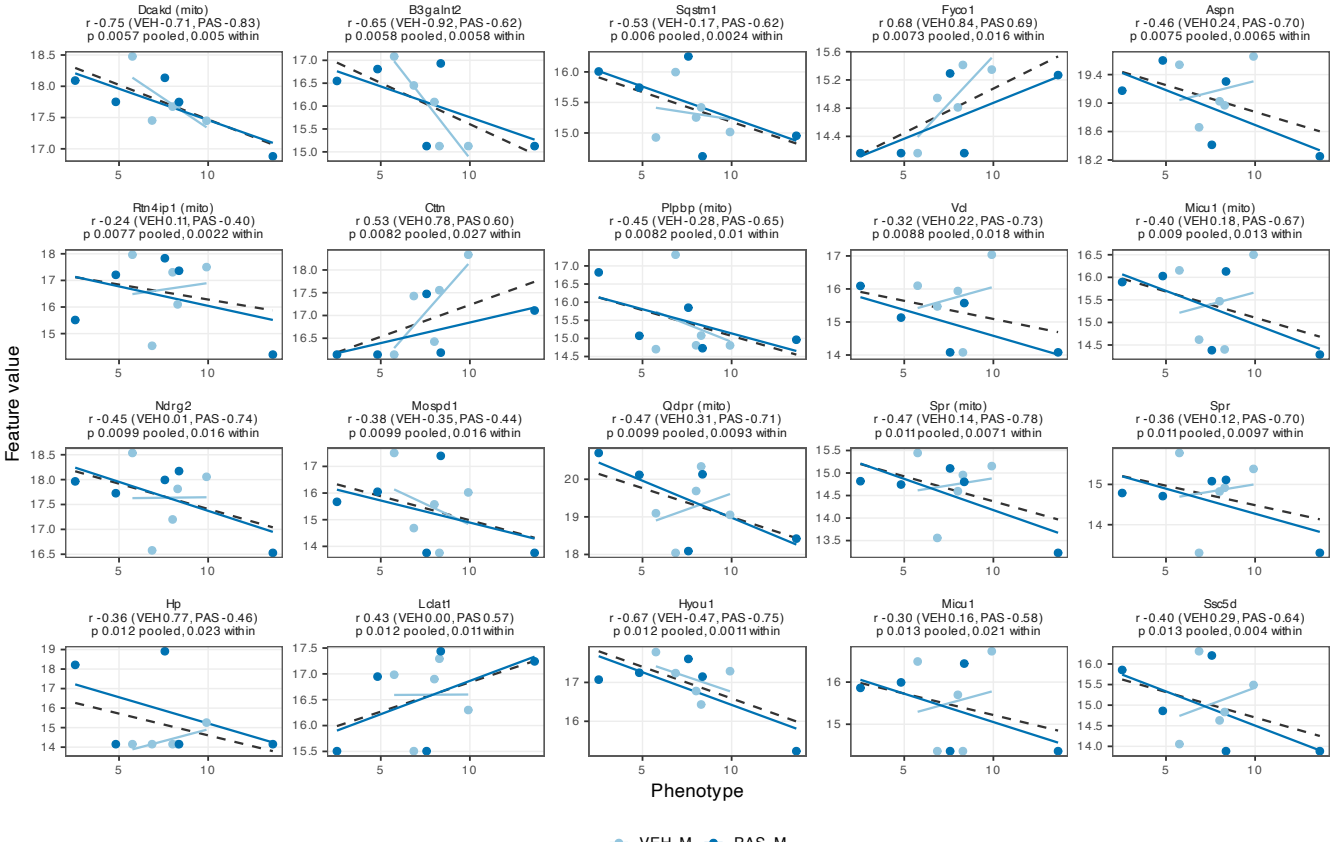
**Male: Proteins associated with Soleus octanoyl, p ≤ 0.05 in either model (n = 182)**

*Phenotype effect of PAS in this sex: -16% (p 0.57), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



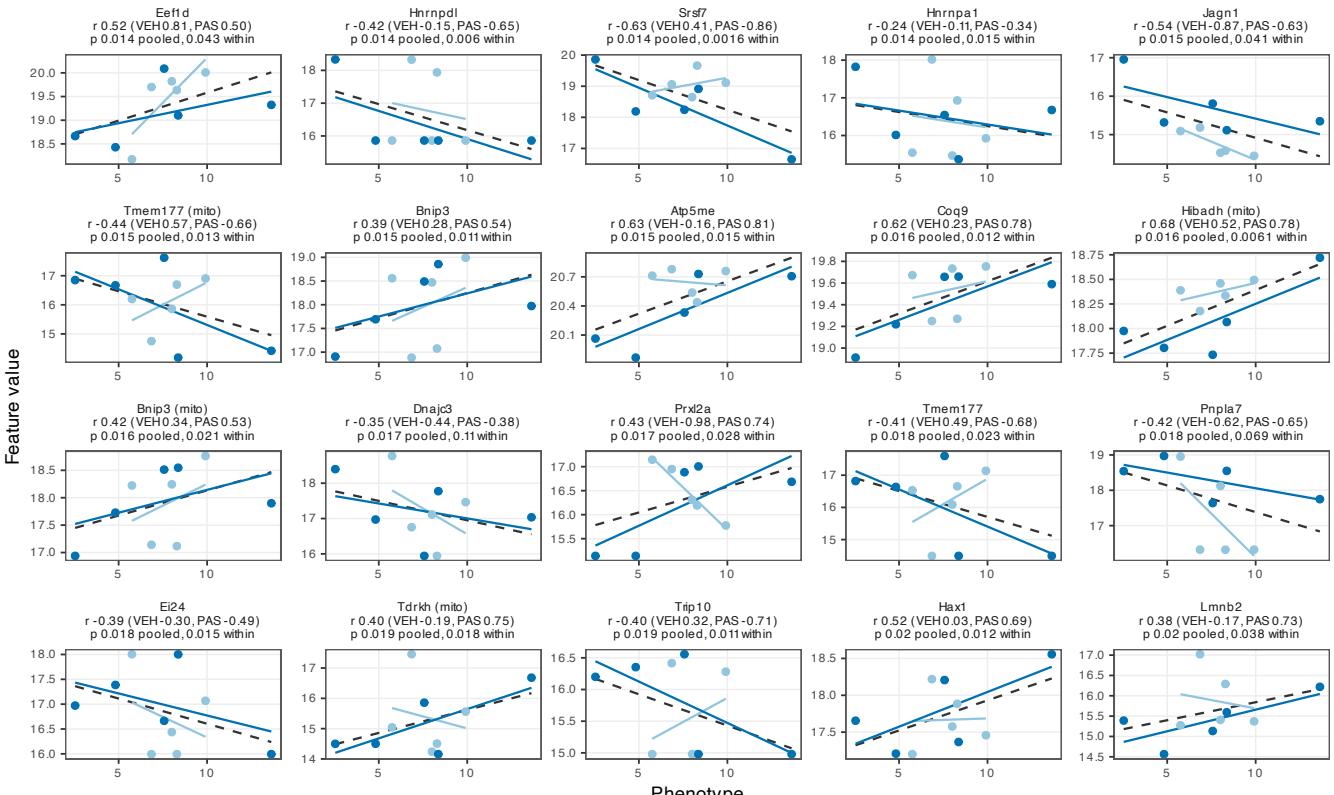
**Male: Proteins associated with Soleus octanoyl, p ≤ 0.05 in either model (n = 182)**

*Phenotype effect of PAS in this sex: -16% (p 0.57), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



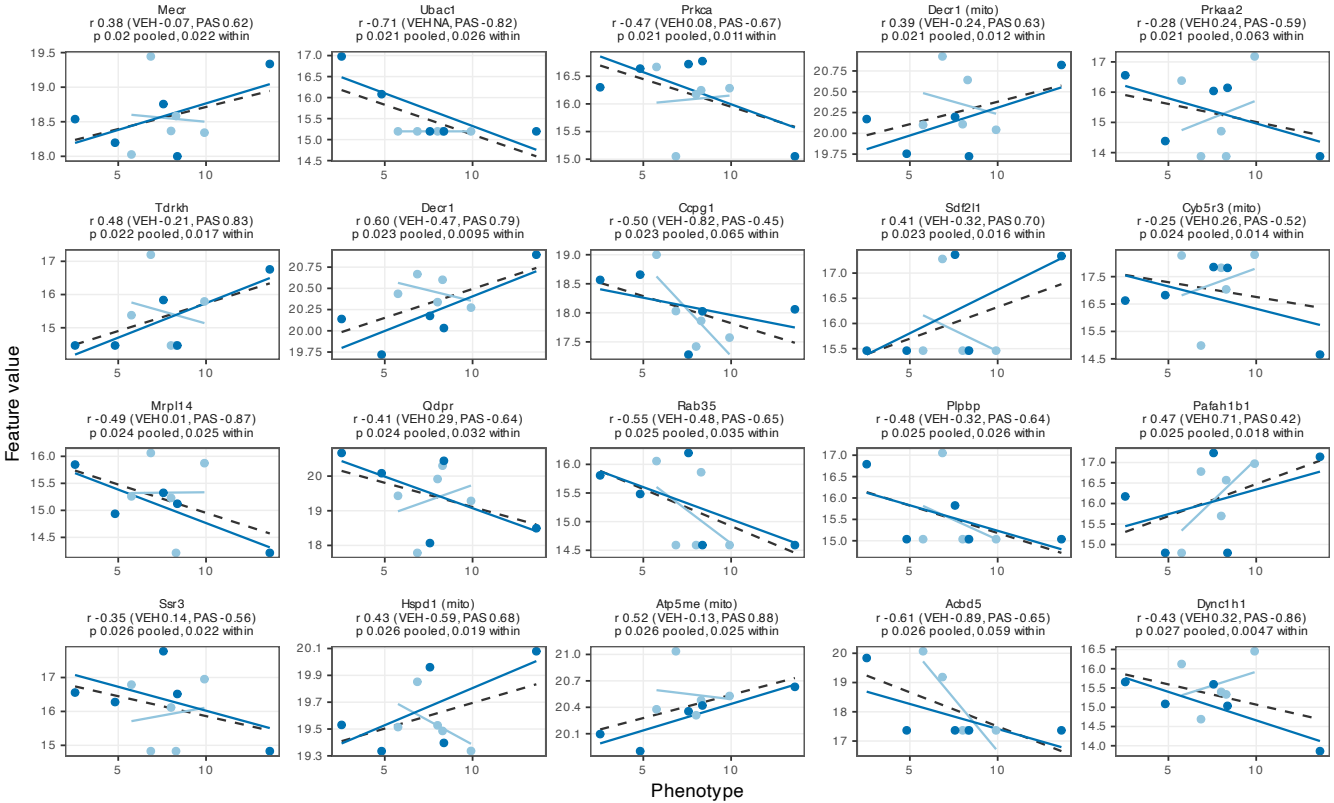
**Male: Proteins associated with Soleus octanoyl, p ≤ 0.05 in either model (n = 182)**

*Phenotype effect of PAS in this sex: -16% (p 0.57), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



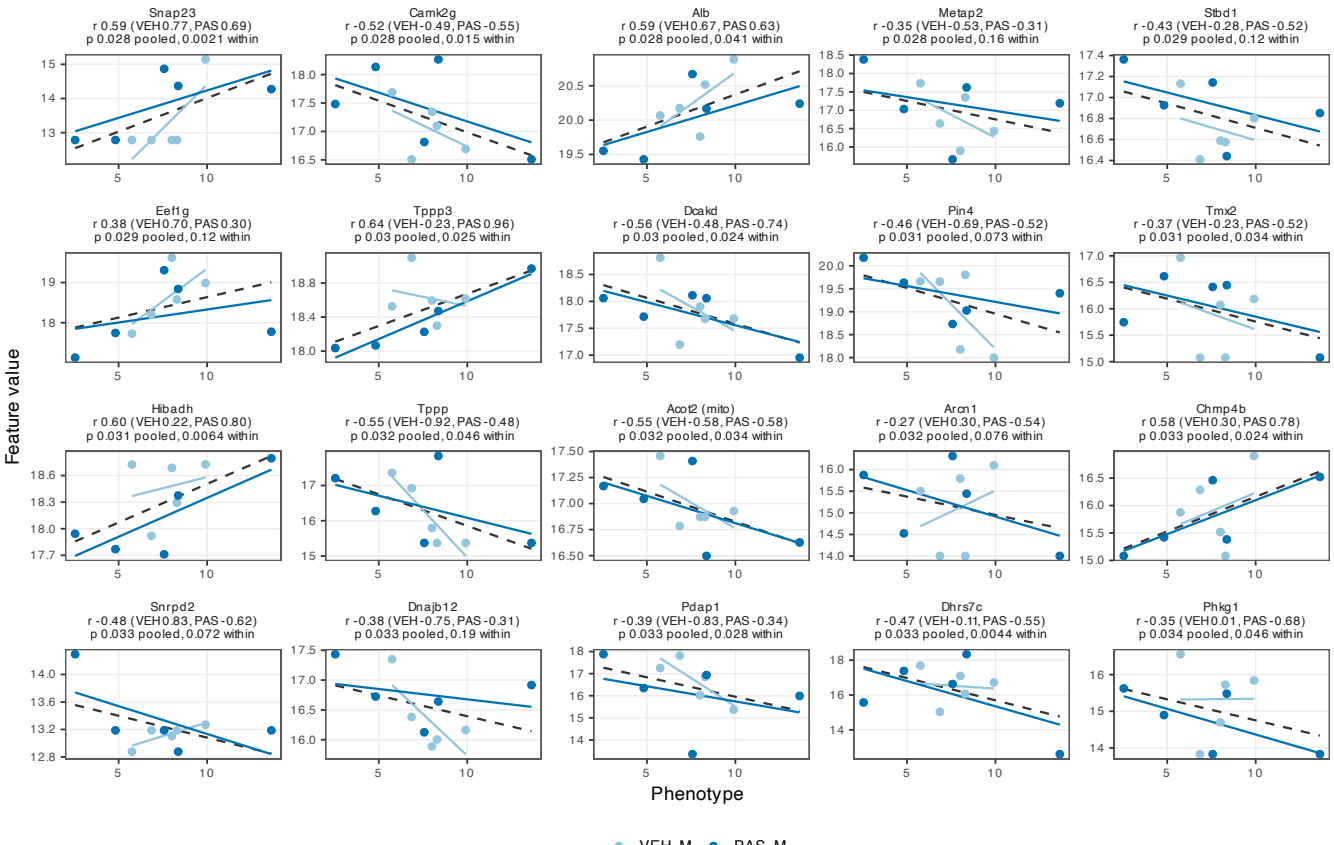
**Male: Proteins associated with Soleus octanoyl, p ≤ 0.05 in either model (n = 182)**

*Phenotype effect of PAS in this sex: -16% (p 0.57), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



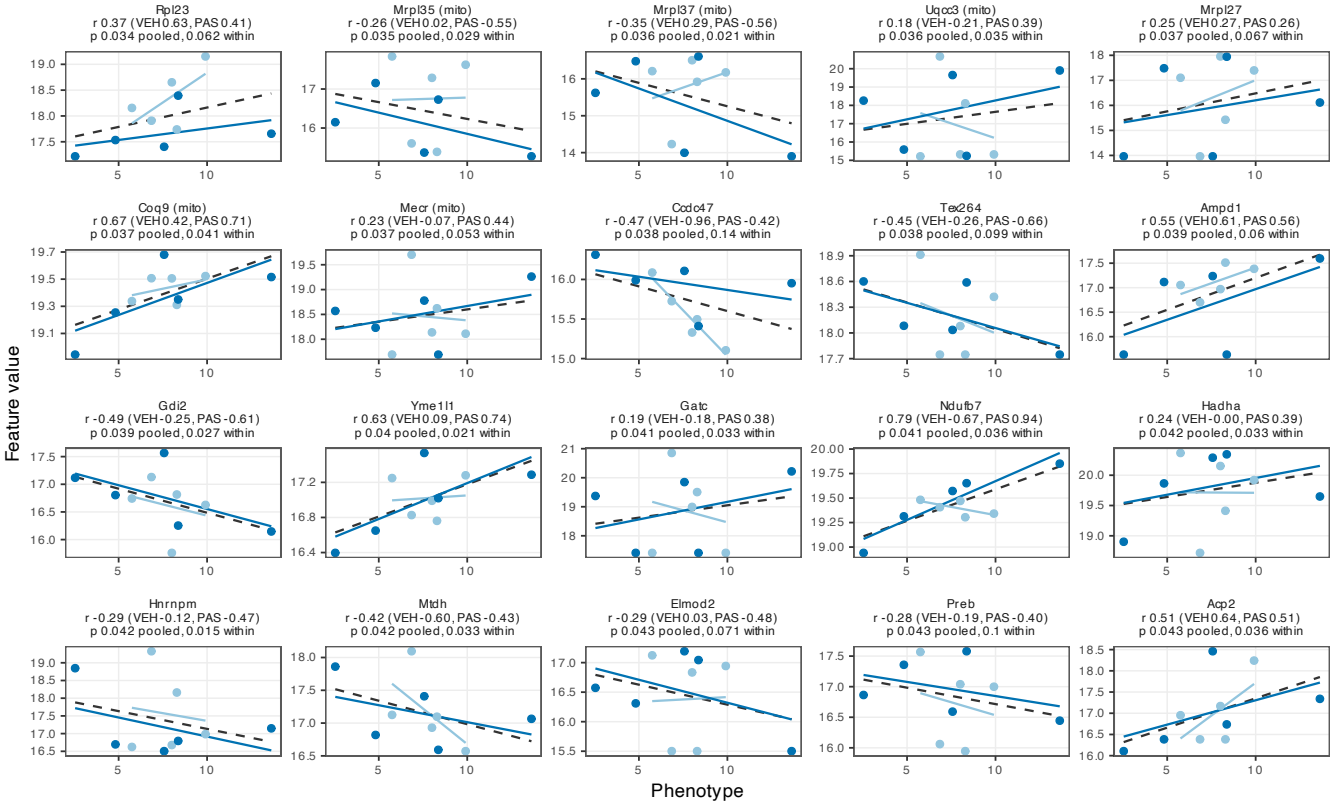
**Male: Proteins associated with Soleus octanoyl, p ≤ 0.05 in either model (n = 182)**

*Phenotype effect of PAS in this sex: -16% (p 0.57), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



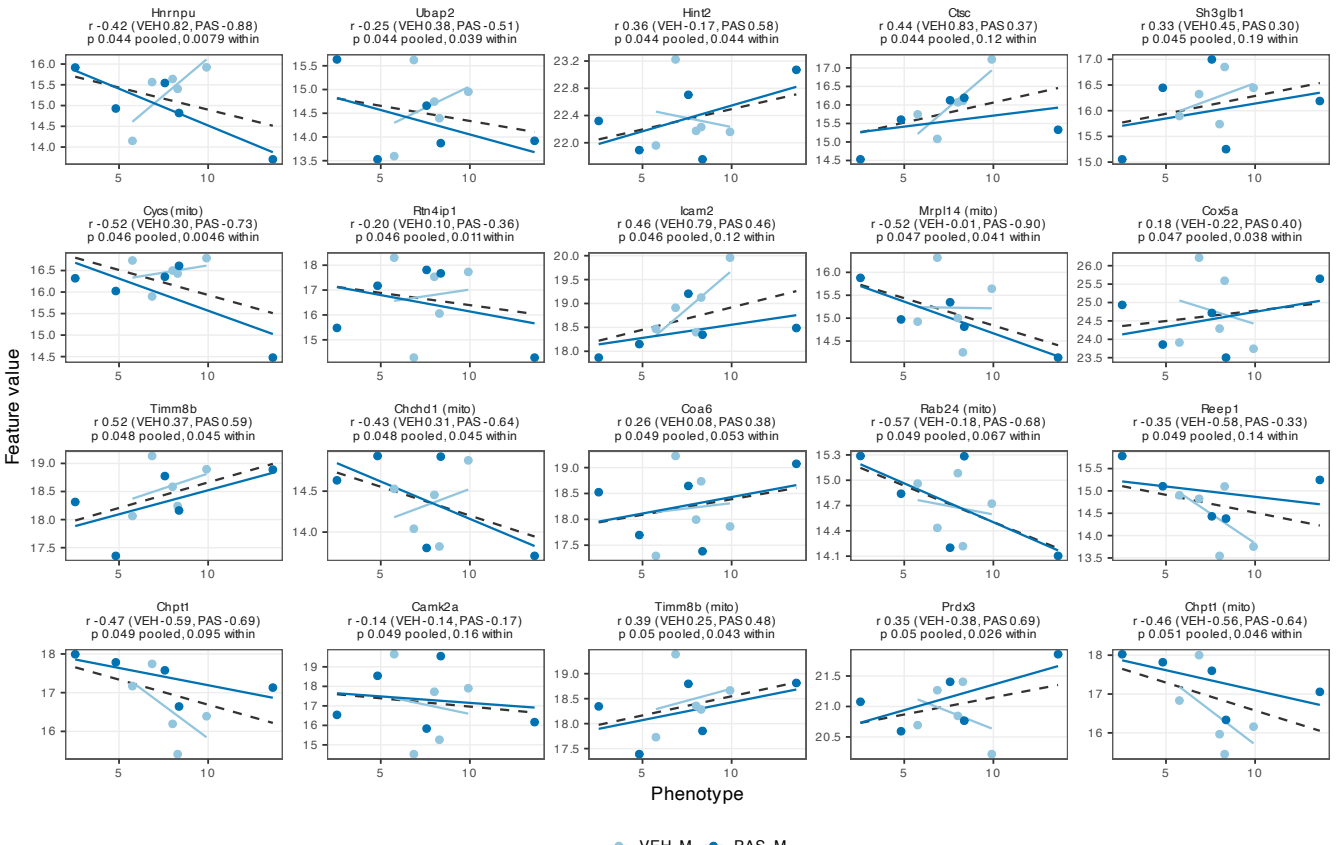
**Male: Proteins associated with Soleus octanoyl, p ≤ 0.05 in either model (n = 182)**

*Phenotype effect of PAS in this sex: -16% (p 0.57), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



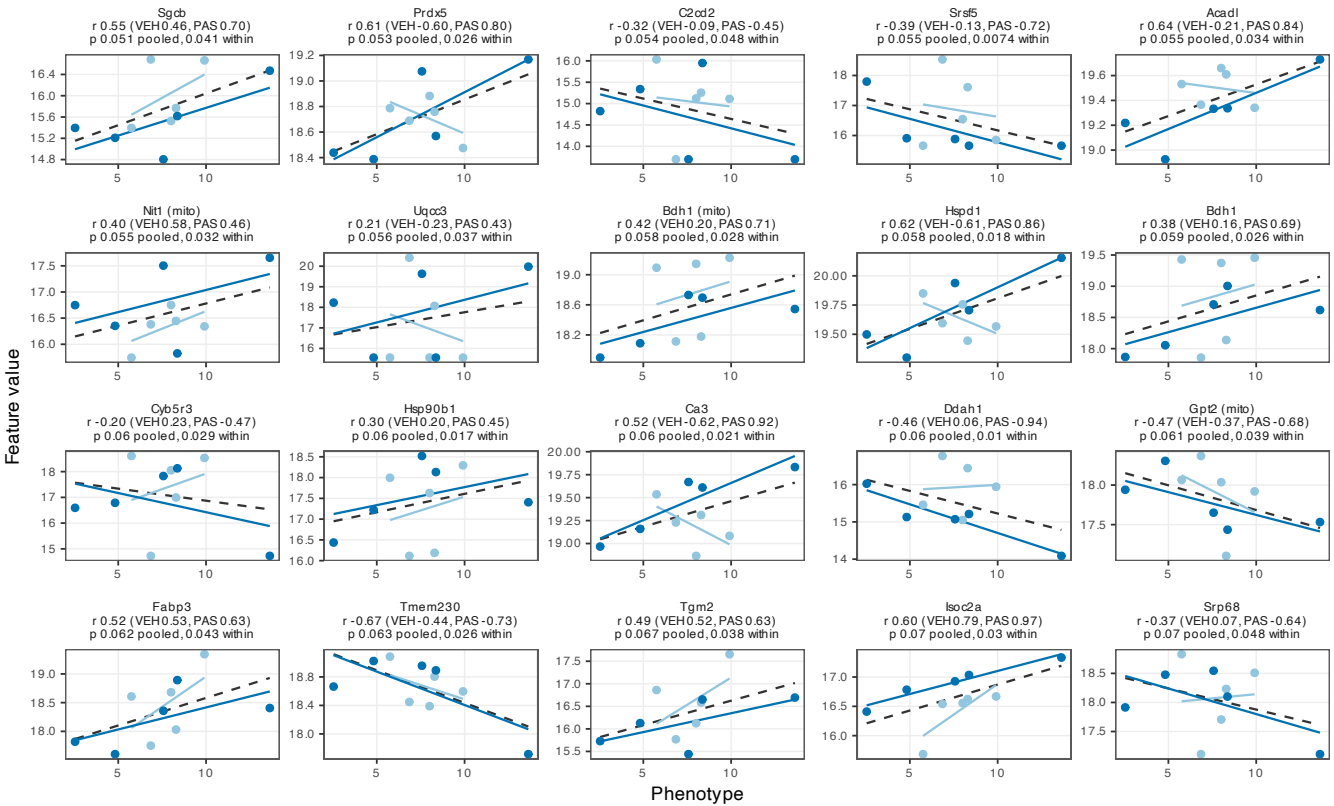
**Male: Proteins associated with Soleus octanoyl, p ≤ 0.05 in either model (n = 182)**

*Phenotype effect of PAS in this sex: -16% (p 0.57), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



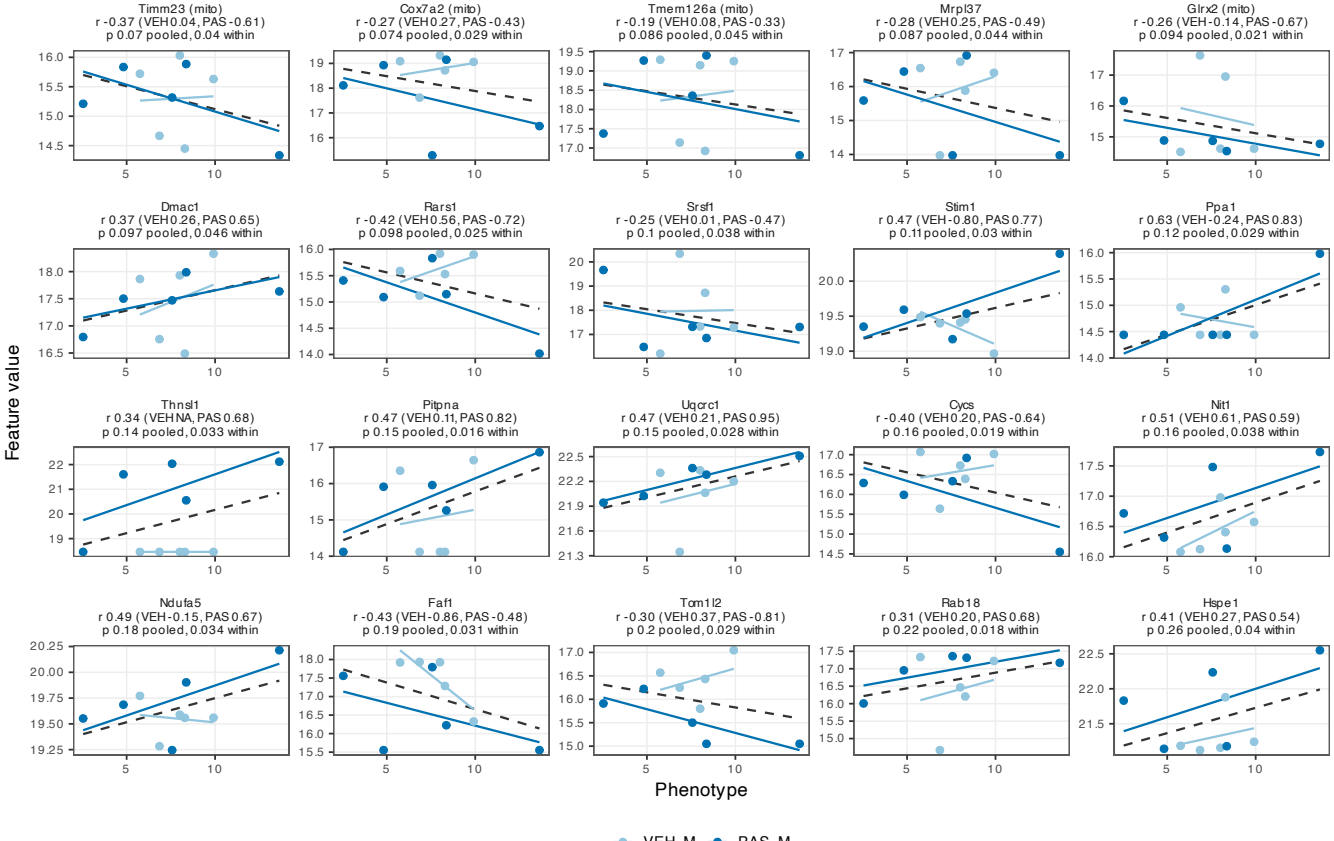
**Male: Proteins associated with Soleus octanoyl, p ≤ 0.05 in either model (n = 182)**

*Phenotype effect of PAS in this sex: -16% (p 0.57), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



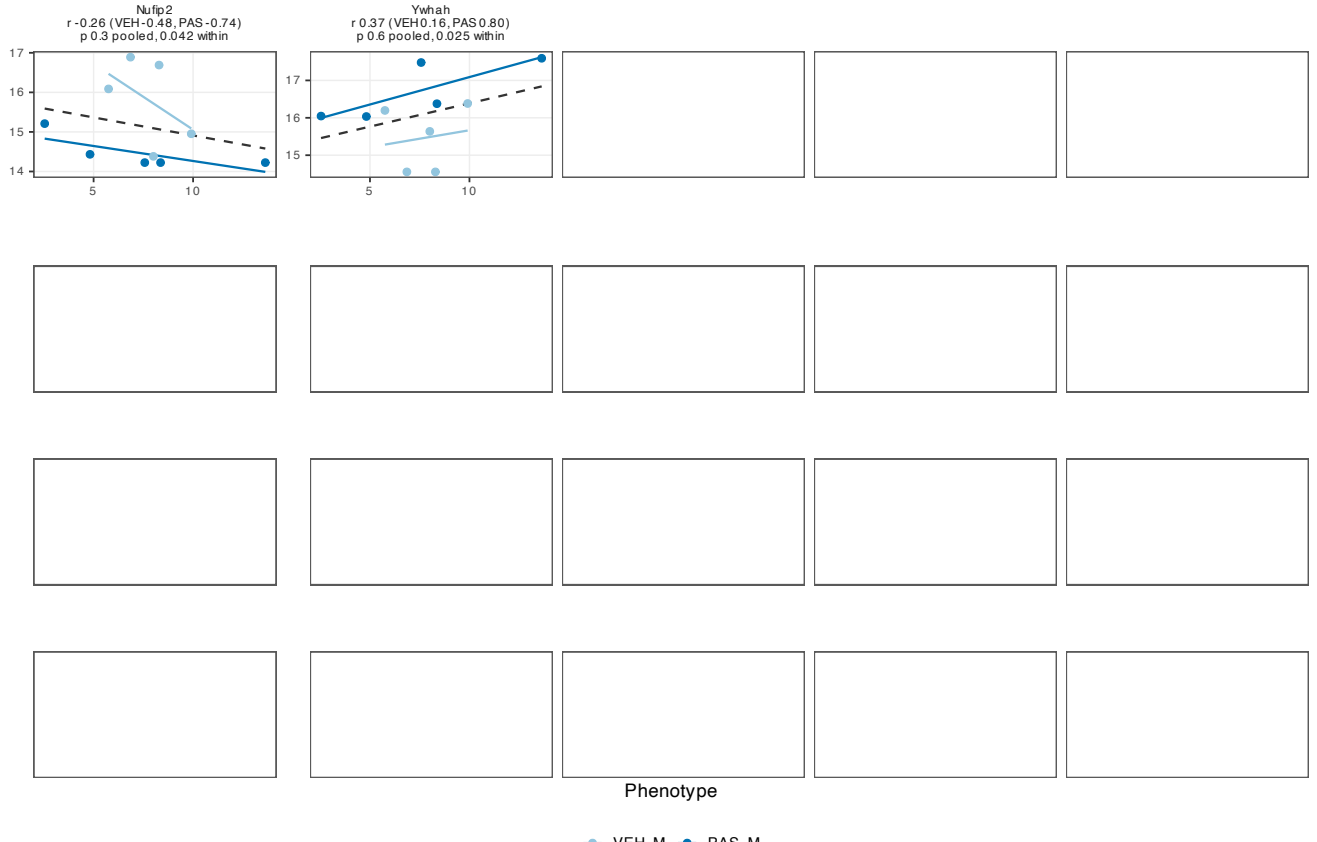
**Male: Proteins associated with Soleus octanoyl, p ≤ 0.05 in either model (n = 182)**

*Phenotype effect of PAS in this sex: -16% (p 0.57), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



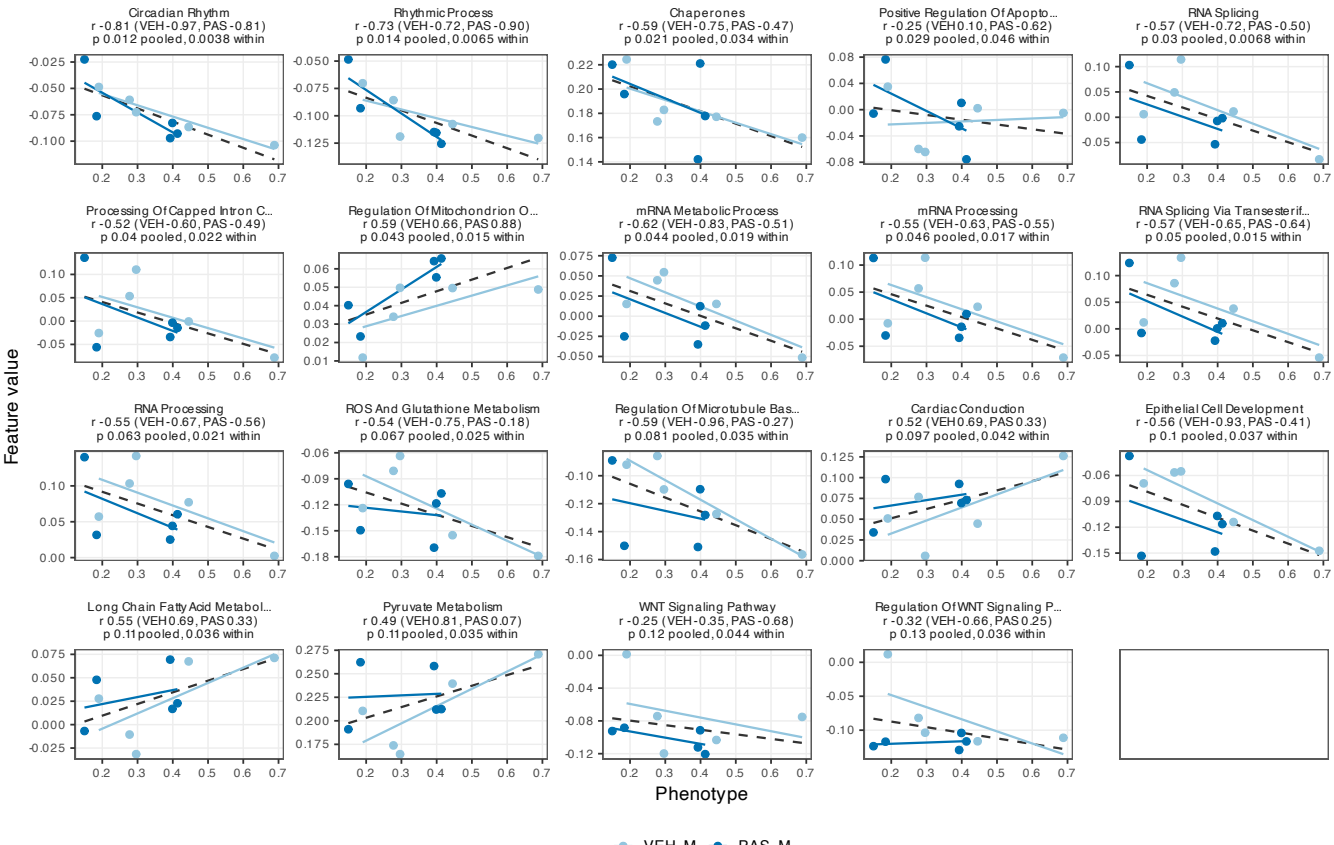
**Male: Proteins associated with Soleus octanoyl, p ≤ 0.05 in either model (n = 182)**

*Phenotype effect of PAS in this sex: -16% (p 0.57), so these associations follow variation between mice, not a PAS response. Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



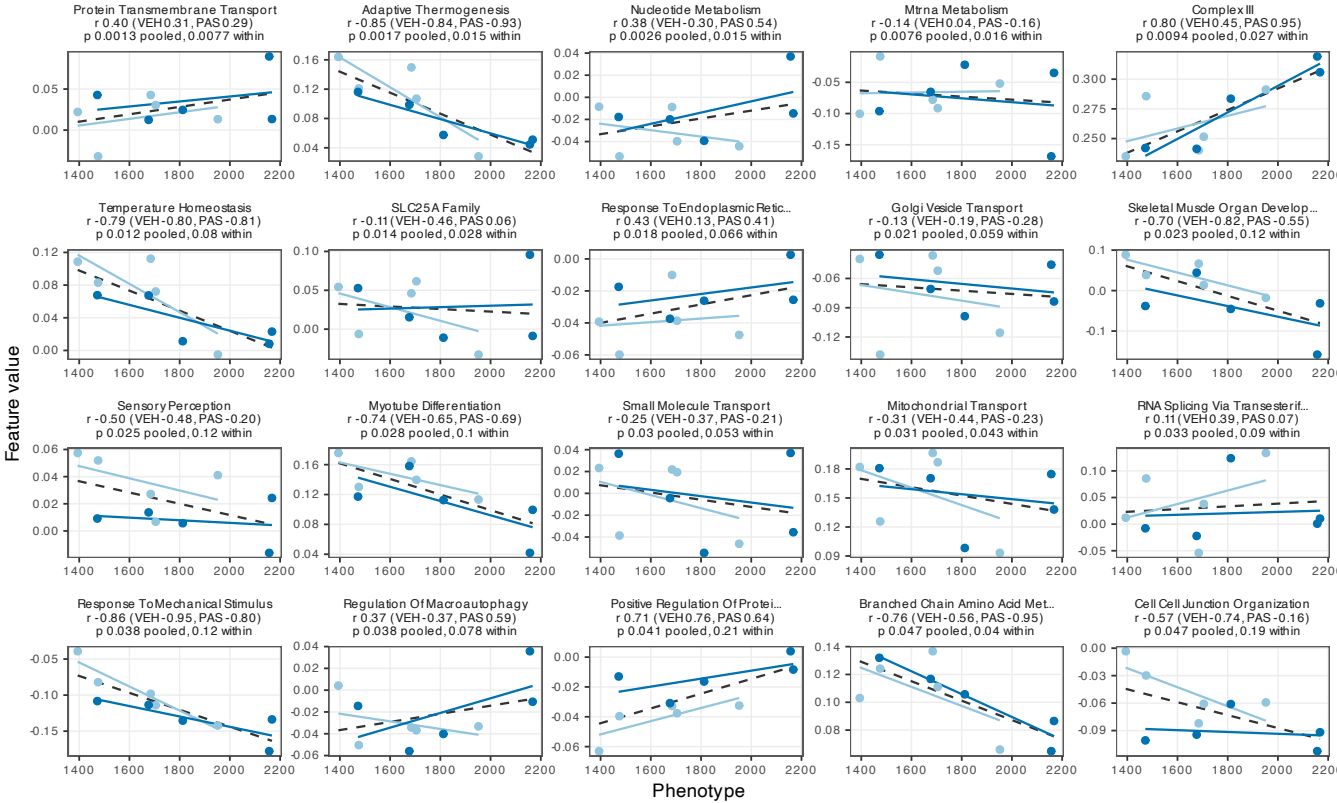
**Male: Pathways associated with Capillary density, p ≤ 0.05 in either model (n = 19)**

*Phenotype effect of PAS in this sex: -18% (p 0.54), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



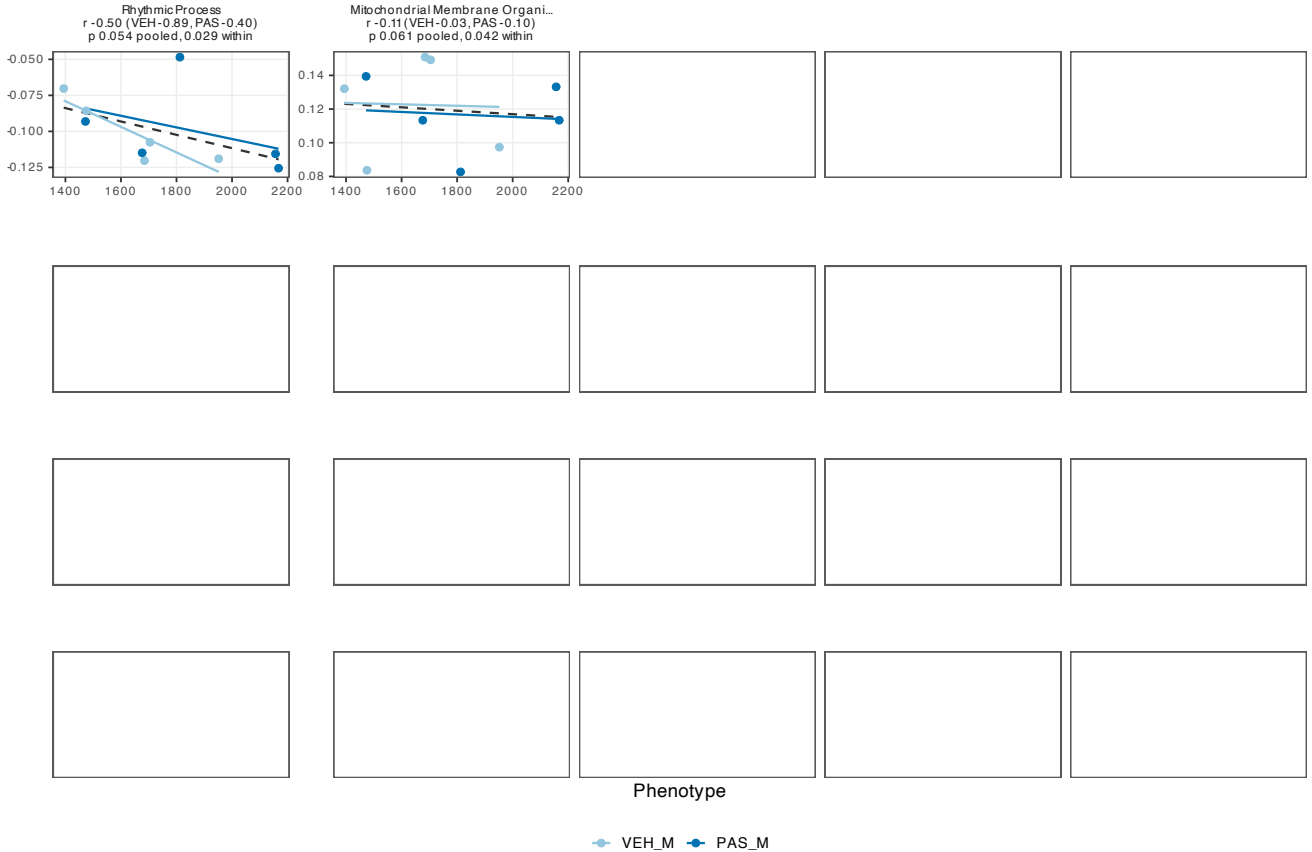
**Male: Pathways associated with CSA, p ≤ 0.05 in either model (n = 22)**

*Phenotype effect of PAS in this sex: +13% (p 0.25), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



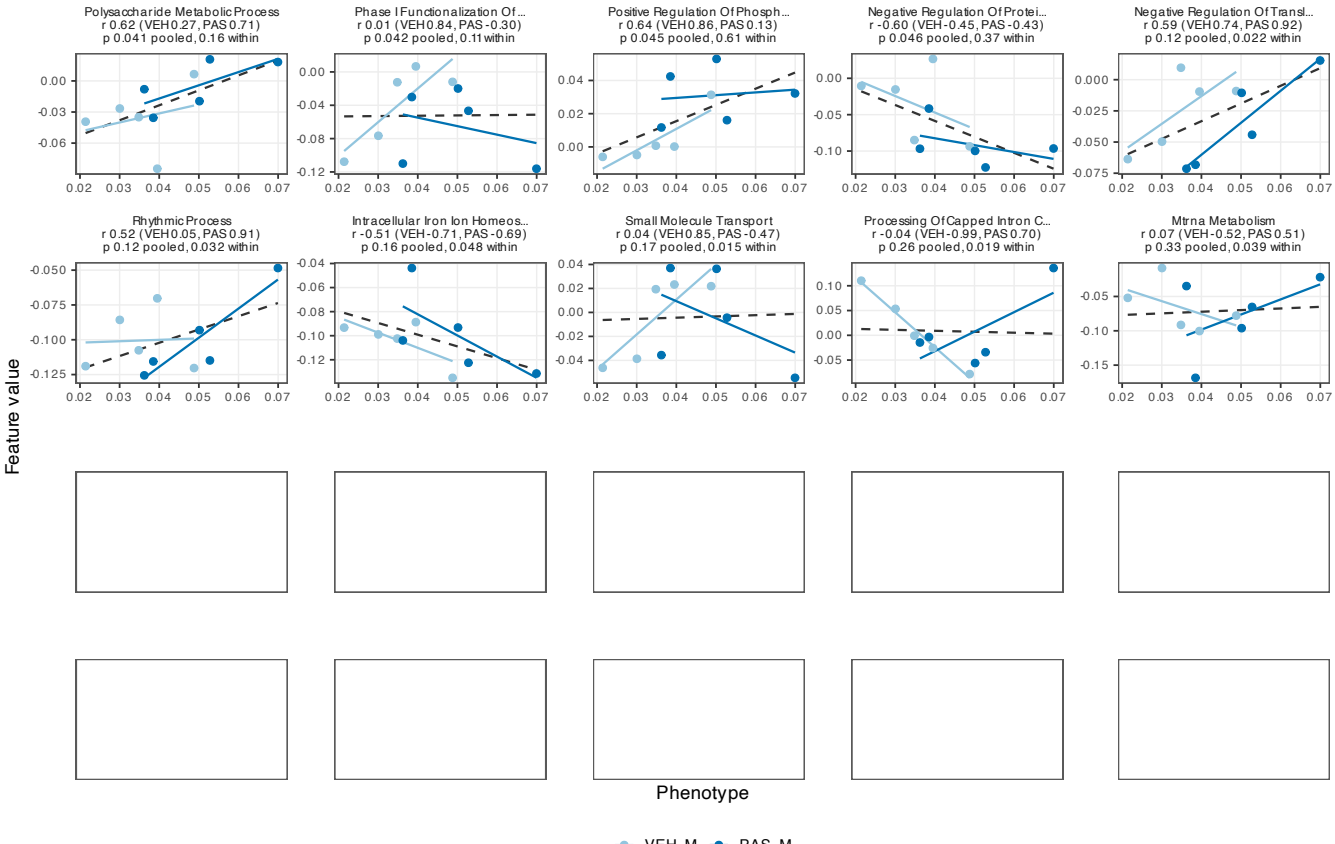
**Male: Pathways associated with CSA, p ≤ 0.05 in either model (n = 22)**

*Phenotype effect of PAS in this sex: +13% (p 0.25), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



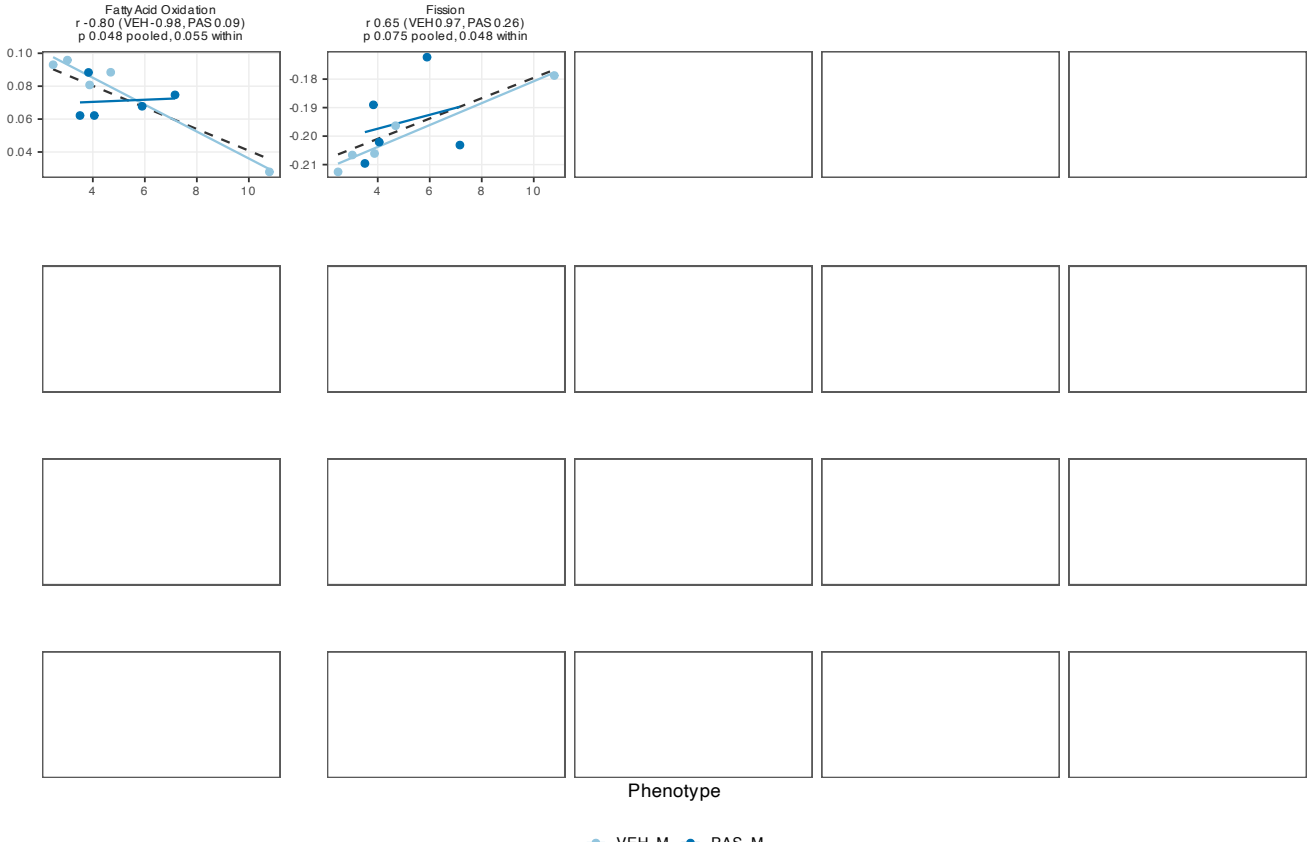
**Male: Pathways associated with Satellite cells / fibre, p ≤ 0.05 in either model (n = 10)**

*Phenotype effect of PAS in this sex: +43% (p 0.086), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



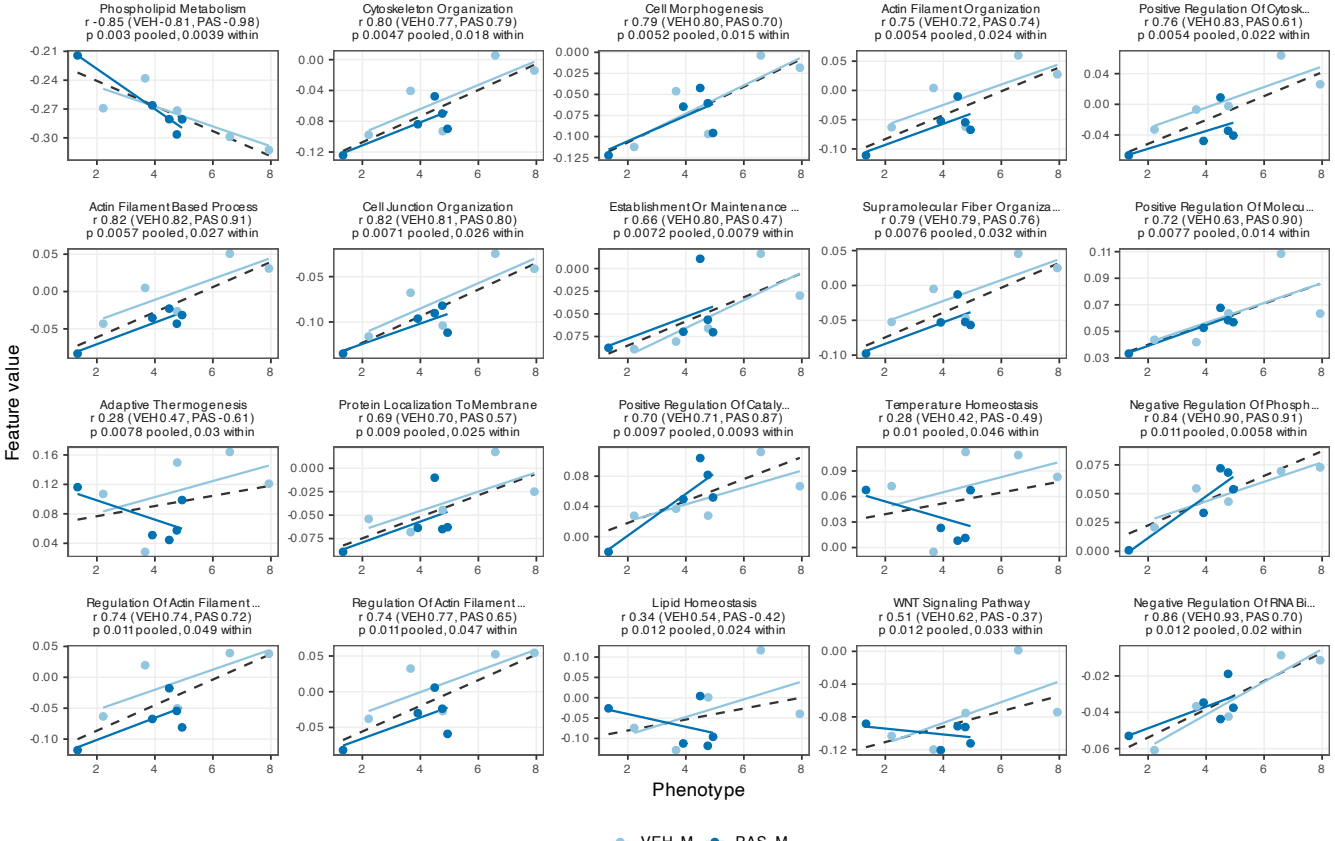
**Male: Pathways associated with Soleus pyruvate, p ≤ 0.05 in either model (n = 2)**

*Phenotype effect of PAS in this sex: +9% (p 0.77), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



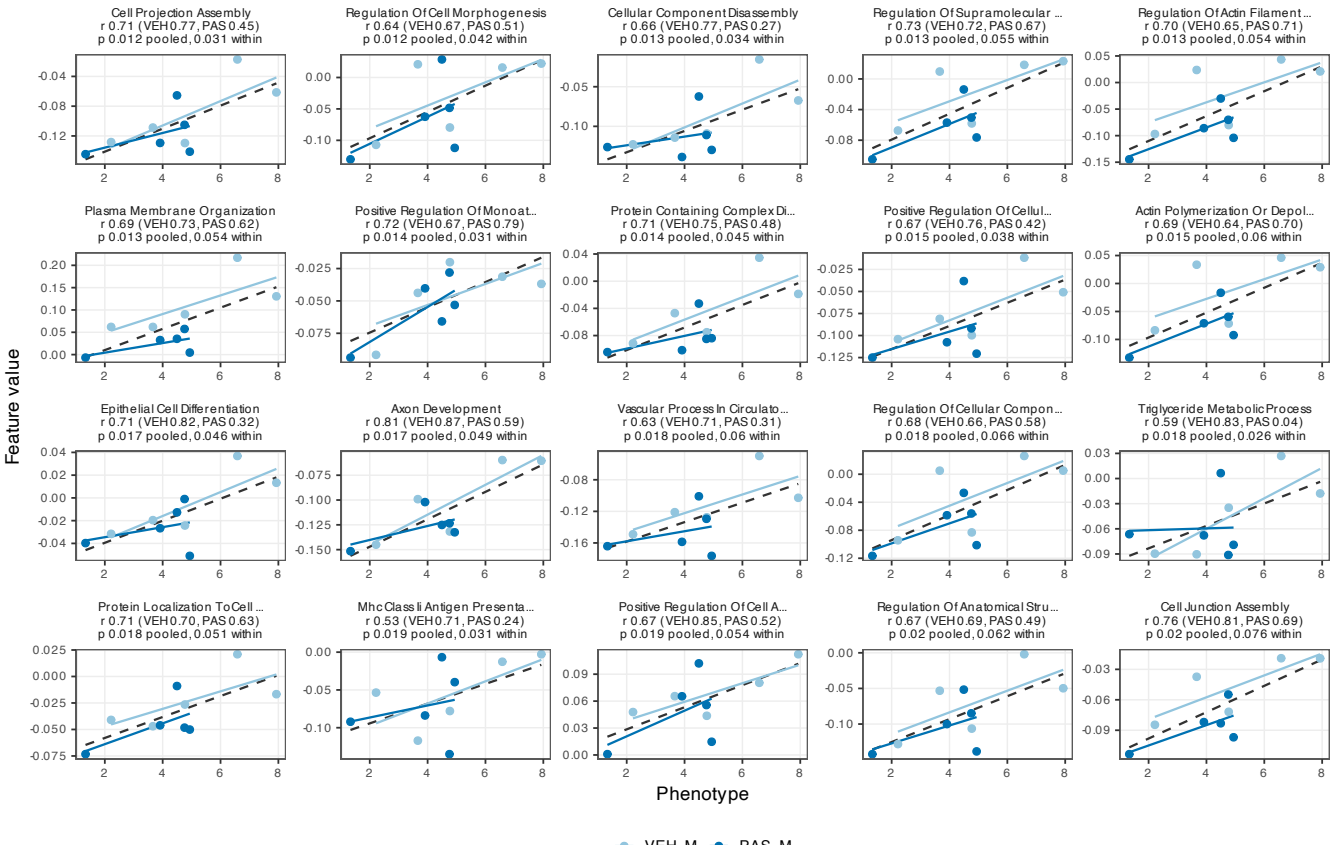
**Male: Pathways associated with Plantaris pyruvate, p ≤ 0.05 in either model (n = 113)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



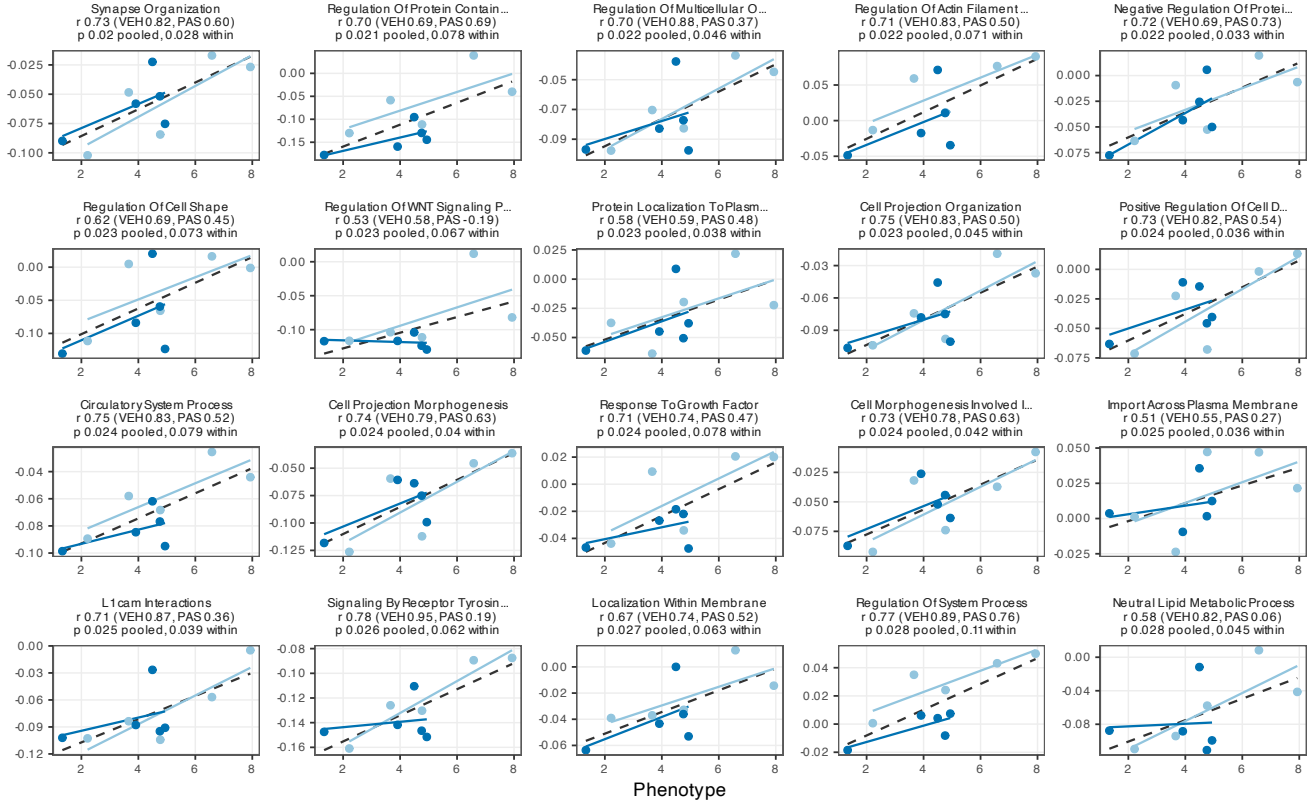
**Male: Pathways associated with Plantaris pyruvate, p ≤ 0.05 in either model (n = 113)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



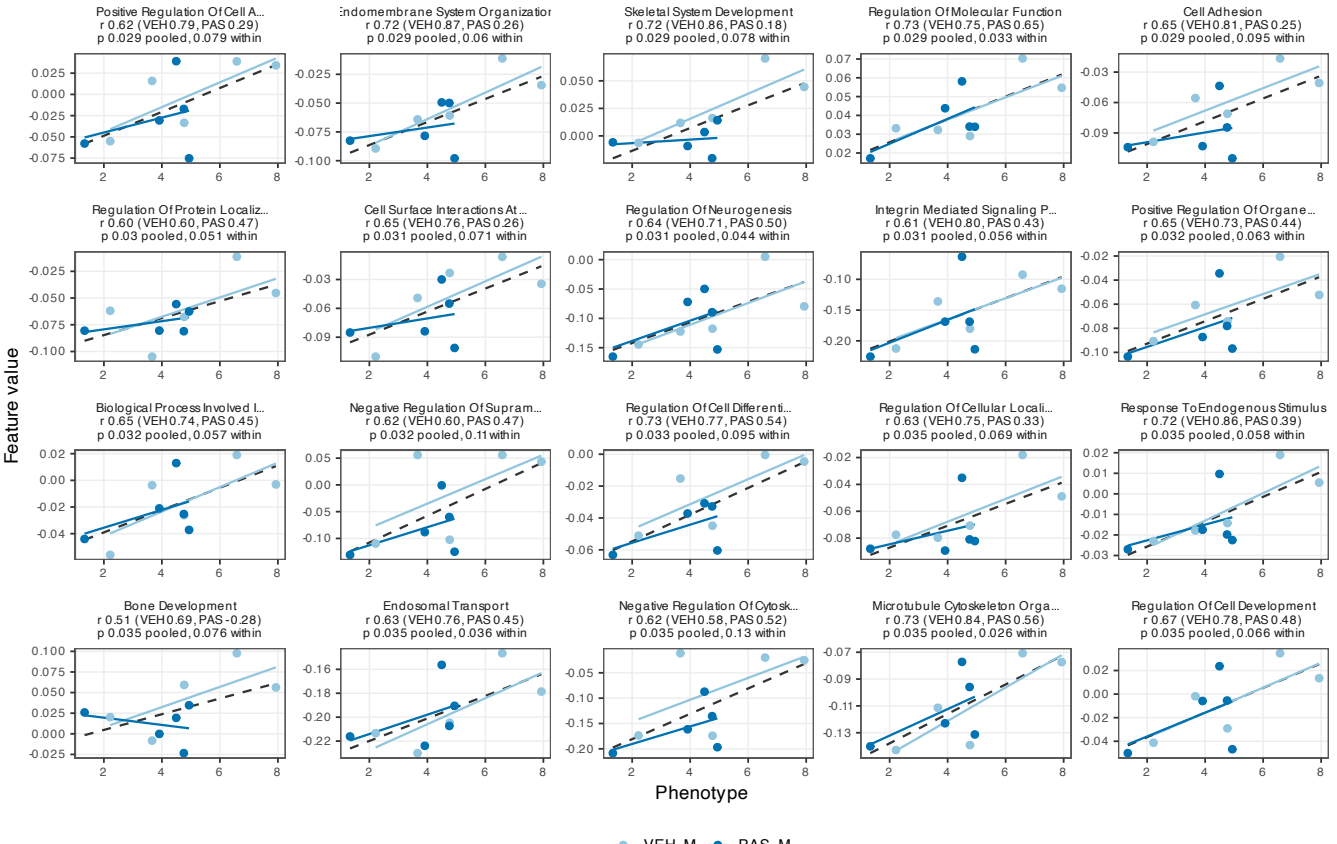
**Male: Pathways associated with Plantaris pyruvate, p ≤ 0.05 in either model (n = 113)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



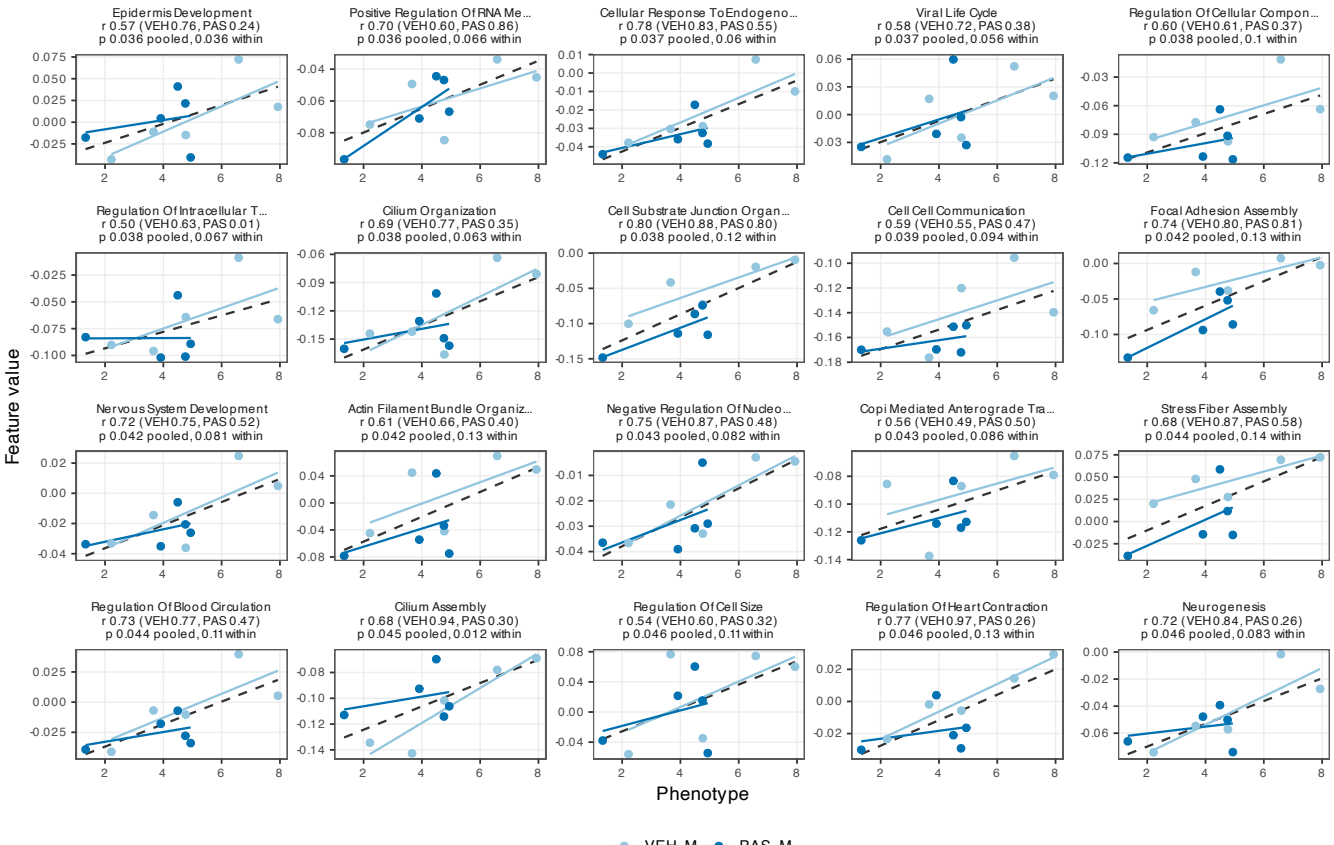
**Male: Pathways associated with Plantaris pyruvate, p ≤ 0.05 in either model (n = 113)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



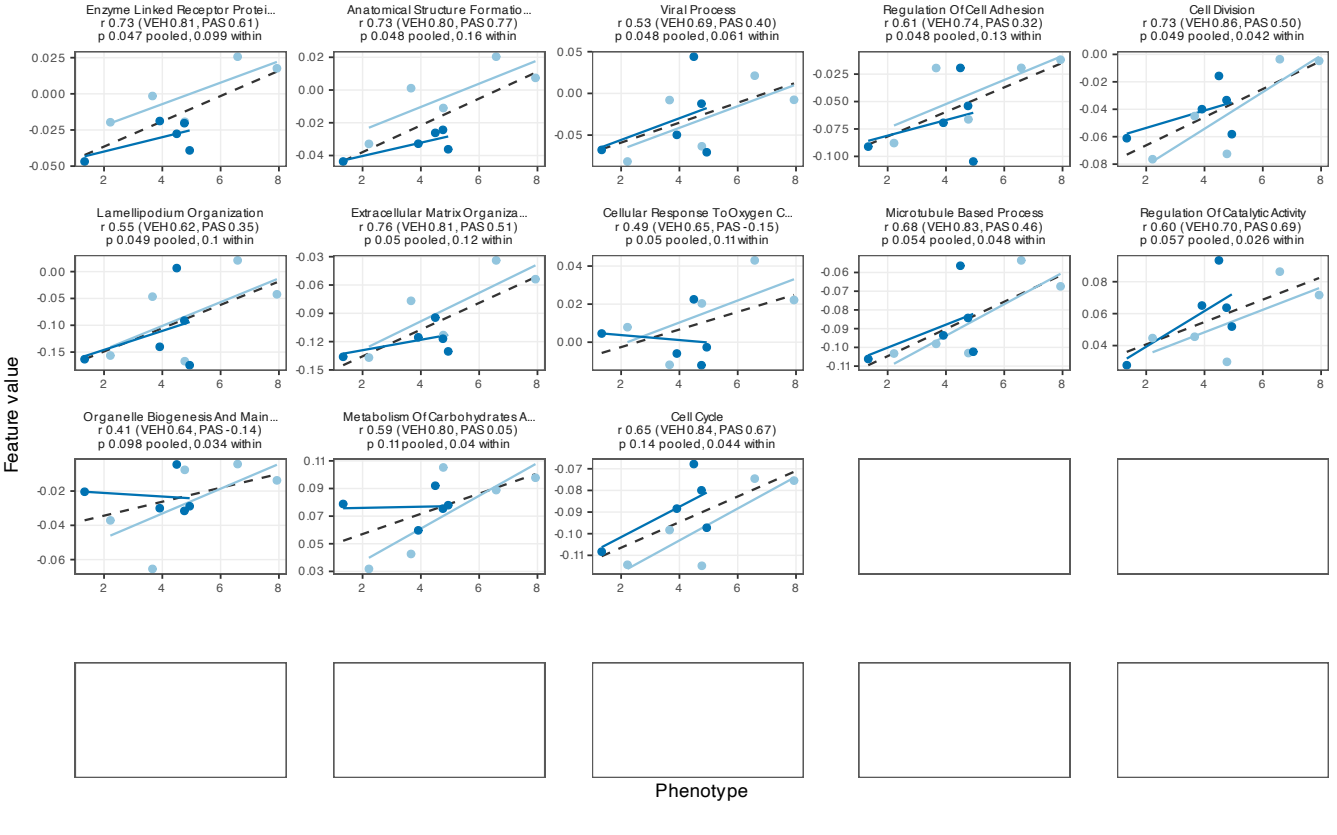
**Male: Pathways associated with Plantaris pyruvate, p ≤ 0.05 in either model (n = 113)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



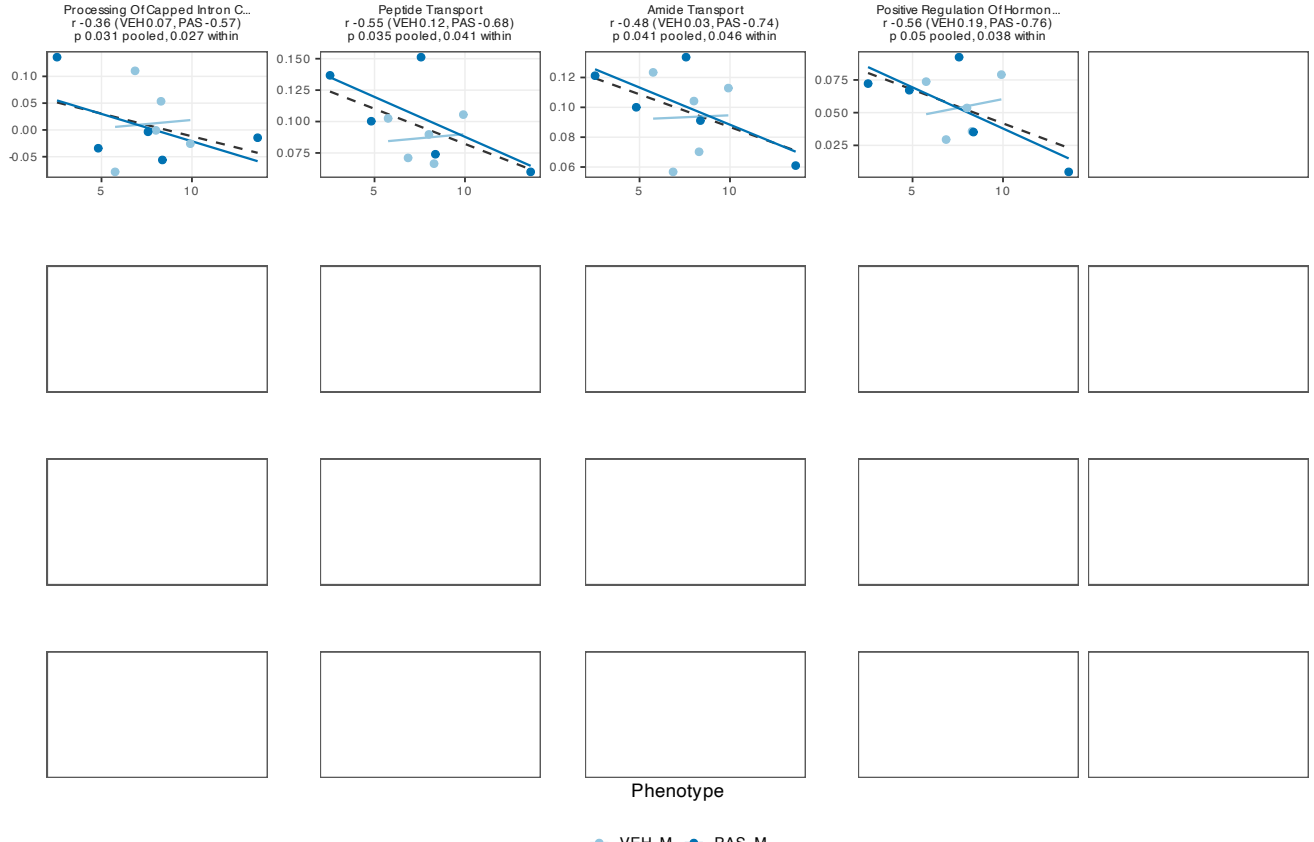
**Male: Pathways associated with Plantaris pyruvate, p ≤ 0.05 in either model (n = 113)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



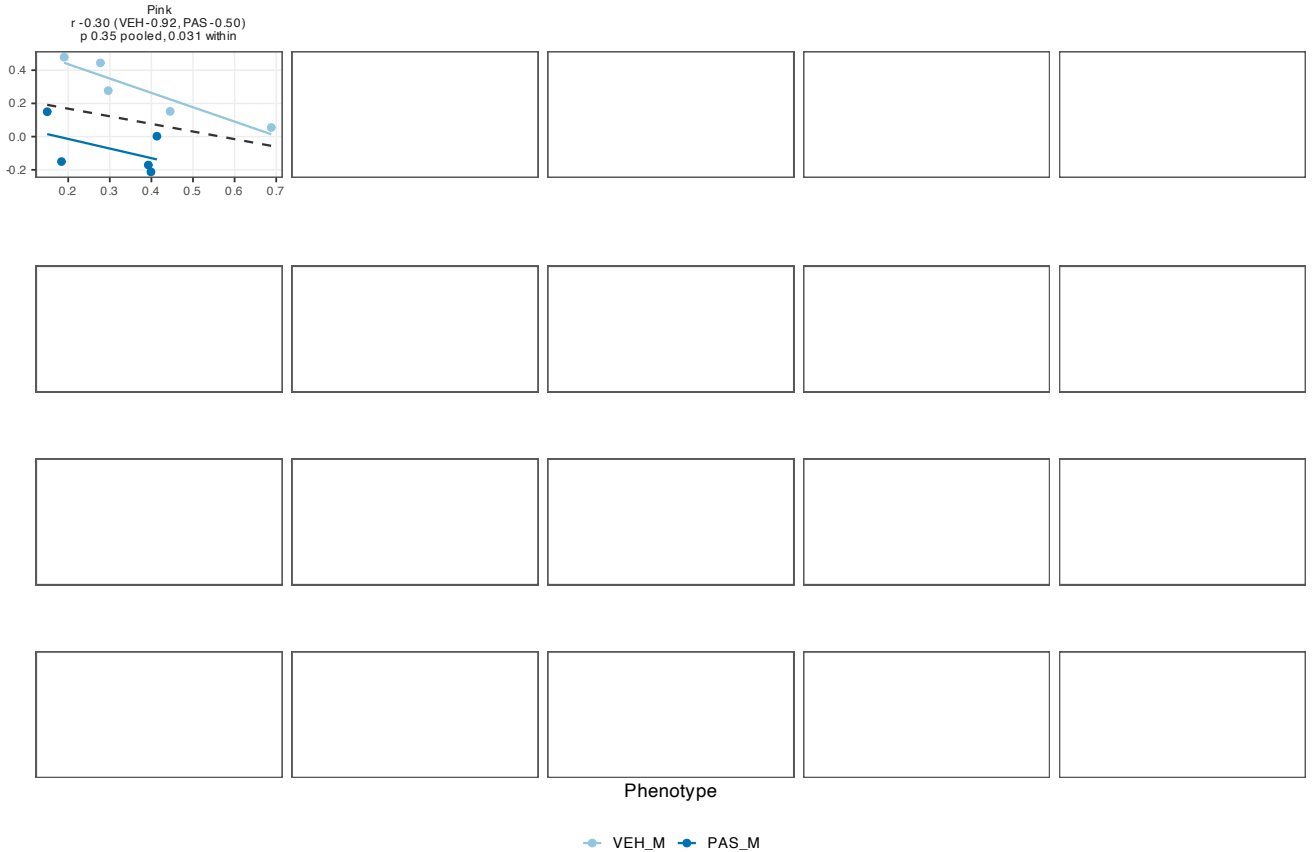
**Male: Pathways associated with Soleus octanoyl, p ≤ 0.05 in either model (n = 4)**

*Phenotype effect of PAS in this sex: -16% (p 0.57), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



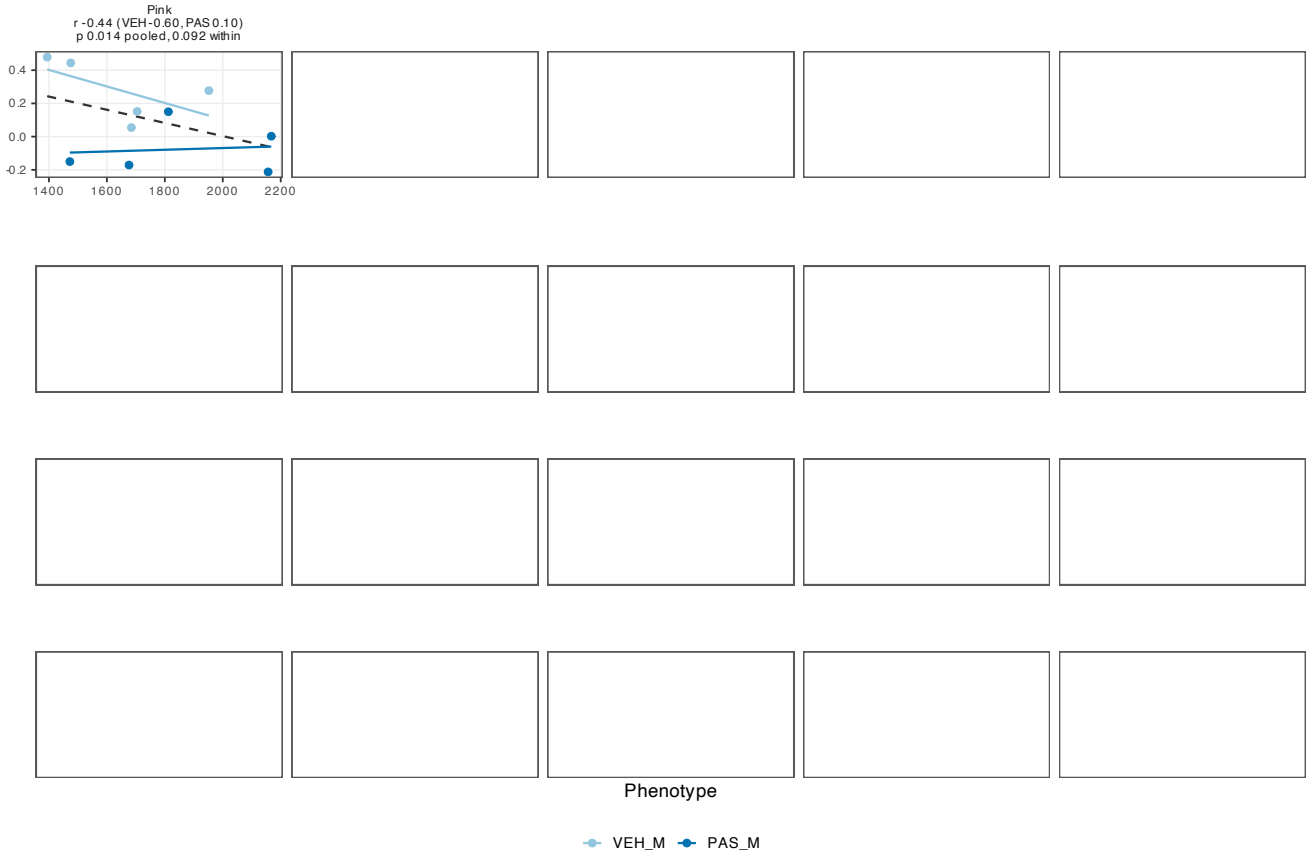
**Male: Modules associated with Capillary density, p ≤ 0.05 in either model (n = 1)**

*Phenotype effect of PAS in this sex: -18% (p 0.54), so these associations follow variation between mice, not a PAS response. Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



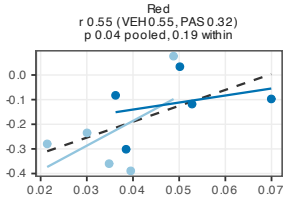
**Male: Modules associated with CSA, p ≤ 0.05 in either model (n = 1)**

*Phenotype effect of PAS in this sex: +13% (p 0.25), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



**Male: Modules associated with Satellite cells / fibre, p ≤ 0.05 in either model (n = 1)**

*Phenotype effect of PAS in this sex: +43% (p 0.086), so these associations follow variation between mice, not a PAS response. Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



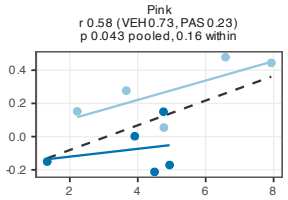

Feature value

Phenotype

VEH_M PAS_M

**Male: Modules associated with Plantaris pyruvate, p ≤ 0.05 in either model (n = 1)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



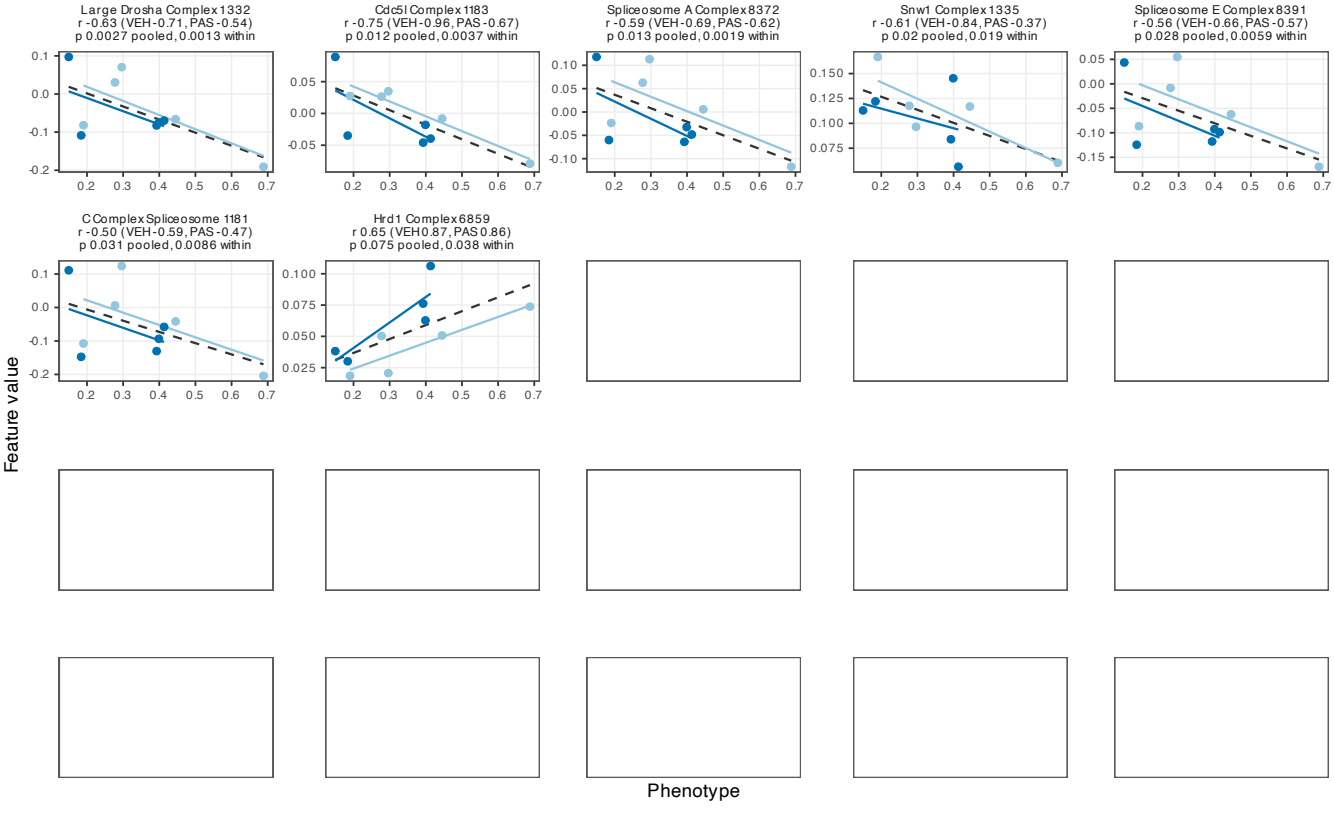

Feature value

Phenotype

VEH_M PAS_M

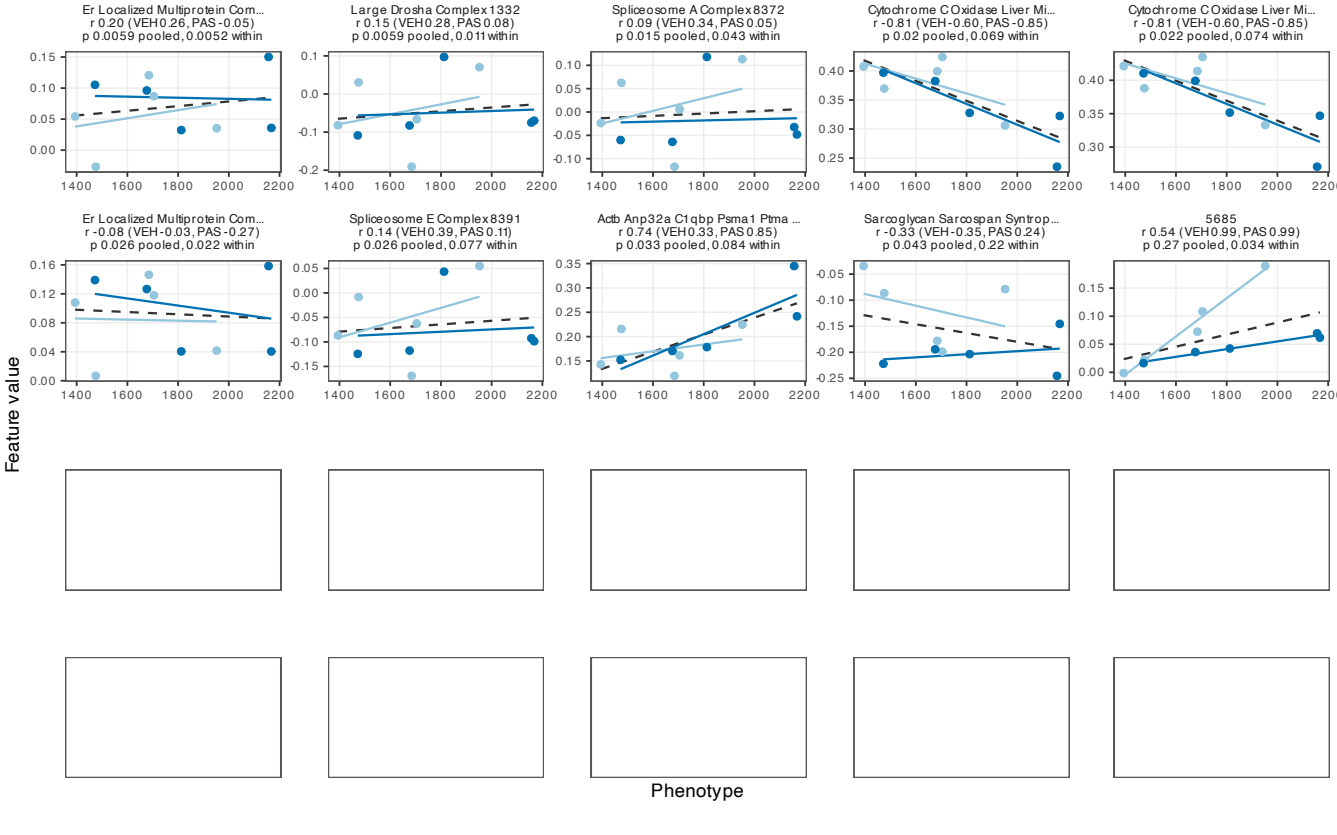
**Male: Complexes associated with Capillary density, p ≤ 0.05 in either model (n = 7)**

*Phenotype effect of PAS in this sex: -18% (p 0.54), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



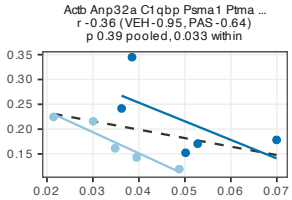
**Male: Complexes associated with CSA, p ≤ 0.05 in either model (n = 10)**

*Phenotype effect of PAS in this sex: +13% (p 0.25), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



**Male: Complexes associated with Satellite cells / fibre, p ≤ 0.05 in either model (n = 1)**

*Phenotype effect of PAS in this sex: +43% (p 0.086), so these associations follow variation between mice, not a PAS response. Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



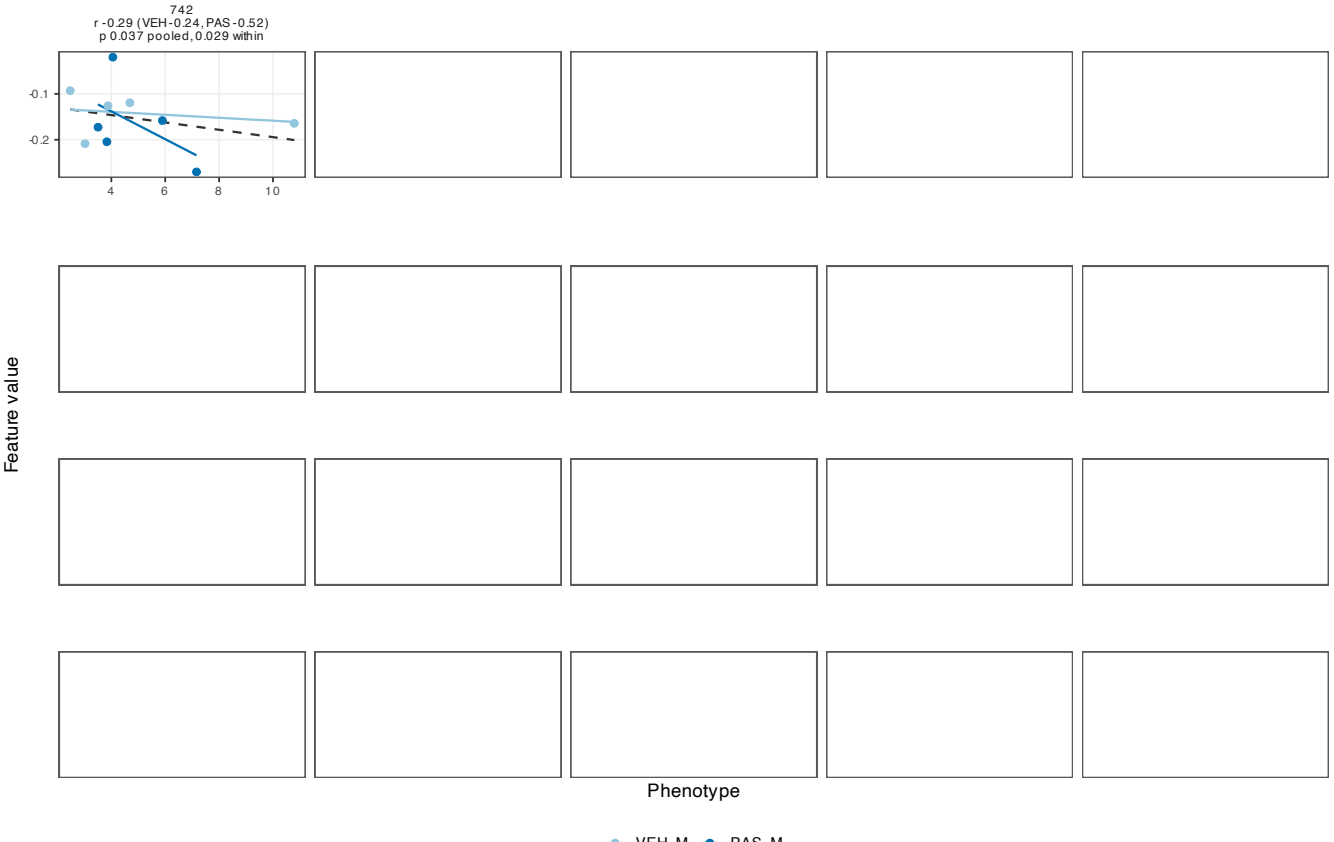

Feature value

Phenotype

VEH_M PAS_M

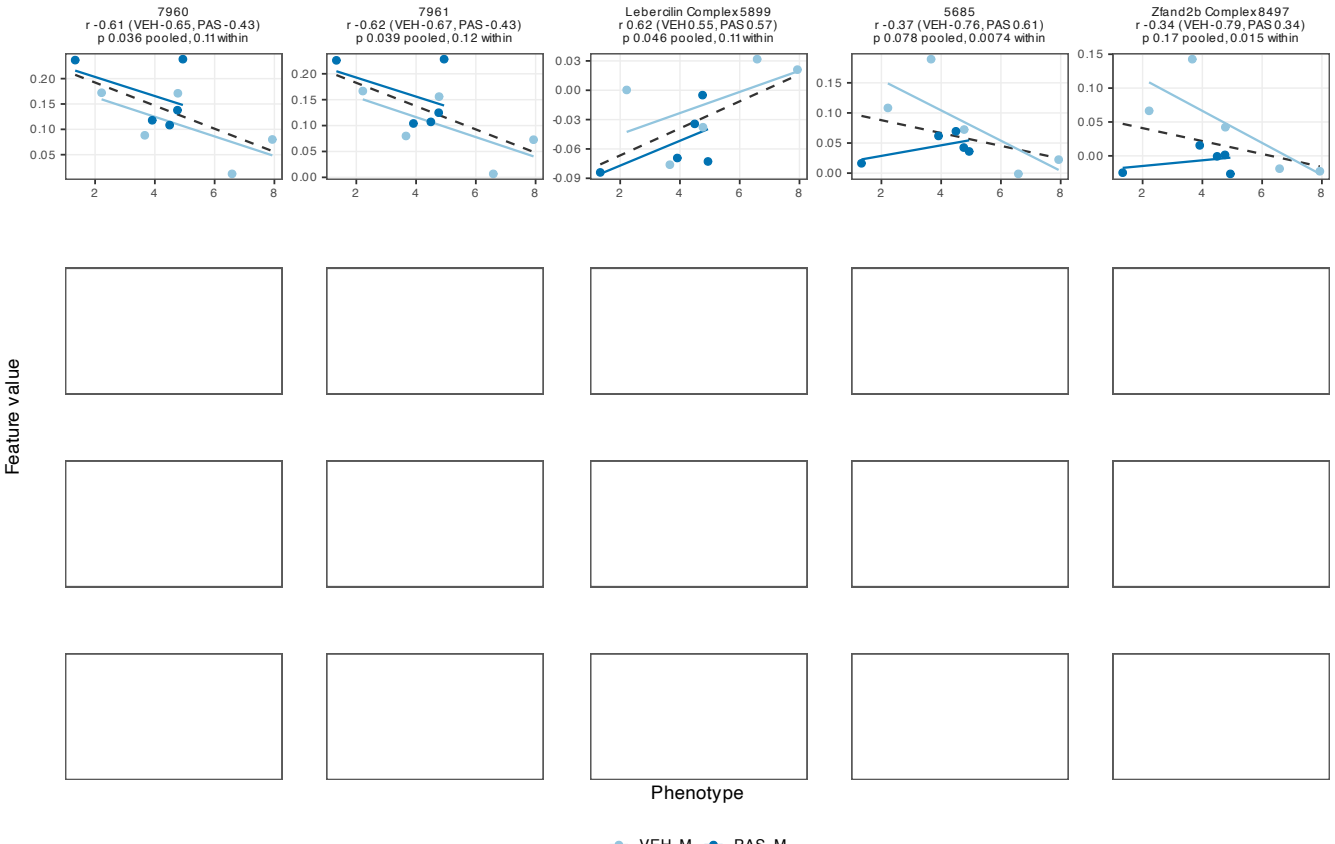
**Male: Complexes associated with Soleus pyruvate, p ≤ 0.05 in either model (n = 1)**

*Phenotype effect of PAS in this sex: +9% (p 0.77), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



**Male: Complexes associated with Plantaris pyruvate, p ≤ 0.05 in either model (n = 5)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*



**Male: Clusters associated with Plantaris pyruvate, p ≤ 0.05 in either model (n = 1)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response. Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model*

